

Econ 230 Globalization

Numbered Transcript

ECON 230 • Dr. Yaron Brook

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Lecture 1: The Global Marketplace

At a glance — Brook on the global marketplace.

- 1.1 I don't know about you guys, but I love my coffee in the morning. Can't get going without probably three cups of coffee, right, sometimes during the morning. I know here in Arizona, I haven't seen any coffee trees out there. No coffee plantations in Arizona, right? And I, for example, I get my coffee from Switzerland. It turns out Switzerland is one of the largest coffee exporters in the world. Coffee in Switzerland, on the Alps? That doesn't make any sense.
- 1.2 But I use something called Nespresso. You guys know the little pods? I love Nespresso, right? You can get all these different flavors and all the different levels of intensity. Just put it in a machine, press a button, and there you have this amazing cup of coffee. So I get my Nespresso, my coffee, from Switzerland. Of course, the Swiss don't grow the coffee, so they have to get it from somewhere. They might get it from Brazil, from Indonesia, from Ethiopia. Coffee was discovered in Ethiopia a few hundred years ago, actually. Starbucks did not even invent coffee.
- 1.3 It was discovered by some goatherder who tasted this thing, and it gave him a little bit of a high, and he liked that. And he took it over to, I think, a monastery close by, and they cooked him, and cooked it, and ate it, and everybody got a high. And they decided, "Oh, this is interesting. Let's do something with this." But today, coffee is produced in dozens of countries. It is then shipped to other countries where it is processed, countries like Switzerland that you would never expect. And it is ultimately delivered then to different coffee shops, different roasters, whatever you do with coffee.
- 1.4 Who knows? I don't care what's done with coffee. This is the beauty of trade. I don't have to know anything about coffee. I just have to buy those little pods that they ship me from Switzerland. And it's a machine that was produced, I doubt that it was actually built in Switzerland, but my guess is the machine probably comes from, I don't know, China, somewhere in Asia, where they can simply, easily, cheaply make that machine. And there you have it. I have my amazing cup of coffee sitting in wherever I happen to be around the world.
- 1.5 And the system of trade has integrated all these factors. The producers, the growers, the producers, the refiners, whatever you do to coffee to get me my cup. And of course, you guys walk into a Starbucks or walk into any coffee shop, and there it is. The coffee is just sitting there. Somehow it got there. From somewhere around the world, again, we don't grow coffee in America. And yet, we all consume it all the time and benefit enormously from that consumption.
- 1.6 I love my iPhone. My iPhone, this is an amazing product. You have to think about life without it. I look around, you guys are mostly too young to remember what life was like

before the iPhone, but this is just an amazing piece of computing power. This is more powerful than the computers that sent man to the moon. This has on it, I have access here to every piece of music ever produced in human history, access to knowledge through the Internet. I have GPS or maps. I still remember coming to Arizona and having to pull out big maps and try to figure out where I was going. No more. I'm all connected, my phone. Everything's right here.

- 1.7 How did I get the iPhone? It says Apple. It's an Apple product. But where does this come from? Well, it turns out that, into this phone, they've placed about 100 different products from 100 different countries. About 100 different countries. The main components of the phone, just the main features of it, come from a global supply chain. You go through some of these. The processing powers, basically, are made by TSMC in Taiwan. The display, this thing, the display itself is from Samsung or LG from South Korea. The flash memory in here is from Toshiba in Japan and Samsung in South Korea.
- 1.8 DRAMs are from a Japanese company. The glass, the glass that doesn't break, right, gorilla glass, this is from a company called Corning, which is headquartered in the United States. Corning is an American company, but they don't make the glass here. They make the glass in Taiwan, Japan, maybe other places. Battery from Samsung and some batteries from China. Camera, Sony; these things, Sony makes them, right? Audio chip is Cirrus Logic, again based in the U.S., but nobody actually makes any logic chips in the United States. So likely TSMC or Samsung in Korea.
- 1.9 Cellular networking, the 5G modem in here, is another American company called Qualcomm. Don't build the chip in America, though. Qualcomm designs the chip. The chip is made, TSMC or Samsung in South Korea. And go on and on. The compass. There's a compass here, Japanese. LiDAR scanner, Sony. Microcontroller or eSIM is from Switzerland. And the DisplayPort interface, I don't even know what that is, but that's from the Netherlands.
- 1.10 And of course, each one of those components is made from stuff that comes from a bunch of different other countries. And each one of those components has to be transferred to where they need to be with ships and planes and other stuff that's made in other countries. So this iPhone, the American iPhone, made by American Apple, is assembled where? Assembled in China. But where's the real value added of this? We'll talk about this in a future class. We'll actually break it down. But who adds the most value to the iPhone?
- 1.11 Chinese who assemble it? Taiwanese who made the chips? Because if you look at the iPhone from a trade perspective, we import it from China. But where's the most value added into this product? Most value added into the product is Apple. It's the design, it's the idea. It's the design, both aesthetic design, but more importantly, the design that's underneath the hood, that makes it work. It's the engineering. And then, of course, it's Apple, so it's the marketing.

- 1.12** But the value added, almost all the value of this, comes from Apple. All the other stuff is relatively cheap. But this is considered a Chinese product for the sake of import and export because it is assembled in China, which is the least valuable piece of the whole equation. Assembly is easy. So Apple has, in a sense, a trade deficit with China, even though Apple contributes more to the trade deficit, to the value of this thing, than China does.
- 1.13** Trade is all around us. Everything we do involves products that are made elsewhere, designed elsewhere, and that we're constantly trading for, internationally and domestically. And today, because the iPhone is relatively cheap and because it has lots of competitors and lots of imitators, almost everybody in the world has an iPhone. We all share a culture of smartphones. Almost everybody in the world today drinks coffee. Comes from a little country in Africa, Ethiopia. And yet today, everybody in the world drinks coffee. Coffee is a huge source of trade.
- 1.14** All of this is done at low cost. Coffee, the iPhone, are just examples of the millions of things that are involved in what we call today globalization. Globalization is just a name we give to the massive amount of trade, of movement of people, movement of goods, movement of ideas, movement of culture. How much music do you listen to that wasn't produced in the United States? How many movies do you see that weren't made in the U.S.? Or maybe they are nominally made by an American studio, but actually filmed in Canada or filmed in Europe, or filmed in Morocco?
- 1.15** Almost everything that we do today is a composition of influences from all over the globe. We live in a globalized world. And the economics of globalization, in some respects, are simple, in other respects, are complicated, very contentious. One of the most contentious topics in America today is trade barriers, trade deficits, tariffs. And as we are taping here in July 2025, there's a lot of discussion about trade deals and tariffs, and this is not new. It turns out, as we'll see in the course, that the second most contentious issue in 19th-century America, only second to slavery, was trade and tariffs.
- 1.16** So in this course, we're going to be talking about all that. We're going to be talking about the essential nature of globalization. What is this about? What happens? How does it work? We're going to be talking about the essence of trade itself. Is trade good or bad? How does it work? We're going to talk about the economics of it, detailed economics of it. We're going to talk about the politics of it, the culture of it, the history. Is it good? Is it bad? Hopefully, by the end of this course, you'll have a good sense of the nature of trade, the nature of globalization.
- 1.17** Now, globalization is more than trade. Globalization is also a culture and ideological influence, and it includes immigration and travel and a lot of different aspects to it. Our focus is going to be primarily on trade, although we'll touch on some of these other issues a little bit. But the focus primarily will be on the issue of trade and how it works. Now, let's just delve into a few more examples about globalization.

- 1.18** People often criticize globalization by saying things like, "All right, so you get cheap T-shirts from Vietnam. Big deal." But the reality is, it is a big deal. Throughout human history, clothing, for example, has been really expensive. We've had to spend a lot of our resources as individuals trying to make or get or trade for good clothes. The fact that today you can buy a T-shirt, which is decent quality, for almost nothing, almost nothing, is a massive achievement.
- 1.19** If you walk around the malls, if you go shopping for clothes, you find store after store after store of clothing stores that promote good clothes, nice clothes, fashionable clothes, hip and cool clothes, that are very moderately priced. It's not true now that in order to wear fashionable stuff, you have to spend thousands of dollars. Indeed, the malls are filled with stores right now. And they change all the time because our fashion change and our taste change. Different things might be here, but almost all those clothes are made in Vietnam or in Bangladesh or in other places around the world.
- 1.20** And yet, we get to now wear really nice clothes made somewhere else, and that means that we get to keep more of our own money, look nice. That's important, right, get to look nice. And then we get to keep some of that money to do other stuff with. As compared to 50 years ago, suddenly 100, 200, 1,000 years ago, when we'd spend a significant amount of our disposable income, if we had any disposable income, on trying to figure out where to get clothes. And they were mostly unbelievably uncomfortable.
- 1.21** Until, really, the revolution in cotton, which happens only about 250, well, maybe 300, 400 years ago, we wore wool. Not the kind of soft, nice wool that you get today, like the coarse, scratchy, horrible wool that existed back then. That was clothing. That was standard stuff. And indeed, the whole textile revolution, if you think about it, is a trade revolution. Who came up with soft, really light, comfortable clothing? What was the material?
- 1.22** Originally, there was the luxury, particularly in Europe, well, silk, which you had to get all the way China. The Europeans had no idea where silk came from, how to make silk. They finally figured it out and stole the idea from the Chinese, and other places could make silk. But silk and then cotton. But these were big technological revolutions that had to be imported into Europe. The British learned how to do cotton and create fabrics that were soft and pleasant from the Indians, from India.
- 1.23** So the fact that we can get cheap T-shirts from Vietnam is really, really, really cool. I mean, this jacket is labeled, the label is, this is a Hugo Boss jacket, so it's German. But you look at the label, it was made in Turkey, and sold in, I don't know where I bought it, probably California. Wow, that's cool. It doesn't cost that much. Light, well made, comfortable. And if you think about that in every realm of our lives, you walk in a grocery store. Here in Arizona, we can't grow anything, because there's no water and no real soil. I don't know how you guys live here.

- 1.24 You walk in a grocery store, any time of the year, there are these beautiful strawberries that come from, they might come from California, but likely they're coming from Mexico these days. You get kiwis from New Zealand. You can get Arctic char from, I don't know, Iceland or something. You get salmon, maybe from Chile, maybe from Maine, maybe from Scotland, maybe from Alaska. Who knows? It depends on the season. It depends on what's available. And you can usually pick, right? Most of the time, you have salmon from three different places, and you can decide if some of it's farm-raised and some of it's wild salmon.
- 1.25 And you can choose what you like flavor-wise, what you think health-wise. If you think about going to get some food, you can choose between these wide variety of different foods from different cultures from different parts of the world. It makes our lives so dramatically more interesting, pleasant, tasteful. And again, people will pooh-pooh, "Oh, yeah. Globalization. You get Vietnamese restaurants. Who cares?" Well, yeah, but it's not just Vietnamese. It's all these other things. And we all should care because it makes life more interesting.
- 1.26 What's life about? Part of life is wearing nice clothes and eating at good restaurants and using your iPhone and drinking coffee. And all of those, every single one of those, depend on the ability to adopt influences from outside, ingredients from all over the world, and integrate them together into our lives. Isn't that what matters in the end? Our lives, the quality of our lives, how interesting and fun and exciting and cheap, in a sense, that we have disposable income to do other things?
- 1.27 I remember a long time ago going to London, and British food, I think the technical term for it is sucks. It is horrible. You can't eat this stuff. I was a little kid in London in the 1960s, and I'm still traumatized by some of the food that I ate there. It was that bad. Today, you go to London, and it's one of the best places in the world to eat food. It's unbelievable how good the food is. There is so much variety. There's so many good restaurants that it's stunning.
- 1.28 My favorite restaurants in London are West African, like Nigerian and Liberian food. I didn't know there was such a thing as West African food. I assumed they ate, but I didn't know there was food-food. Now, these restaurants have one and two Michelin stars. These are really high-end, fancy restaurants treating us to the flavors of West Africa. I don't know if I'm ever going to go to West Africa, but I can eat their food in London. There's amazing Thai restaurants, of course, Indian restaurants, best in the world outside of India, probably in London. And on and on and on.
- 1.29 The diversity, the quantity and the quality. And of course, what has that done, what has that done to British food? It's made British food much better. They finally know how to cook. They've learned from all these people from all around the world how to cook. And their food has been elevated. They use more spices. They bring in more interesting ingredients. They do stuff with the food that they couldn't have imagined in the past. Life in London is a gazillion times better because of that.

- 1.30** As we said, think about film and TV. Netflix now does an amazing business in TV shows from out of the country. Some of my favorite shows on Netflix are from South Korea. And I'm not even talking about Squid Games, which was one, like the biggest show on Netflix, and yet South Korea. Music now is global. And of course, it goes the other way. American films and American music dominate the world.
- 1.31** One of the arguments against globalization we'll talk about in a future class is the idea that the world has become monoculture—that is, Western culture, American culture, dominates the world. You go anywhere in the world, you get a McDonald's, you get Starbucks, you get rap, and you get, I don't know, superhero movies. And there's truth to that. Because even though we might, our culture, we're not that impressed with it, they are. It turns out that there's some value in McDonald's, Starbucks, and our music and our movies. The rest of the world appreciates them.
- 1.32** I once was in Milan when they opened up the first Starbucks in Italy. Like Italy, coffee is Italian culture. Getting that espresso in Italy, that was special. And like American coffee, Starbucks, commoditized coffee, they looked down their nose on that. And yet, at the Starbucks in Milano, the line was around the block when they opened, in order to get in and experience whatever it is that Starbucks is selling that makes it unique. And it's a whole experience.
- 1.33** And it's American culture impacting Italy in what it defines as its culture, which is coffee. But of course, Italians, 500 years ago, didn't know what coffee was. Coffee is a new product for Italians brought in from Africa through the Islamic world, and Italians adopted it, now it's Italian. That's globalization. And you notice it's products, but it's also culture, it's attitude. And indeed, one could argue that part of the globalization around the world is indeed much, again, much more than just the goods.
- 1.34** Think about sports, right? Who are some of the best baseball players today in the world? Japanese. We keep bringing them to the United States to play baseball here. Japanese baseball is incredibly competitive, and the Japanese love baseball. We're getting basketball players from China. And the Chinese love basketball, and they watch basketball, and they adore American basketball players. Again, American culture influencing the rest of the world in terms of their sports.
- 1.35** The rest of the world tries to influence the United States, but soccer doesn't seem to stick over here. Not really. A little bit, but not really. But you see, even in sports, kind of a globalizing influence. Where was soccer invented? Maybe in England. But who dominates soccer today? Southern Europeans and the South Americans. So again, all these ideas, these applications, they all spread.
- 1.36** And as we said, goods are a big part of it. And one of the things that happens over time is that, because goods can travel all over the world, and they can travel all over the world at a relatively low price, the consequence of that is that, in different cultures, the prices of goods start to be about the same. And it's not just goods. Over time, wages

start to be about the same. And what you get is, we get richer because we've got cheaper stuff. So our disposable income goes up. We can spend it on more different things. The people making the stuff in these other countries are getting richer because they're making stuff and they're selling it to Americans for dollars, and that makes them richer.

- 1.37 So the world generally is becoming wealthier and more economically equal over time. We'll talk about inequality across countries, which is, again, globalization. The claim is that it makes inequality worse, but it makes inequality, to the extent that inequality matters, it makes inequality much better because it helps raise up poor people in relatively poor countries because now they're producing for rich people. If you produce for rich people, you make more money. Rich people benefit because they get cheaper stuff. Poor people benefit because they get to sell the rich people's stuff. They get their money in return.
- 1.38 Globalization also involves immigration. There is massive immigration going on in the world right now, but there has always been a lot of immigration. People have traveled. We all probably started in Africa and then spread from there. If you follow the history of human movement, we've always moved around. There's a huge similarity between the German language, what people speak in Germany, and what people speak in northern India. They speak a very similar, there's real similarities in language.
- 1.39 And if you look at their genes, they have very similar genes. Might not have the same color of skin or color of eyes, but they have very similar genes, because they both started from the same tribe in, probably around Ukraine, Ukraine, Russia, in the plains there. One went west to Germany, the other one went south to India. Both have the same origins, but we are migrating species. We've always migrated. We've always immigrated.
- 1.40 And immigration, of course, even more contentious maybe than trade in today's world, not in the 19th century. The 19th century, immigration was pretty much open into the United States. Trade was the big issue, not immigration. In the 21st century, immigration is a contentious issue. It's restricted, but it's still happening on large scales all over the world. People move to where there are better jobs, better opportunities, better quality of life, better standard of living. There's a constant mingling of people in the world.
- 1.41 And ultimately, that solves economic problems. It enhances, it revitalizes cultures. It has the potential to do negatives as well. We're not going to talk much about immigration in this course. Maybe that's another course for the future. But it's part of this globalizing tendency, and you go to New York and you see how many people from how many different countries live in one city and produce and create and build. Any city in the United States today has this amazing diversity of immigrants in it.

- 1.42 And of course, part of globalization is not just immigration, but it's also travel. We live in a world where, 150 years ago, many of our ancestors came to America. They got on a boat and they left Europe, and they knew they would never see their family again. It took them three weeks to get here, and they had no idea where they were going because they'd never seen a video, they'd never seen a photograph, they'd never seen anything. Maybe they read some story about America, but they had no clue where they were going.
- 1.43 And they arrived, and indeed, many of them would never see their family in Europe again. Today, we can get on a plane. We can hop over to Europe for a weekend. You get a little bit of jet lag if you do it for a weekend, but you can certainly do a week, and you can go anywhere in the world today, almost anywhere. Don't want to go to North Korea. But other than that, you can pretty much go anywhere in the world today—travel, stay at hotels, eat the local food, interact with local people.
- 1.44 And there's a real—you know, people do it in large quantities. Tourism is going off the chart because it's enjoyable. It's part of what people use their disposal income for. It's fun to see the world. It's fun to be exposed to different people, different cultures, different settings, to learn about human history firsthand by being places. Travel has gotten so bad, so bad, so good, so plentiful, that in Europe, they're really complaining. Like many cities in Europe are now starting to restrict tourism because it gets so crowded there, and Airbnbs are driving up, supposedly, driving up rents, and people are complaining.
- 1.45 But the reality is that it's, tourism is huge because we enjoy it and we get to see the world. And it's fun to see the world. It benefits our lives. And of course, we talked about culture, music, fashion, food, so on. But the other aspect of culture that gets exported and moved around the world are ideas. Ideas. Well, it used to be that if you talked about Western civilization 100 years ago, what you meant was a few countries in Western Europe and the United States.
- 1.46 And those ideas of science and engineering, and the idea of individualism, the value of the individual, the worth of the individual, the idea of the pursuit of happiness, the individual pursuit of happiness, the idea of political liberty, political freedom, those were ideas limited to a small group of cultures, a small number of people. Western Europe, United States. Today, Western civilization, the ideas of Western civilization, are global ideas. You travel to any continent, and the people in those continents who are studying science and engineering, and value science and engineering, and value human reason, and they care about their own lives.
- 1.47 They want to pursue their own happiness. They want to do cool stuff with themselves. They're not just about the tribe. And there's political liberty. There's more political liberty today around the world than any time in human history. Well, maybe 10 years ago, there was a little bit more. But generally, over this period, there's been a huge increase. And if you go to a place like Tokyo, you go to a place like Seoul, South Korea,

or Hong Kong, or Singapore, yes, they're different in the sense that you can't read the type, you can't read the street signs. But in so many ways, they're the same.

- 1.48** Same skyscrapers, same clothes. Everybody's wearing basically the same clothes that we wear. They're all doing the kind of activities, the kind of work that we do. They're all drinking coffee. They're eating Japanese food, but we eat Japanese food here, so we know what Japanese food is like. Globalization has taught us that. And they look like Western, vibrant places. And we know that they elect their governments, and they value their freedom, and they value their own lives, and they value their own happiness.
- 1.49** And the Japanese weren't like that. Political liberty is not inherent to Japanese culture, or to Singaporean culture, or to Korean culture. They learned that from the West. Globalization has had this incredible impact on the world by filtering out the bad ideas and elevating the good ideas. Now, not always. Sometimes the bad ideas get elevated as well, and we export bad ideas as well. China probably wouldn't have come up with communism if it hadn't been invented by Karl Marx in the West. They probably wouldn't have come up with it themselves.
- 1.50** But over time, hopefully, good ideas went out, and over time, different cultures adopt the better ideas, no matter where they're developed. It doesn't matter who developed them. Is the idea good or not? Is the clothing good or not? Is the food good or not? Who cares where it came from? Does it enhance human wellbeing? Does it make life better? So globalization is everywhere. Globalization is in our lives and in the lives of pretty much everybody in the world. There's nobody not touched by this. There's nobody who doesn't live it every single day, and it's not part of their lives every single day.
- 1.51** But some people are going to tell you that this is bad. And there are a lot of reasons why they think this is bad. Some of them, economic reasons. We're told regularly that when we trade with Vietnam for the T-shirts, we lose because we're giving them money and we get the T-shirts in return. And somehow giving them money, they are better off, but we are somehow worse off. The only reason we give them the money is to get the T-shirts and we value the T-shirt more than the money, right? That's why you do it. Why do you go to a store and buy a T-shirt? Because it cost you 20 bucks, because the T-shirt is more valuable to you than the \$20 you gave up.
- 1.52** Well, if that store happens to be in Vietnam, it's still the same principle. You're not giving up your money unless you're getting something of greater value in return. And yet we're told that if the money somehow goes outside of the country, it's bad. One of the questions you should ask yourself, and we'll talk about, is where does that money go? You can't use dollars to buy stuff in Vietnam. So what happens to those dollars? Where do they go?

- 1.53** We're told that our culture gets diluted by these foreign influences. Well, isn't it our responsibility to keep the good stuff? And there's some stuff that other people do better than us. Suddenly, the Brits improved their culture by importing good chefs and good cuisine from other parts of the world. And where would America be without pizza? So is it really diluting our culture, or is it an opportunity to actually improve our culture? Is there no area in which our culture could be better?
- 1.54** We're told that we're poorer because of globalization. Hard to tell by looking at the numbers. We all seem quite a bit richer. We'll look at some of those numbers. We're told that we lose jobs because of globalization. Does that make sense? When we trade, do we lose jobs? Or do we gain work? Do we gain jobs? We'll check that out. So there are a lot of arguments against globalization, more so right now than ever, because clearly, the Trump administration is hostile to global trade, to international trade. So you get bombarded these days by stories about how terrible trade actually is. We're going to look at the details of that and try to figure out if that is true or not.
- 1.55** Just a few points before we get into that in greater depth, a few principles. People can be self-sufficient. You can indeed live on a farm and grow all the things you need and sew your own clothes and do all that. And for much of human history, we did that. And we were really, really, really, really poor. And we've existed in small tribes, where the tribe did everything and produced everything and created everything, and very insular and controlled. And the culture was very insular and controlled. And that's all we did.
- 1.56** And indeed, there's still tribes out there—you can find them in the Amazon, maybe—where they haven't had any exposure to the outside world and they still live like they did 10,000 years ago. And they're really, really, really, really, really poor. So it is possible to create a world in which we're self-sufficient, but most of us will be dead. The population of planet would be a lot smaller. There'd be a lot fewer jobs, and we'd be a lot poorer, with a lot fewer opportunities and a lot less fun in life.
- 1.57** You'd spend all your time, every single moment of every single day, trying to produce the clothes and the food and all the things that you need in order to live, and it still would suck. And you'd have to have 12 kids to help you. You'd have to have a lot of kids so that they could help you. People think the Industrial Revolution invented child labor. Child labor has been with the human race from the beginning of time. One of the reasons to have children is so that they could labor for you, so that you could survive, particularly as you got older.
- 1.58** But most of these tribes ultimately exposed themselves to other tribes, to the rest of the world. They migrated, they interacted. Trade started, really, with the beginning of time. We've always been traders, more effective ones, less effective ones. And yes, there's still isolated tribes that are not engaged in the world. But the successful ones, the ones that got rich, are the ones that actually traded.

- 1.59** And trade has been a feature of human history since the beginning of civilization, since we started to produce in scale, anything in scale, whether ideas or buildings or agriculture, food, stuff. Trade has been essential to that. Trade among us and trade internationally on the scope that was possible at any given point in time. So, trade is essential for the kind of life, kind of world, in which we live, for the standard of living, for the quality of life in which we live. And that has been true, and that is true, throughout history.
- 1.60** One other point about trade that is going to be a theme throughout, that I want you to always remember, and hopefully, if you take one thing out of this class, course, this course, this is the one thing I want you to take, because I think, if you get this, and it's pretty simple, if you get this, then all the noise around trade that people talk about, you can boil down to what's essential. All trade happens between individuals. Countries don't trade. Individuals trade.
- 1.61** I buy my coffee from the local store. The local store, which is run by individuals, owned by people, import that from Switzerland, from a company in Switzerland called Nespresso that has shareholders and has employees, people, individuals, who work to produce these little pods for coffee. They buy their coffee from individual farmers, somewhere in Africa, probably. And every single one of those transactions, every single one of them, is a win-win transaction because you wouldn't have it any other way.
- 1.62** You're not going to go into a clothing store and say, "I'm here to lose. I don't want a good deal. I want to overpay for the stuff I'm going to get." No, every transaction is win-win. Now, people make mistakes, and there is exploitation in the world, and people are enslaved and people do bad stuff, but for the most part, the overwhelming majority, 99.9999, you can take it out quite a bit, of trades that happen are win-win trade. And all globalization is doing is extending the number of individuals I can trade with.
- 1.63** Usually, I can trade with a few people in my tribe. Now I can trade with 8 billion people around the world. And that is super cool. That is super cool that we can do that. So trade is always between individuals. We might talk in terms of nations, but that's just an easy way to express ourselves. Behind that is this idea of, there's an individual in Vietnam making that T-shirt. And there's a whole chain of trade that happens. The government is involved in regulating and all that, but the government doesn't touch the T-shirt before I get it. I'm not buying it from the Vietnamese government. I'm buying it from the Vietnamese person who made it.
- 1.64** OK, so we're going to be delving deep into many of these questions and issues over the next few classes. And what I want to start with is, because I want to get it out of the way at the beginning, is to talk about economics, to talk about the economics of trade. Then once we understand the principles behind the economics of trade, why trade makes sense from an economic perspective, then we can start getting into the politics and into the objections and into everything else.

- 1.65** But we have a common language, which is, how does this work? And what is it about trade, how is it that trade benefits us, if it does? So I think we should start in the discussion of the economics of trade with an understanding. Really, if you think about economics, economics is a study of two things, really, at the end of the day. It's a study of production, how we produce the values that we need in order to survive or to thrive, to be successful. How do we produce T-shirts and food and coffee and cameras and everything else?
- 1.66** And it's a study of trade. Really, if I define economics, it's that field that studies production and trade. Because those are the two fundamental activities that human beings engage in, in order to sustain ourselves materially and arguably even spiritually. We need to produce. You produce the values; they don't just appear. We can be gatherers. Even hunting is production. You need a strategy. You need tools. You need weapons. You need stuff to hunt. You need to produce to hunt.
- 1.67** Maybe you can gather berries and nuts, but human beings cannot survive for very long on berries and nuts. I've tried it. It's just not—don't try it. So production and trade are the fundamentals. And in order to trade, you have to have stuff to trade. So the real fundamental, the essence of human beings' economic activity in the world, is production. You have to be able to produce. You have to be able to make stuff. You have to be able to make stuff. You have to go to work.
- 1.68** You can't consume, you can't trade. You have nothing to trade for. In a broader society, where we're trading goods for goods, you have to have the goods to be able to go and trade for other goods. So you have to produce the goods first, and then you can go and trade. So, production is the primary in economics. It is the building block of everything else in economics. And trade comes, in a sense, after production. We start with production, and then we get to trade stuff.
- 1.69** One of the observations made by, in many ways, the father of economics, or certainly the father of modern economics, Adam Smith, in *The Wealth of Nations*, which was published in 1776, one of the observations he makes is that, as producers, we are much better if we specialize. Specialization and a division of labor, where we each do something different, where we each specialize, in where we have an advantage, that specialization improves our productive capabilities dramatically. We can make a lot more stuff if we specialize than if each one of us tries to produce everything, if we're all generalists.
- 1.70** And there are a number of reasons for this. Over time, we become better at the thing that we're doing. Practice improves our skill set. And pretty much anything you do, practice will make you better at it. So specialization allows you to do it over and over and over again, the same thing. And you figure out all kinds of ways to make that thing that you do better, more efficient, more productive. So specialization makes you dramatically more efficient at something.

- 1.71** And it then encourages us to take the next big step forward, which is, figure out how to mechanize this, how to innovate, how to replace ourselves with a machine, which is the ultimate, is if a machine could do what we're doing. So innovation comes from people who know a particular task and they say, "I wonder if there's a way to do this significantly better. What if I introduced a machine here that helped me do this piece of it? What if I innovated?"
- 1.72** A lot of medical devices are invented by whom? By doctors, who do a certain procedure and they go, "Huh. It would be really helpful if I had this particular type of clamp, or if I had this particular robot that I could send into the human body and do the surgery for me. I just do this joystick." I mean, that's where they come from. They don't come from some technologist sitting at home thinking, "I wonder how I could improve surgery in human beings." They usually come from surgeons who actually do it and say, "There are better ways to do this."
- 1.73** The more you specialize, the more you see what's going on, the more you're likely to figure out ways. In Adam Smith's time, it was just the beginning of mechanization, the beginning of the Industrial Revolution, the beginning of starting to use machines. 1776 is the year where the steam engine was first used for commercial purpose. Before that, its only real use was to drain mines from water, so they were used as pumps. After that, they started being used in textile mills in automating textile productions.
- 1.74** But it's only because somebody specializes in textiles that they're going to figure out and think about, "Oh, there's a steam engine. I wonder if I could use it here. How do we do that?" And you do it. So Adam Smith is very, very early on in the whole idea of mechanization, and yet he sees it. He sees that specialization is going to bring this. If you're a blacksmith, you're more likely to innovate tools for blacksmiths, new tools to make for that.
- 1.75** But if you just dabble, if you're just doing everything, who has time to innovate? And you don't know any one thing well enough to be able to innovate in that one thing. So with specialization, observation that Adam Smith makes and is absolutely 100% true in the world, output rises as specialization increases. And as specialization increases, what else has to happen? If I'm just a blacksmith and you're the farmer and you're the doctor, what else has to happen? Because I'm not doing everything. So what has to happen now? We have to trade.
- 1.76** So if I'm going to get the food and I'm going to get medical services, I have to trade my blacksmith abilities for these other things. In another course I have here in the Peterson Academy, we talk about money, and money serves as a mechanism by which we do the trade. But ultimately, as specialization increases, trade increases, because now I need the things that I was doing before, but now I'm specialized. I'm not doing them, so I need you to do them. You're specialized in them. I trade with you.

- 1.77** And then as trade increases, there are more opportunities to specialize. If I'm on a farm doing five things, and now I can buy four of them, and I can specialize just in watermelons, I can just become the best watermelon farmer in the world, then I can get my other foodstuff from other people, then the fact that this trade allows me to specialize—and this is a spiral, right? It's hard to tell, what starts, the chicken or the egg, right? But there's a spiral. Specialization encourages trade, encourages specialization, encourages trade.
- 1.78** And both of those constantly increase total economic output. We become richer because, again, specialization makes it possible for us to become more efficient. And we're producing more stuff per unit. We innovate even more stuff per unit, and we become richer. How we specialize in the economy is going to be based on a comparative advantage, on the relative advantage we have in producing this particular thing relative to other people.
- 1.79** In next class, we'll be going through an example of comparative advantage. It's maybe the most important concept to really fully understand, specialization and trade, and yet it is very misunderstood. It's hard to hold because there's a certain sense in which it's counterintuitive. It doesn't quite work in an intuitive way. A lot of the confusions out there about trade, I think, come from the fact that people don't fully understand comparative advantage.
- 1.80** But if we all, as we'll see in the example, if we all work in what we have a comparative advantage of, everybody is ultimately better off. And you produce more stuff. From an economic perspective, you can show, and we'll try to show that as we go through the next class, jobs. Jobs are created, not destroyed. The more people specialize and the more they trade, the more time there is to do other things; the more time there is to invent new things, to produce new things, human desires, human wants.
- 1.81** You'd even say human needs are actually infinite. Every time we think, "No, can't want more than this," somebody invents some product, and said, "Oh, yeah, I want one of those." And there's no end to it. There really is no end to it because we're a conceptual being, because we can imagine unbelievable things. Right now, it's really cool to go on vacation to Europe. But in 50 years, it might be really cool to go on vacation to Mars. And who knows, 50 years after that, where it might be cool to go on vacation.
- 1.82** There's no end to what we aspire to, what we desire, what we can achieve, what we can want. And this is not a bad thing. It's like, you climb a mountain, and then you could just sit there and die of boredom, or you can go climb the next mountain. Human life is about constantly climbing the next mountain. And that's a good thing, not a bad thing. In spite of what many psychologists will tell you, it's a good thing. You need challenges. You need to keep pushing the boundaries and keep going.
- 1.83** So, the fact that we specialize and we're now richer because we're more productive opens up massive opportunities for us. There's so many things that didn't exist before

we got rich. Like spectator sports—they existed a little bit in Athens and stuff, and in Rome, they were pretty ugly in Rome, right? But then they disappeared. There's no spectator sports in the Middle Ages, or even in the Renaissance, there's barely any. And today, think about how much time you guys probably consume watching football, baseball, basketball, and everything else. I mean, huge quantities of time.

- 1.84** Where did that time come from? From the fact that you specialize in trade. Otherwise, you don't have that time. Restaurants are a completely modern invention. There weren't no restaurants 400 years ago. You had inns for people who had to travel from place to place and to stop overnight and would grab something to eat. But going out to eat, the idea of going out to eat, is a phenomenon of the last—mostly, it's a phenomenon of wealth, of specialization.
- 1.85** I now don't have to worry even about cooking. I don't have to learn how to cook. I can eat much better by people who are professional cooks. This is what they do for a living, and it's not that expensive, given the quality of what you're getting and given that you have time now to specialize. Travel, vacation. The concept vacation is a completely new concept. Up until 100 years ago, you worked every second of the day to feed yourself. Suddenly, we have spare time. We get time off. We can travel. Travel, wow. It's unheard of.
- 1.86** This is generally culturally true. We have no appreciation for how rich we are, how comfortable our lives is, and how many amazing things we can do with our lives that just a few decades ago were not possible, certainly 100 years ago were not possible. Nobody went out to eat a hundred years ago. I mean, maybe a few aristocrats, like the really, really, really rich, went on vacation. Yes, J. P. Morgan went on vacation in 1900. Like, maybe the top 1%. Nobody else did.
- 1.87** Today, everybody does. Almost everybody goes on vacation. Maybe it's a simple vacation, maybe it's a really tough, maybe it's an amazing vacation, but really expensive one. But everybody has the ability to go to the beach. You can fly to the beach. That was impossible. So free trade, trade, makes it possible for us to have more free time. And think about it—the more people you trade with, the more specialization happens, the more wealth you can have, the more free time you can have, the less stuff you have to do for yourself.
- 1.88** So if you're in a group of 100 people and they're all specializing, OK, there's a certain amount of trade that can happen then. That's good. Now it's a thousand people. Now it's a million. Now it's eight billion. The amount of wealth you can attain from eight billion is dramatically higher than the amount of wealth you can get from a hundred.
- 1.89** Here's a simple example, which I think is easy to understand. If you're a basketball player, professional basketball player, professional NBA player, and you're the best in the world, what is it? Thirty-five years ago, calling Michael Jordan, right? Best basketball player in the world, back then. He made, when he was the best basketball

player in the world 30 years ago, a lot less in inflation-adjusted dollars than LeBron James makes today. Why?

- 1.90** Is LeBron James a better player than Michael Jordan? Well, that's a big argument going on about whether he is or isn't, but not by that much. But he makes a lot more. I mean, LeBron James is a billionaire while he's still playing. I mean, Michael Jordan invested well and became a billionaire, but he didn't become a billionaire from playing basketball. LeBron James became a billionaire just because of how much money he makes every year, orders of magnitude greater than Michael.
- 1.91** Why? Because in the '90s, the NBA was watched primarily by Americans. Had an audience of 350 million people. You can make a lot of money off of 350 million people. But today, basketball is watched by billions of people. You can make a lot more money off of billions of people than you can from 300 million. So LeBron James, even if you collect the same amount of money per fan, is going to make a lot more money than Michael Jordan because there are just so many more fans.
- 1.92** Any product that you make today, if you can sell to a global audience of 8 billion people, versus if you can only sell it to your local village of 100 people, where are you going to make more money? 8 billion. And you're going to make it cheaper because economies of scale. You can now make 8 billion of them, instead of just 100 of them. 8 billion are going to cost you a lot less to make per unit than 100. So your profit margins are going to be higher. In every respect, a big market is better than a small market. So globalization in that sense makes us all richer. Free trade.
- 1.93** Now, it's true that trade destroys certain jobs. Some people lose, lose jobs in free trade. All the people who made textiles in North Carolina in the United States 50 years ago, those people are ultimately out of a job. And today, all those textiles are being made, let's say, in Vietnam. But since we all got richer, we all are buying other stuff. Many more jobs are created than are destroyed. Many more jobs are created than are destroyed. And you can see that unemployment rate in the United States is not high. It's very low.
- 1.94** And since the jobs in North Carolina went away, many, many more jobs have been created than went away. And all that capital that was invested in making T-shirts is now invested in, I don't know, NVIDIA stock, maybe, in building AI? So we add value, we have more jobs. I mean, ultimately, jobs are going to be destroyed because technology evolves, because people specialize, become better than we are.
- 1.95** A lot of jobs, for example, in the Rust Belt, people talk a lot about the jobs in the Rust Belt going away. You know where they went? People talk about China, but no, most of those jobs went to South Carolina and Alabama, unionized states where they don't have unions, because it was more efficient to build there. So there are lots of reasons why jobs might be destroyed. But in the big picture, more jobs are created when they are shifted to more productive activities. More productive people are doing them.

- 1.96** In this case, even within the United States, jobs are shifting all the time. A lot of jobs were destroyed, by the way, in the buggy industry in the early part of the 20th century. Buggies, remember buggies? A lot of jobs were destroyed there. So what? Jobs are going to be destroyed all the time. That's what progress is about. Constant jobs disappearing, but more jobs, better jobs, being created. And that is what's happened in the 20th and 21st century. Indeed, trade results in no net job reductions, either in the home country or anywhere else.
- 1.97** And then finally, as we'll see in the next class, trade deficits are meaningless. They're not good, they're not bad. They don't mean anything other than an explanation for the flow of money and goods. I have a massive trade deficit with my grocery store. Massive. I go there, I leave my dollars, I get groceries. They've never hired me to come and speak. Like I said, they never give me the money back. I never get the money back. I give them the money, I get the groceries, and I walk. That's a trade deficit.
- 1.98** We go to Vietnam, we buy T-shirts. The Vietnamese don't buy much stuff from America. Why? Because they're poor. We'll get to that. But they don't buy much stuff from America, so we have a trade deficit. It's the exact same trade deficit I have with my groceries. And with my barber, and with my doctor, and with everybody I interact with. Now, Peterson Academy has a trade deficit with me, because they pay me. I haven't bought stuff from them.
- 1.99** But that's the nature of trade. They're not reciprocal, that we buy exactly the same amount from each other. That's the beauty of a complex economy. Trade deficits do not mean really anything. They're certainly not evidence of economic or policy problems. They can be. Economic and policy problems can cause a trade deficit. But then the problem is the economic policies, not the trade deficit. We'll talk about when that can happen and what the problem is.
- 1.100** We got a little touch of the economics of trade. We're going to dig now deeper into a few of those areas. The next class, we're going to talk about comparative advantage, and we're going to go through some numbers and some examples of how it actually works. We will be digging deeper into trade deficits and what they mean and how they function. And yeah, a lot still to come.

Lecture 2: Why Trade Works

At a glance — Why trade works.

- 2.1 So last class, we talked about the general principles around globalization and trade, some of the broad ideas around this, and some of the economic—we started into some of the economic principles, and we mentioned the idea of comparative advantage. What I want to do now is really dig a little deeper into comparative advantage, and particularly give you an example of it. We'll get to that example in a second.
- 2.2 But the principle behind comparative advantage is that the fact that you are capable and able to produce something, and maybe even capable and able to produce something really, really efficiently and well, does not mean you should. You might be even better at producing something else. So what you choose to specialize in—first of all, specialization is good. Second, what you choose to specialize in is going to depend on, from an economic perspective, on your comparative advantage and your ability to trade for the alternative thing.
- 2.3 You could have people out there who are good at everything. They're really geniuses. Whatever they touch is going to turn to gold. Should they touch everything? The answer is no, because then they get diluted. They should specialize just like everybody else. And comparative advantage gives us some of the principles on how to specialize, what to specialize in. And it shows that the fact that they specialize and they'll produce everything benefits themselves and benefits those around them as well.
- 2.4 So, the fact that you have the technical ability to produce a product is irrelevant for determining if you should produce that product yourself, or you should produce something else and then trade for the product. And you always have to think about this concept in economics. There's a concept, an important concept in economics, we'll go deeper into it in a little while, called the opportunity cost. You have to always think about, what could I be doing with my time?
- 2.5 And in finance, we have to talk about, what could I be doing with this money? What is my opportunity cost of doing this, rather than that? So it's an important idea in economics. You're not just, "I'll do what I'm good at." Well, what else could I be doing in? Am I even better at something else? And of course, this subtracts away from what you're interested in, which is also pretty important. But this is just about the productive value.
- 2.6 David Ricardo, who's an economist in the early and mid, really the late 18th century, early 19th century, a British economist, was the first scholar to identify and articulate the idea of comparative advantage. He did it in the book, *On the Principles of Political Economy and Taxation*, in 1817. If Adam Smith thought specialization is what's important, and comparative advantage comes out of specialization—you really, really

get good at something—Ricardo thought, because we have comparative advantage, we specialize.

- 2.7** That is, he put comparative advantage first, specialization second. Adam Smith did it the other way around. As we'll see, they're both correct. It's a continuous spiral. You're constantly, it feeds off of each other. It's a virtuous circle of increased productivity. Specialization begets greater comparative advantage, greater comparative advantage begets specialization.
- 2.8** So let's do an example, and I've taken this example. I tried to come up with one myself, but in the end, a guy named Don Boudreaux has this great example in one of his essays. I just figured I'd just take his, because it covers a lot of this material, and he does it so well. Don Boudreaux is an economist at George Mason University, often writes in the Wall Street Journal. He often writes, recently he's been writing a lot with Phil Gramm, the former senator from Texas. They write a lot about trade, and one of his expertise in economics is on trade.
- 2.9** So let's run through this example of comparative advantage. Let's say you're a radiologist. We'll call you R. You're a radiologist, and you make \$240,000 a year. Now, you want to build a deck for your home, and you're really good at deck-building. You're like a world-class deck-builder, in addition to being a world-class neurologist or radiologist, world-class radiologist and a world-class deck-builder. You could build the deck in one month, right? One month. But you would have to take time off. You would have to not earn anything in radiology that month.
- 2.10** So it would cost you to build a deck, it would cost you \$20,000. That is the cost of the deck if you built it yourself, because you'd have to take a month off. So you think the deck, in terms of value to you, to your life, the benefits you will get, you think the deck is worth to you, \$21,000. Economists, we try to translate everything, any value we have for something, into dollars. Based on this, you would build a deck. The deck is worth \$21,000. It would cost you the month off of working as a radiologist, \$20,000. Good deal. You're up \$1,000 in a sense. You're out \$20,000, but you're up value, utility, of \$1,000 because of the \$21,000 value.
- 2.11** It turns out that you have a neighbor. Neighbor is a bookkeeper. Your neighbor, we'll call him N, makes \$84,000 a year. Now, he's not a very good deck-builder, but he's willing to come and build your deck. And it would take him two months to build it. It would take you one month. You're really efficient. You're really good at this. It would take him two months to do it. So he would have to take two months off of work. So it would cost him—this is \$7,000 a month—it would cost him \$14,000, which is two months.
- 2.12** So you would have to pay him more than \$14,000 to get him to come and build the deck. But that's a lot less than you taking the time off and building it for \$20,000. So you decide to hire your neighbor. And even though you're a better deck-builder, it'll be

faster. You're going to hire the neighbor and you're going to pay him \$14,000. I mean, this is just for illustration, and \$1. So he's going to be \$1 richer, and he's going to build your deck, and you are going to have your \$240,000. You haven't given up anything.

2.13 You're going to get a \$21,000 value, the deck, for \$14,001. Your neighbor's better off. He's made a dollar more than he would have otherwise. And you're better off because you didn't have to spend \$20,000. You only spent \$14,001. So even though you are much better at building decks, you shouldn't build it. There's somebody else who could build it cheaper, not because he's better than you. That's what's interesting about the example. He's worse than you. But his opportunity cost, his work, he makes less money. So he's willing to do it for less than you, even though it'll take him double the time. Everybody's better off in this example.

2.14 So, now let's assume there's some advantage to specialization. You now, you've got more time to spend on radiology. So you now are going to, your wage has gone up to \$252,000. So it's \$21,000 a month. If you built the deck yourself, it would take you one month and it would be 21,000. And at this point, it's like, you're in different—it's not that good of a deal anymore to take the one month off, because the deck is only worth 21,000.

2.15 So you become a better radiologist. But in a sense, you become a worse deck builder. Because before, it cost you \$20,000 to build a deck. Now it's going to cost you \$21,000 to build a deck. It's more expensive for you to build a deck, even though now, because you're much better at radiology, your wages went up in radiology. And the better you do in radiology, the less it makes sense for you to ever go and build a deck.

2.16 So Jones, he still is same salary, same amount. But he now can do it, this 14,000, is from your perspective even cheaper. Because the difference between 21,000 and 14,000 is higher than the difference between 14,000 and 20,000. Here it's 21,000 a month. So you could actually afford to pay Jones more because your radiology skills have improved. She can afford to pay more than 14,001. He could negotiate. He now knows you're making more money. He could negotiate even a higher wage.

2.17 So the fact that you became a better radiologist actually makes it possible for Jones to earn more, even though he hasn't become a better deck-builder. And see, again, everybody's better off, and the world is better off in a sense. Your clients are better off because they get a better radiologist. And you can spend all your time on the things you're good on radiology.

2.18 Now, imagine Jones, or imagine your neighbor, your neighbor, imagine he lives across the border in Canada. You live in northern Michigan. He lives in Canada. And he's going to cross the border and come and do the job for you. Does that change anything here, the fact that he's Canadian and you're American? Does America lose? Does Canada—I mean, two people trading, exchanging. At the end of the day, you get a deck. He gets a

little bit of income that's above the income he makes in Canada. But nothing has changed. The fact that he's a foreigner makes nothing.

- 2.19** But let's assume now there's a third person. There's a third person. He's a Canadian. And this Canadian is a neurosurgeon, and he makes \$360,000 a year. He's not very good at deck-building. He's very similar to your neighbor. So it would take him two months to build the deck. So if you wanted to buy two months of the Canadian's time, it would cost you \$60,000. And you're going, "No way. I can build it for \$21,000, and my neighbor can build it for \$14,000, or somewhere between 14 and 20. It depends how we negotiate. No way am I hiring a Canadian to build it for me."
- 2.20** But what if the Canadian government stepped in and said, "Canadian, if you go over to America and build this deck, we'll give you 50,000 bucks." What do we call that? Call it a subsidy. We want to develop in Canadians the deck-building skill, for whatever Canadian reason we might have. Hard to understand Canadians, generally, what drives them. I'm kidding. So you get a subsidy, \$50,000, that the Canadian government gives him.
- 2.21** So, how much are you going to have to pay him in order to come and build the deck for you? Well, basically, you have to pay him 10,001. He's cheaper than your neighbor. Good deal. And maybe he'll negotiate 11,000, 12,000. But as long as he's less than 14, he's better off and you're better off. The Canadian is better off. If you hire him, you're better off. What about your neighbor? He's no worse off than he was originally. Still making 84,000. He just doesn't get the extra buck or the extra whatever from building the deck. But he's still fine.
- 2.22** Who has lost? Anybody here lost? Who's lost? Canadian taxpayers have lost. Canadian taxpayers are subsidizing your deck. But you're better off for it. And nobody's lost anything. They've lost an opportunity, but they haven't really lost anything. And you actually now have saved quite a bit of money, right? What would have cost you 14 is costing you 10. You get to save \$4,000. You go and spend that \$4,000 on something. So this \$4,000 are going to go to creating a job for somebody somewhere, somehow. And it's a win-win and neutral. Nothing happened.
- 2.23** And you can see here that the people who pay the subsidy, the people who pay the subsidy, are the taxpayers of the country that is subsidizing. They're the losers. They're the ones who should be upset. I mean, who's going to be upset here? Well, Canadians should be. And this is true of subsidies, generally. We complain about other countries subsidizing their industries. Yeah, the people in those countries should really complain. It's terrible. It's bad economics. It's central planning. It's a distortion of their own economy.
- 2.24** We just benefit from it. We buy cheap stuff from countries that are subsidizing their industries. We get cheap stuff. We get it cheaper than otherwise we could get it. And that means we have more money to spend on other things. And again, it's not ideal

because it's not ideal for Canadians to be subsidizing this. It makes no sense economically, but not your problem. All you want to do is build the deck. And if you will, America gets decks at the expense of Canadian taxpayers.

- 2.25** It doesn't strike me that Americans should be complaining. It strikes me that Canadians should be complaining. It's a unique feature of politics that the response to something like this is, "We should start subsidizing our deck-makers over here." It's not fair that Canada is subsidizing deck-makers. We should subsidize them here, which just makes no sense. Somebody else does something stupid, so we want to mimic them, instead of, in a sense, taking advantage of it—not even taking advantage in a vicious sense, just taking advantage of it because they're willing to trade with us.
- 2.26** So we don't have any negative effect from that. Now, what actually would happen in an example like this is, you don't get your neighbor to take off two months from work and come and build your deck. What do you get? What happens to somebody, one of your neighbors? One of your neighbors figures out:
- 2.27** "There's a bunch of people who want decks. I'm going to go out there and make decks. And my opportunity cost is this \$84,000. I could be a bookkeeper, but I'm going to do this. And if I do it many times, I'll get really good at it, so that it won't take me two months in the future. It'll only take me a month, or it'll only take me two weeks. I can get better than the radiologist at building decks, and then I can build lots of decks. And I can, again, specialize and therefore become more productive and therefore be able to offer these services as a consequence."
- 2.28** And you see that, again, specialization begets these are specialized, begets comparative advantages here. Now, he could specialize, and that would beget a comparative advantage that he now would gain in deck-building on a broader scale. And even the radiologist specializing benefits the neighbor, the deck-builder. The more specialized the customers, the ability of the customer to be more productive and make more money, the more they're willing to spend up to 21,000. They won't spend more than 21,000. The more they're willing to spend on somebody else building the deck for them.
- 2.29** And this is how an economy works. This is why we have deck-builders. We don't have accountants who take some time off and build decks. Now, some people will take the time off and build their own deck, but that's because they're having fun doing it. They're not going through an economic calculation, but that's a minority of people. For most of what we need, we find somebody who's really good at it, somebody who's specialized at it, somebody who has a comparative advantage—not an absolute advantage, notice.
- 2.30** Who has the absolute advantage for building the deck? The radiologist. He can do it in a month. All these other people take two months to do it. So he's much more efficient. It doesn't mean he should do it. Just because you're good at something, doesn't mean

you should do that something. That happens every day in our lives. We have somebody else do something that we could do for ourselves.

- 2.31** I love Uber or Waymo. Waymo is even better because it's just a machine doing it. But yeah, I mean, I could drive. I'm pretty good at it. I like driving. It's much cheaper, much more efficient, much more productive, much better, much better use of my comparative advantage if I let an Uber driver or Waymo do the driving for me. And I can do other stuff while it's driving. I can think about it. I don't have to rent a car. I don't have to go through the whole process. I don't have to do all those things.
- 2.32** And you can find a million things that you do on a regular basis. Much better to outsource it. And think about it, a lot of the conveniences of life, a lot of things we buy, machinery, a lot of things that we mechanize, right, I remember, it's not that long ago. This will make me sound really old, but first year when I was married, we did the laundry by hand. We didn't have a washing machine. Literally, soap, clothes, rinse, hang, the whole thing. I outsourced that. I got somebody to do it a lot cheaper, so that I could focus on the things that I'm good at. It's called the washing machine. It has a comparative advantage relative to me doing the laundry.
- 2.33** Same thing with dishes. Old days, we used to do all the dishes, dry all the dishes, all of that. Then somebody specialized in dishwashing and realized, "I can make a machine to replace it." So we're constantly looking for ways to offload things that we don't have a comparative advantage in. Our comparative advantage is in our career. That's where our comparative advantage is. Or in raising our kids. And ultimately, just because of the fun of it, in doing fun stuff. And we offload, whether to people or machines, the things that we don't have a comparative advantage in.
- 2.34** And of course, this is true for individuals. It's true for states. It's true for international trade. Again, states don't trade, governments don't trade, only individuals trade. If you could find a personal assistant who can help you schedule stuff and book your travel and do all that, and it's going to be remote anyway, because today everything's remote and you're doing it via Zoom or whatever, why do you care if they live state next door, five states next door, or a different continent, assuming the time zones you can work out? It doesn't matter.
- 2.35** It doesn't matter for you as an individual. And it could be actually incredibly beneficial because the person in this other state might not be making 360,000. Imagine if this Canadian was only making, because Canada is a poor country, is only making 50,000. Now, his ability to make decks is even cheaper. So if I can find somebody in India who speaks good English and is fluent and is willing to abide by my time zones, I benefit from it, if they're far away, because they're cheaper.
- 2.36** And now I have more money to spend on other things. And if you aggregate that across all Americans, then you say, trade between these countries, it's exactly the same principle. The United States have a comparative advantage in doing what? What is the

comparative advantage of the United States when it comes to the iPhone? It's in design. It's in the aesthetics. Nobody can build it this beautiful. And it's in the internal design.

- 2.37** It's figuring out, choosing what chips. Apple now uses its own chip. What does that mean, its own chip? It means it designs the chip. It doesn't build it. But TSMC has an overwhelming comparative advantage in building the chip. If it gets a design from somebody else, it can't design the chip. It can only build a chip other people design. So when NVIDIA, now the biggest company in the world, wants to—what does NVIDIA do? NVIDIA doesn't manufacture anything. It designs.
- 2.38** Engineers on their computers, using American-written software, software companies, American software companies that are world-class at chip design, sit there and design a chip, and then they send a design to TSMC, and TSMC makes it, makes the chip. With, by the way, equipment that comes from where? Netherlands. ASML is the only company in the world that makes the machines that make it possible to make chips. ASML is a company from Netherlands who got a lot of its IP from Intel, who was looking to sell IP at some point and sold it to ASML, who built the machine.
- 2.39** You can see that it goes all over the place. But the comparative advantage Apple has is in design, design of chips, design aesthetic, design of the functioning of the phone, and with ideas on what a phone could do, the imagination. I mean, the genius of the iPhone is the imagination of thinking you could take a phone and turn it into, I don't know what this is, a do-all, a do-everything machine. That is the genius of Apple.
- 2.40** And that's all based on comparative advantage. It just turns out that a chip designer who was really brilliant came up with the idea that he was going to specialize. TSMC was an idea of one engineer who worked at Texas Instrument, and his idea was, "I am going to specialize in building the chip, in the physical process of taking silicone and turning it into a chip. And I am going to accept designs from everywhere in the world. I am just going to specialize. I'm only going to do one thing, but really be world-class at it."
- 2.41** So he started a foundry, which is what they call a chip-producing manufacturing plant in Taiwan, because in America, nobody was interested. Like, Intel had always designed and produced the chip and then sold the chips. And he said, "No, let Apple design the chip, and I'll produce it. Let Intel design the chip, and I'll produce it. I'll just become the world expert at producing chips." And he specialized, again, specialization, value, and gained comparative advantage in it. And now, basically, there are two companies in the world that make almost all the, at least the logic chips in the world, and that's TSMC and Samsung, with TSMC dominating.
- 2.42** So everybody does—I mean, Apple has no comparative advantage in glass. So when Steve Jobs first imagined the iPhone, he went to a glass manufacturer, Corning, and he said, "Hey, can you make me a tough glass that's actually, that also is going to be

sensitive to—I can put a layer that's touchscreen on it?" And Corning designed a glass especially for Apple, because it has a comparative advantage in glass, because it's specialized in glass for 100 years or longer.

- 2.43** Comparative advantage knows no borders. We all continuously act economically, based on comparative advantage, and we don't care where the stuff comes from or shouldn't care where the stuff actually comes from. And as we saw here, even when another country subsidizes it, it's not harmful to us, the fact that they are subsidizing. Indeed, other countries' subsidies improve their comparative advantage. He now suddenly has a comparative advantage because he can do it at 10,000. But it's at the cost of their own people. And the argument is not, "We shouldn't buy his stuff because it's subsidized." The argument should be, "We Canadians don't want to subsidize."
- 2.44** Most industrial planning, most central planning, most is an understatement; really, all have a huge cost and almost always have negative consequences to the country where they are applied. Other countries can benefit from that, at least in the short run, until the other country implodes because they're over-subsidizing. And you'll see that in China, for example; it subsidizes a lot of its industries. Over time, that's not sustainable. Over time, they have to make money. Over time, they won't have the money to keep subsidizing these things. They will implode at some point, unless they change the model.
- 2.45** You could be the best at everything and still not—it should never be in your interest to do everything. I know a lot of people who struggle with the fact that they're really, really good at everything, or really, really good at a lot of things, and yet they find it difficult to outsource, to let other people do stuff for them. Big mistake. Big mistake.
- 2.46** All right, I want to move now to another concept that we talked about that's relevant for trade, relevant to all of this really here, and that is the concept of the opportunity cost, the cost of not doing what is most profitable for you, and sometimes the hidden, the way in which opportunity costs are hidden, and you don't really see the cost of what you're doing or not doing, as the case may be. And here I want to go back to a famous story in economics, a fallacy in economics.
- 2.47** It's called the broken window fallacy, which Frédéric Bastiat in the 19th century articulated. And it's a really useful concept in economics to hold because it exists everywhere. Once you see broken windows, once you understand the broken window fallacy, you see broken windows everywhere. And the debate about international trade, to a large extent, is an example of a broken window fallacy—not in an obvious way, but it's an example of it.
- 2.48** So here's the broken window fallacy. Economy is sluggish. Things are not going that well. So I get a group of kids and I pay them some money. No, I just ask them. You don't have to pay the kids money. You just tell them, give them a bunch of rocks and say, "I want you to go to the baker's and I want you to smash his window." And kids would

love that, right? To throw a rock, smash a window. That's fun. So they do that. They go smash the window. And what happens?

- 2.49** Suddenly, there's a bunch of economic stuff that happens. The baker now has to replace the window. He finds a glazier. The glazier has to buy raw material, contract with a glass company. They have to make a glass. Two specs, exactly. Then the glazier comes in now and he installs the window, and the baker pays him money, and we smashed the window and there's a bunch of economic activity going on. Isn't this cool? We've actually increased economic activity by breaking the baker's window.
- 2.50** And there are a lot of economists and a lot of people in economics, a lot of people in politics in particular, who think this is fantastic. This is the way you get an economy going. Like, hurricanes are fantastic because they smash a bunch of houses, that creates economic activity. You suddenly have to rebuild the houses. Like, an earthquake is great. Any natural disaster, and indeed, to the extreme, wars are good. We're told often that wars end depressions. We just go out there and break stuff, then you have to rebuild it. And that's good because you create economic activity.
- 2.51** The fallacy is what you don't see. And what don't you see? Let's say this window cost the baker \$1,000. Well, what you don't see is what the baker would have done with \$1,000 if he hadn't had to build, to buy a window which he already had. So he could have done other things with the \$1,000. He could have bought a better oven to become more productive, so to increase his specialization in bakery.
- 2.52** He could have gone out and built a deck for himself, bought the wood and built a deck. I don't know. He could have saved the money and put it in a bank. And the bank could have taken that money and loaned it to some new, aspiring entrepreneur who wanted to build a business. The \$1,000 are not buried in the ground. They were going to be used. And instead of using them for something that would grow the economy, in other words, add something new that didn't exist before to the economy, they're being used instead to replace something that already existed in the economy, which is zero-sum.
- 2.53** No, nothing changed. The only thing that changed is, we get the same window. The baker is out of \$1,000. The glazier has \$1,000 minus the material costs. And some of the money went to the material producer, whoever made the glass for him. The \$1,000 are being redistributed to all these other people. But we still have the same piece of glass. Everything's the same. And the baker is \$1,000 poor, and he hasn't had the opportunity to use that \$1,000 for what he believed would be better for him.
- 2.54** So the point is that destruction does not create wealth, ever. You don't create wealth by destroying stuff. All you get is the same as you had before. And there are opportunity costs that are often unseen, because there's no parallel universe in which the window doesn't get broken, and we can follow the baker to see exactly how he used the \$1,000.

- 2.55** And we tend, generally, in life, to not pay attention to that which is not seen. To pay attention to that which is not seen requires thinking, requires effort. For that, you need to be incentivized, motivated. You need to be a thinker, which not many people are, in terms of these things. So if we go back to trade, if trade is good, making both American workers and consumers better off by making them worse off by eroding a division of labor, by stopping trade, it seems like, if I stop trade, if we stop making chips in TSMC, if we don't let TSMC make chips, Apple, if you want to make chips, you have to make them in America.
- 2.56** So make them in America. Build a factory here in the United States and build the chips here. So let's say Apple stops sending money to TSMC to bring chips here and invest in building a new plant to build TSMC, to build equivalent chips here in the United States. Then, yeah, we'll have a factory and we'll employ people and everybody will say, "See, this is great. We've got economic activity going on".
- 2.57** By the way, this is what Biden did. Biden, and with the help of Republicans, passed something called the CHIP Act. The CHIP Act basically subsidized—remember the subsidies? We Americans subsidized chip manufacturers who built chip factories in the United States. And everybody's going, "It's worked." Here in Arizona, TSMC is building a chip factory. People are going to work there. We've got employment, we've got a factory. It's in America. This is amazing. This is great.
- 2.58** But the jobs that are going to go into this factory cost a lot of money. The subsidy, and it's more expensive to build a chip manufacturing plant here in Arizona than it is in Taiwan. So every one of those jobs cost a lot of money. The chips coming off that line are going to be significantly more expensive. So Apple and other companies, NVIDIA, others, will have to pay more for the chips that come off the line in Arizona, have less money for other things, like, I don't know, developing the next big thing, the next great thing, hiring the best programmers they can for whatever it is that they're working on, the next thing.
- 2.59** Instead, they're buying chips that they could have bought from TSMC over in Taiwan, but now they're not allowed to, let's say, and have to buy them here at a higher cost. And again, there's the opportunity cost, the unseen. What could have Apple, what could have Nvidia done with all their money if they hadn't spent it on more expensive chips built in Arizona, instead of over there?
- 2.60** What could have all the people who are working in the TSMC factory here in Arizona have done with their lives, specializing in what they have a comparative advantage? They don't in chip manufacturing, that's why they need the subsidy. What could they have done? How could they have been better? So there's a massive opportunity cost. And we're losing higher value, more specialized, quote, "American" jobs or jobs done by Americans, by doing this.

- 2.61** Now, we can't see these costs, and that's the trick. This is how they can get away with it. What we can see are the benefits. There's a factory. There are people going to work. What we can't see is that, "Oh, there should have been a factory over there. A lot more people would have gone to work." That is unknown to us. So parallel universe in which you get that as an example. So it's those unseen costs.
- 2.62** But the net effect of protectionism and of subsidies is always going to be fewer high-quality jobs, less domestic specialization, less wealth, a smaller—I hate this analogy—economic pie, but I hate economic pies. Just smaller wealth, less wealth, less prosperity. And you can't measure that, as compared to what? The economy in the U.S. has done really well over the last 30 years. And yet, I think the opportunity cost was massive. I think it could have done a lot better over the last 30 years.
- 2.63** Everybody's saying, "Look, our economy is doing well. Therefore, everything must be great." And I'm saying, "It could have done so much better." That's a hard argument to make, hard argument to make, because it's unseen. People are very perceptual. They like to see the stuff. Imagination is hard, and there are lots of options. How do you know that that is what would have happened? Well, basic economic principles, I would say.
- 2.64** There's a quote from a 19th-century economist called Henry George, and it relates to all tariffs and all these restrictions on trade that exist out there. He says, this is the quote, he says, "Protectionists seek to do to our nation in times of peace what our enemies seek to do to us in times of war." Because he said, when an enemy in a war, what does an enemy try to do? Think about 100 years ago, 200 years ago. What do they try to do? They try to blockade you. They try to isolate you. They cut off your ports. They don't allow you to trade.
- 2.65** Why? Because they know that a country that can't trade is a country that's going to get poorer. It's a country that's going to have a harder time defending itself. It's going to be a country that can't produce the stuff that it actually needs, or at least not at the standard of living they're used to. We today do the same thing. We don't like Iran as an enemy. So what do we do? We embargo them. We don't trade with them, because we know that trade is good for both sides.
- 2.66** So we don't want Iran to get better, so we don't trade with them. And we ask all our allies not to trade with them. So think about protectionism and tariffs as a way of building a wall around ourselves. That's exactly the same thing. We're hurting ourselves. Just like we're hurting Iran when we're not allowing them to trade, we're hurting ourselves when we don't allow ourselves to trade. So protectionism, tariffs of all sorts, anything that limits trade is self-destructive.
- 2.67** All right. So let's turn now to another economic concept that people seem to get all upside down. And you hear this a lot, again; particularly in this administration, you hear it a lot. But we've really heard it for years and years and years now, because we've had

this for years. And that is that there's something problematic, there's something wrong with having a trade deficit. Trade deficits are bad. And what is a trade deficit? Trade deficit means we import more than we export. So we import \$100, we export \$80 worth of stuff, so the trade deficit is \$20. And that's supposed to be a bad thing.

2.68 The other country is getting all our money. So they're bilking us somehow. You've heard this a lot from President Trump about Canada, about other countries. We're subsidizing them because we're giving them money, but we get goods in return. You're not subsidizing your grocery store when you go buying there and have a constant, ongoing trade deficit with them. So let's be clear about a few things.

2.69 The fact that you have a trade deficit with somebody doesn't mean you're in debt to them. This is not about debt. It's just about the fact that you sell them more stuff than—you buy from them, sorry, more stuff than you sell to them. And we'll talk in a minute about what happens to that extra \$20 that is the gap. Where does that go? Because they have \$20 that they haven't used to buy stuff from me. I bought \$100 worth of stuff from them. So they have \$100 bill, then they use \$80 to buy my stuff. And then what happens to the \$20? So where does that go? We'll talk about that.

2.70 But the reality is that all trade is, all trade on an international scale is a lot of individual trade, a lot of the little trade deficits that each one of us has. Lots of exchanges between consenting adults, voluntary exchanges, win-win relationships, lots of them. That's a trade deficit. And the fact that one party buys more than they buy from is exactly the same as your trade deficit with your grocery store. Measuring exports and imports on a national scale gives you some aggregate, but it's completely pointless. I wish they just didn't report it.

2.71 It's an unrelated story, but I think very relevant. After World War II, Hong Kong was under British rule. They appointed a governor to rule Hong Kong, and he was a real economic liberal, so a very, very pro-free market, real free market guy, governed Hong Kong. And he had a policy of not reporting, or for that matter, collecting any data on the Hong Kong economy. There was no GDP, there was no inflation, there was no trade balance. It's not that there wasn't any. They didn't report it, and they didn't aggregate it, so the government didn't have it.

2.72 And his policy was laissez-faire, his policy was, "We're just going to leave the economy alone. Let it do its thing. We're not going to—" And people said, "Why aren't you reporting? Why aren't you collecting data? Why aren't you reporting?" He says, "Once you collect data, you start guiding policy to the data. We don't need to. We just leave people free. They'll produce, they'll create, they'll build."

2.73 Often when we aggregate data like trade, like nobody would know we had a trade deficit in America if the government didn't tell us we had a trade deficit, and then it's a deficit. Deficits sound bad. That's not good. It's not good to have a deficit. So maybe we should do something about it. So now we have a number, so we have to do something

about it. If we don't have the number, we won't do anything about it, which is ideal. So there's no, there's no issue other than people tell us there's an issue about it.

- 2.74** Because what actually happens with the money that we send overseas? We send dollars overseas. First of all, it's a win-win. I got the goods, they got dollars. I think in that deal, I did better. I'd rather have goods than dollars. Why do we have dollars? Why do we accumulate wealth? Why do we accumulate? Why do we save money? To spend it. The point of money is to spend it. Consumption is the ultimate goal.
- 2.75** I mean, put aside if you have kids and stuff like that, everybody should aspire to die with zero balance in their bank account, if you could target it just right. Because money is worthless other than as a means to getting something. It's a store of value so that one day you can use that value to get stuff. But it's a medium of exchange. It's for the purpose of exchange. So the fact that I'm buying stuff and sending my dollars overseas or sending my dollars next door, I always feel I'm better off because they get the piece of paper, I get stuff. I love stuff. Much better.
- 2.76** But now they have the piece of paper. What do they do with that piece of paper? Well, they can buy American goods, but if we have a trade deficit, it means they're not buying as many American goods as I bought from them. So what happens to that excess money? Because if you're in Vietnam and you have \$20, you can't go to the store and use it. What do you do with those \$20? Well, you either buy American stuff or you invest in America. There's just nothing else you can do.
- 2.77** Now, theoretically, what the Vietnamese could do is, they could buy oil from Saudi Arabia, which you can use dollars for. And then the Saudis will invest that \$20 in the U.S. But at the end of the cycle, the dollars come back to the U.S. It's the only place where dollars are useful. So if you look in aggregate, if you look in aggregate, then what you find is, the orange part is as a percentage of GDP, that is the trade deficit. And the purple part, I think that's purple, something like that, is as a percentage of GDP than an investment in the United States by foreigners.
- 2.78** What you find is that every dollar, in a sense, that you send overseas has come back in a form of investment. That didn't come back, so the bottom is a deficit, is the difference between inputs and exports. And the top is, again, the net investment. So Americans also invest in foreign countries, so this is net. Net, foreigners invest in America a lot more than we invest in foreigners. We don't invest in foreign countries. We send them dollars and get goods. We buy stuff from them.
- 2.79** Now every dollar is accounted for. It's pretty cool graph. And you can see that nothing's happened. It's not that dollars go away. Every dollar comes back. And is it better that the dollar come back as buying stuff from us or invested in us? And by the way, investment means they're buying our government bonds, so they're funding our wonderful government deficit. They are buying our stocks. They're buying real estate. They're buying factories.

- 2.80** They're opening stores, like Ikea is in there. You all buy from Ikea? Ikea is a great store. I think they're headquartered now in Denmark. They used to be in Sweden, Sweden, Denmark, something like that. They open stores in the United States. Well, their investment in the United States is right there somewhere. Is that good, bad? That's good. Is it bad to invest and to buy goods? I don't know. It seems that way because that's what we're doing. It seems like that's our inclination, at least right now.
- 2.81** And you can see that when we had a tiny little trade deficit, we had a tiny little trade deficit in 1983, it looks like, something like that, we had a tiny little investment surplus. And if you look at 1981, we actually had a trade surplus. And what happened to investment? It was negative because they have to offset each other. The dollars all have to balance out in the end. So is it good, bad? Looks pretty innocuous. And this is where they put their money.
- 2.82** Each one of these is a different investment category. So in the top left, you have total U.S. securities. The gray is, yeah, foreign-owned is on the left-hand side. U.S. is the light. That's a U.S. investment, the greatest foreign investment. And that line is their net, the percentage of all investments that are done by funds. And you can see, so the next one is how it's composed. Then in the middle, you've got treasury debt, you've got agency debt. Agency debt is like mortgage-backed securities. And then you've got corporate debt and equities.
- 2.83** And you see that foreigners are participating in all of our investment vehicles in a significant way. Some are up, some are down. They're buying less government bonds. They're smart. They're buying less government bonds and buying more corporate debt right now. They're buying more equities, steadily increase the amount of equity they bought. And they're getting out of the government market because our deficits are just insane and they're worried about default or inflation. But the money all comes back.
- 2.84** Trade deficits are just not an issue. They're not relevant to anything. It just says, at any given point in time, if more goods are flowing in versus money going out, or money being brought back in to buy in exchange for goods flowing out, none of that means anything. Again, each one of us, think of all the trade deficits you have and all the trade surpluses you have. And it all balances out because, at the end of the day, you're better off.
- 2.85** But generally, you have a huge trade surplus with your employer. You get the money, they get the goods, goods being labor in this case. And you have trade deficits with pretty much everybody else. And what happens to the difference between your surplus and your—what happens to the difference between what you gain from your employer and what you spend on everybody else? Invest. You invest that money. Just like it works in international trade, there's no difference. So again, trade deficits offset.
- 2.86** So a few trade deficit myths that are out there. They tell us trade deficits reduce employment. But unemployment is very low in the United States, and we have massive

trade deficits. Indeed, unemployment has been very low, with the exception of the financial crisis, since the 1970s. We've had lower and lower unemployment and been stable. Unemployment in the United States is measured at such low rates that if you talk to economists 10 years ago or 20 years ago, they would have said that's below what they call the natural rate of unemployment. It's much lower than anybody expected it to be.

- 2.87** So it doesn't look like trade deficits correlated with unemployment, quite the contrary. And when you have a recession, trade deficit didn't go away over there when we saw the numbers in the 2007-2008 recession. So unemployment is uncorrelated, but generally I'd say, the more—oh, I think I've got a graph. Yeah. I mean, if anything, the yellow is net imports. The purple or whatever is unemployment rate, and it doesn't—I mean, people have studied this. There's just no relationship there. There's nothing there. If anything, when trade deficit goes up, unemployment tends to go down. And that makes sense. So if anything, the trade deficit and unemployment are inverse, not positively correlated.
- 2.88** Trade deficits are evidence of economic policy problems. So the fact that we have a trade deficit means that there's something wrong with our economy. Economy is not functioning well. I mean, this is trade deficits versus GDP. Completely uncorrelated. Trade deficits go up, they go down, and GDP pretty much stays the same. Its growth rate is pretty much constant, somewhere around 2%. I mean, it's low in my view, way too low, but not because of trade deficits, not because of net export.
- 2.89** We overregulate, we have a tax, we do a lot of bad things, but there's no significant statistical correlation. And this goes back, you can take this back to 1890. I mean, this chart is only '75, but you can take a graph like this back to 1890, and there's no correlation between this. People have done that. Phil Gramm and Don Boudreaux have done the analysis going back.
- 2.90** It's the other way around, really. Countries that do well from an economic policy perspective, countries that have been OK in terms of economic growth, attract investment. You don't want to invest in a country that's screwing up. You want to invest in a country that's doing well. Well, what's a net investment in the United States? Very high. People love investing in America. Is that because we're screwing up our economy and we're doing really, really badly? Well, no, they wouldn't want to invest in us.
- 2.91** Indeed, you could argue that the reason we have a trade deficit is because people want to invest in America. So we buy stuff from them. They have these dollars and they go, "We could use these dollars to buy stuff from Americans, or we could invest." And a return on investment from their perspective is much higher value than buying stuff from us. And they just like our economy. Our stocks have done well. We have a stable, relatively speaking, a stable currency. They like dollars. Our stock market's done well. Our bonds always pay. We haven't defaulted on our bonds.

- 2.92** Our companies are world leaders in so many different segments of industry. Foreigners want to invest in America, so they sell us stuff. It's actually the other way around, right? They want to invest in us, so they sell us stuff because they know, by selling us stuff, they'll get dollars. That'll make it possible for them to invest in us. So it's actually a sign of confidence in the U.S. economy that people are willing to sell to us.
- 2.93** Imagine a people, which maybe is happening right now—people hate America. They don't like America. They don't want anything to do with America. Then they wouldn't want to invest in us. They don't want to invest in us, they're not going to sell us stuff because they don't want to be stuck with those dollars. So they find other people to sell their stuff to. So arguably, the bigger the trade deficit, the more confidence foreigners have in America. It's the exact opposite of how people think about it.
- 2.94** Now, it is true that you can imagine a scenario where you have a trade deficit because the citizens of the home country are irresponsibly spending like crazy. They're buying stuff like crazy, and they're taking on a lot of debt. And as a consequence, they're taking on—they have a trade deficit. And in a sense, what's happening is, the foreigners are investing in the country by lending us money, and then we're taking the money they lend us, and we're going buying stuff.
- 2.95** Now, two things would happen here. One is, the foreigners would have to be fairly stupid to keep lending us money if they don't think we can return it—and if we keep just using it irresponsibly just to buy stuff, instead of invest and do stuff with the money. Second, if you're going to complain about anything in that case, then you should complain about people borrowing money and spending it, not about trade deficits.
- 2.96** So right now in America, you could argue, and maybe foreigners are figuring this out, that our government is out of control. They're spending money like there's no tomorrow. They're spending money left and right. They're running massive budget deficits, a very different type of deficit than a trade deficit. They're spending money they don't have. In order to do that, they have to sell bonds. They have to borrow money. Some of that borrowing happens from Americans. Some of that borrowing happens from other people.
- 2.97** And the ability to buy American bonds to fund our government's spending is what stimulates the trade deficit. And maybe that's all right. Maybe that's true. But where's the problem? That government spending like there's no tomorrow. Not in the trade deficit—it's just a symptom of that disease. The disease is government running massive deficits, constantly.
- 2.98** And you still have to think that the foreigners believe, and I think we do, too, the people who buy the bonds, believe that the government will pay it back, ultimately. In other words, that the government will be responsible because, based on history,

American governments have always been. I don't know if that's a good assumption or not, given our governments recently, but at least that's what people believe.

- 2.99** All right, let's go through a couple more of these examples of myths. People believe we have trade deficits that are caused by unfair trade practices by foreign governments. So the foreign government restricts its own citizens from buying American goods, and that's why they're not buying as many goods from us as they otherwise would.
- 2.100** Well, the real victim here is the foreign citizens. The dollars still have to flow back to America. So they can buy our goods. They still have to invest. The benefits of that investment still accrue to us. If a foreign government is using subsidies to make their goods cheaper, so we buy them, again we've seen the example of that. That's their problem. It's a big problem for them. So again, trade deficits are not relevant.
- 2.101** Another story is, we don't save enough in America, and therefore we need the Chinese and the Vietnamese to save and to invest in America because we're not investing in America. Well, again, that doesn't really make sense. They're investing in America because it makes sense for them to invest in America. If we invested more, it would still make sense for them to invest in America. It wouldn't change the fact that this was an attractive place to invest.
- 2.102** So it's not a lack of American saving that's creating a trade deficit. There might be other reasons to worry about a lack of American savings, but that is not the cause of the trade deficit. The cause of the trade deficit is just their willingness and their interest in investing in the United States. We saw that, if foreigners invest in a variety of different things, some of it is funding our debts or lending us money, but a lot of it is in our equities and in our real estate and in Ikea, in factories and buildings and stores.
- 2.103** If you go to the mall, a bunch of the hot fashion stores are not American stores. They're foreign stores that have invested in America by opening a store. That's an investment. We don't think about that, but that's offsetting the trade deficit that we actually have with them.
- 2.104** Finally, there's one aspect of this that I think is important to note. Take the iPhone. We talked about the iPhone. The fact is that most of the value of the iPhone is something that Apple contributed. But when we evaluate the trade deficit, the whole value of the iPhone, the \$1,000 and something, is attributed to China because they're the last ones that export the iPhone to us. We're not even really recording the deficit or the trade correctly.
- 2.105** Because if we took out of every iPhone that comes from China here, if we took out of that the contribution of Apple to that iPhone in terms of value, the trade deficit would be a lot smaller. And think about that not just in terms of Apple, but in everything else. We could approach this in terms of capturing all the stuff. The whole chip industry—we're importing chips from TSMC, but the reality is, what's the real contributor to chips? It's the design.

- 2.106** It's what NVIDIA does. It's what CoreComm does. It's what all these American companies do, but it's not captured anywhere. So the trade deficit probably dramatically overstates, given that America is dominant. The dominant thing we do is not manufacturing, not building stuff, but designing stuff, which is the higher end, the higher value. It's considered a service, not a manufacturing thing. Programming is a service job. Designing chips is a service job. And we don't capture that when we think about trade.
- 2.107** So let's do a quick example of a country who had a trade deficit. We went through some numbers, and then we'll wrap up this particular class. So take Vietnam. We have a trade with Vietnam. In 2024, we ran \$123 billion trade deficit with Vietnam. We imported from Vietnam 136 billion-worth of goods and services, but we only exported to them \$13 billion. Very little.
- 2.108** Now, why is that? Why are Vietnamese not buying a lot of stuff from Americans? Well, primarily because they're poor. Stuff we produce is high-value. They can't afford it. Vietnamese are still a poor culture. GDP per capita is a tiny fraction of what American GDP per capita is. We're rich. I mean, if you look at the numbers, we're fantastically rich on a global scale. Now, according to the math of many people who oppose trade, right, they're eating our lunch. We gave them \$123 billion. It's rigged. They're stealing from us. They're taking from us.
- 2.109** But the reality is, nobody's being cheated here. We got the stuff we wanted. T-shirts, I guess, clothes, other products, and they got dollar bills. You know, win-win. They have a huge comparative advantage in making textiles. They're much more efficient at making textiles than we are on a comparative advantage. We might have an absolute advantage. That is, a factory in North Carolina might actually make T-shirts faster and better than they do in Vietnam. But those resources in America would be better used designing chips.
- 2.110** And in Vietnam, it's very, very—they have clearly a comparative advantage on textiles. So we buy from them. We're better off. They're better off. Just to give you a sense of how much richer we are, U.S. GDP is \$25 trillion. This is in 2017 dollars. GDP per capita is over \$70,000. Vietnam's GDP is \$400 billion. \$25 trillion versus \$400 billion. There's a thousand billion in every trillion. It's a lot smaller.
- 2.111** And GDP per capita is \$4,700. Just think of how poor you have to be. Think about an American who makes \$4,700 a year. That's Vietnam. What can they afford to buy that we make? And it's not surprising. But everybody, the Vietnamese don't have a lot of jobs. They're not very well educated. They're not high-end, in that sense. They can't design chips. They can't design iPhones. What can they do? Clothes. They can do clothes great. And what do we get? Cheap but high-quality clothes. Again, win-win. Nobody loses.

- 2.112** And the more they do this, the better they'll become, the more efficient they'll become, the more specialized. They'll benefit from the comparative advantage. They'll become better at it. Maybe at some point, there'll be another country that gains comparative advantage in clothes-making. My guess that country would be, I don't know, Nigeria, somewhere in Africa, where they are just starting to industrialize, and they'll have a comparative advantage in textiles.
- 2.113** By then, the Vietnamese will be richer. They'll be more educated, and they'll switch to an industry that is more value-added where they have the comparative advantage, in particular vis-a-vis Africa. And the textiles will all go to Africa, and we'll start having a trade deficit with Africa. And we'll actually be selling more to Vietnam because they'll be richer and they're going to afford more of our stuff.
- 2.114** So to the extent that America specializes, Vietnam specializes, everybody benefits. Workers become richer. Vietnamese workers slowly secure middle-class jobs, middle-class in their context. They get more educated, and ultimately, they'll start building more high-value products. And we've seen this over and over again in trade.
- 2.115** Look at Japan today versus Japan 60 years ago, what they're producing today versus what they were producing back then. Today, they produce high-end, sophisticated, matched with America in terms of the sophistication of the products that they make. That wasn't true 60 years ago. 60 years ago, they were making the kind of stuff Vietnam is making today. Not clothes, but they were making cheap stuff.
- 2.116** Today, you can't say a Toyota is cheap. It's just as good as any automobile America makes. But in the '50s, in the '60s, you could say they were cheap. They weren't that good. But they elevated themselves as they became more specialized and took advantage of their comparative advantage. So the economics of trade are pretty straightforward. They're actually very simple. Voluntary trade is good. It's win-win. Everybody benefits.
- 2.117** Really, nobody loses, both on an individual level in terms of two people trading with one another, five people trading with one another, but also on a national level. Ultimately, everybody benefits. Jobs are created if we specialize and if we abide by our comparative advantage. And it happens naturally. It's not like people think, "What's my comparative advantage?" It happens naturally. We're looking for deals. Who can build my deck? Really well, cheaper. Where do I buy my textiles from?
- 2.118** Marketplace just makes it happen, functioning based on the principle of specialization, comparative advantage. They feed off of each other. They make each other better. Trade is good. We should encourage trade. We want more trade, not less trade, because we want more specialization, not less specialization. We want to trade with more people rather than with fewer people, because we have more options when we trade with more people. Economics of trade, really, are really, really simple. Trade is good. Free trade is the best.

Lecture 3: Empires of Exchange

At a glance — Empires of exchange.

- 3.1 We've talked about the economics of trade and why trade makes sense from an economic perspective. We've taken a big-picture look a little bit about globalization in our world today. These principles of comparative advantage, specialization, all of those, they've always applied. Again, it's not that theory dictates action; it's that the theory here identifies stuff that's already going on in existence.
- 3.2 And what I want to do over the next two classes is really talk about the history of trade. A little bit about trade patterns, who is trading with whom, over what period of times. We'll get some maps up. We can actually look at some maps, which I think are interesting because you'll see that the choke points of trade 3000 years ago are very similar to the choke points of trade we have today. And so we're going to take an historical perspective. We're going to start with ancient civilizations.
- 3.3 And today we're going to go through up until Adam Smith, up until the end of the 18th century and before the industrial revolution and the explosion of trade that happens in the 19th century and what was called the first era of globalization, which is really the 19th century, up until World War I. Second era of globalization is World War II on.
- 3.4 But really, what we'll see is, globalization has always been happening on a much smaller scale, at a much slower speed, but it's really always been happening because trade has always been happening, at least since the beginning of civilizations as we know them. So if we go back to the history that we have evidence about, much of human history, before 3000, 4000 years ago, we know very little about. We just don't have any evidence.
- 3.5 They didn't leave archeological sites because they didn't build anything. They didn't make anything. Most hunter-gatherers roamed, and they didn't leave much evidence of their existence. Some, but nothing extensive. We know a lot about Sumerian, Babylonian, Assyrian civilizations. In the 2000 BC or so era leading up to the golden age of Greece and then the age of Rome, these civilizations had extensive trade relationships between them, in the Near East, what we call the Near East, what today we call the Middle East.
- 3.6 These are civilizations that were focused on the Nile River, the Egyptians, and then focused on where today modern Iraq and modern Iran are in those areas. These are areas that are relatively fertile. These are areas that adopted agriculture relatively early. Areas, because of the water supply, they had waters, they built canals, they developed agriculture. These are areas that embraced technologies that came in from other parts of the world, in this case, other parts of Asia.

- 3.7** The wheel, for example, was probably invented in the plains of, what would you call it? Southeast Europe, so modern-day Ukraine, modern-day Russia. A lot happened in that area early on in human history. It's a very, very fertile area. It's flat. It's easy to move around. A lot of tribes went back and forth, from deep into Asia into Europe—that route because it was relatively easy. It's just plains and flat lands, easy to commute across.
- 3.8** The Egyptians had the Nile as a major artery, so they could use boats up and down the Nile to trade, trade with deeper into Africa, but then through the Nile in the Mediterranean, trade with the Phoenicians, who would be where modern-day Lebanon and Israel are. And of course, with the Greek city-states that might have been sprinkled through the islands off the coast of Greece.
- 3.9** They didn't venture very far. Somewhat through the Mediterranean, you see some settlements maybe as far away as Spain and even the Atlantic coast of Spain. You see some trading along those routes. And you see that the trade routes are pretty far. I mean, think about it. In those days, you walked, you rode a camel, you rode a horse. That was about it. Maybe you went very simple boats, very unsophisticated boats, to go along the coast. And you see that the trade routes are primarily along the coasts.
- 3.10** Rivers. So you get the Euphrates down where Iraq is. You get the Nile in Egypt. You get rivers in China, and then the rest is walking, horseback, camels. That is the, those are the transportation routes, and they're extensive. This is months and months of traveling to do the Silk Road. And yet, trade was happening. From Africa, you got ivory. There was cotton already. Beads were being transported from Africa.
- 3.11** Throughout the Mediterranean, the Minoans, the Minoans are people of the Greek islands, a very advanced civilization for the time. In the Greek island, it then gets wiped out at some point, if you've ever been to Santorini. One of the ideas was that Santorini is this big—have you ever been to Santorini? So it's an island in the Greek islands. You got to go there. It's one of the most beautiful places on planet Earth.
- 3.12** It's a volcano that erupted. So what's left is the crater filled in by the Mediterranean. And then you get these sheer cliffs, and at the top of the cliffs are these Greek towns. And it's just stunningly beautiful. Anyway, the idea was that when Santorini exploded, that it killed off the Minoan civilization. It basically destroyed the weather, it destroyed climate, destroyed agriculture. But that's probably not true. So that was a theory. It's probably not true.
- 3.13** Nobody really knows why the Minoan civilization disappeared. We know it disappeared. And only later did the Greek civilization arise in its place. The Phoenicians were very skilled seafarers, and they traveled throughout the Mediterranean Sea. They established colonies as far away, as I said, Spain. They also spread a Phoenician alphabet, so writing was new here, and the Phoenicians probably had the most advanced alphabet.

- 3.14** So you're seeing here the beginnings of globalization—trade, language, alphabet, which is a big innovation. Different products that you can't get anywhere else. Cotton in Egypt, silk in China. And of course, the very beginnings of a spice trade. One of the most valued things in trade up until the 19th century that people valued the most was spices. And the spices primarily come from the bottom right-hand corner. They come from Southeast Asia, from a place like Indonesia and Malaysia and the Philippines and different parts of that part of Asia.
- 3.15** Trade is also, as I said, a cultural conduit. You get spread of technologies, different belief systems. At this point, there's no real systematic religious set. Every one of these little places had their own religion, their own gods. But there are similarities. The gods have very similar roles, jobs to play in the cultures. A lot of the founding stories are very similar. So there's a lot of influence across the different cultures, in spite of the fact that they're so far away and they're so difficult to get to.
- 3.16** Trade is also how diplomacy is done. A lot of times, kings of different empires communicate, send each other gifts, and the gifts were ways in which they handled diplomacy and who got what territory and how to avoid wars and how to prevent wars. But complicated trade routes and complicated forms of trade are documented in letters going back and forth between the different kingdoms and the different empires.
- 3.17** Even they, at a very, very simple level of products, understood the importance of being able to trade with other cultures and with other places. And during this period, technology does spread. I mean, one of the things that spread here was things like agriculture, which is a technology. It was invented somewhere and then slowly spreads through these different regions and adopted by different peoples in different places around the world.
- 3.18** One of the most long-distance, significant long-distance goods came from northern Europe. Now, northern Europe doesn't appear here on the map. The trade was minimal with northern Europe. There was not a lot there. It was very hard to get to, although there are rivers that flow through Europe that connect northern Europe to the Black Sea and through the Black Sea to the Mediterranean and to these cultures.
- 3.19** And there were certain things that were valued. For example, amber came from the Baltic region, and you can find amber in Greek graves, in Greek sites, archeological sites. And there's no amber in the region. They had to get it from the north. And this is 1000, 1500 BC. This is a long time ago. Amber was something that came even from northern Europe, where there were not a lot of people. Very harsh climate, very harsh environment. They came down the rivers.
- 3.20** You can find amber in Italy, in ancient ruins. You can find it in Egypt. But primarily, it appears that in Mycenaeans, in Mycenaean graves, you can find it, and in Egyptian tombs. So it traveled; it traveled, this, what is it, tree resin with different pretty colors that involved. It was obviously valued, and people were trading for it.

- 3.21** In central and western Europe, there was some trade between different tribes, Celtic tribes and other tribes. Not very advanced. They were still pretty primitive. There was no civilization in Europe at this point. All the civilizations were really there, where I mentioned. So the trade is pretty simple, pretty basic. They're trading salt and tin and iron and textiles, maybe animal skins, things like that. You've got Germany, France, British Isles. Of course, none of these countries exist. It's just roaming tribes. There are no borders. There's nothing there.
- 3.22** And again, where you find, where you find backwardness, you find very little trade. Trade is a feature of civilization. And of course, it's again one of these reinforcing loops. The more you trade, the richer you become, the more civilized, the more influences of the better things you take from different cultures, the higher the civilization becomes, the more trade is needed, and it's a continuous reinforcing mechanism. But trade is one of those essential things, one of those essential characteristics of a civilization.
- 3.23** We find in northern Europe, you find, maybe what they exchanged for the amber was weapons. So the Greeks, Mycenaeans in those days, were very good at building swords and weapons, and those traveled north; amber traveled south. So you find in ancient grave sites, in other pre-Roman, northern European sites, you find bronze weapons, pottery, beads, and other kind of artifacts.
- 3.24** So, we don't know a huge amount about what was going on in this world because there was very little writing going on. We tease it out of archeological sites. We look into graves; we can date them pretty well, and then we look at what exists there, and we can evaluate that. So for example, we don't really know what happened in Mycenae. We don't know why it disappeared. We know it disappeared. We know the culture disappeared.
- 3.25** We know, for example, that this era of civilization, particularly in the Greek islands, was followed by Dark Ages, a dark age in which there was very little trade. We suddenly don't find any of these things from faraway places that might have traveled 100 years earlier. Now it's not traveling. So, Dark Ages are almost always associated with the death of trade, death of money, death of that which facilitates trade.
- 3.26** And you can see the evidence of that in the archeology. It's just like you can see evidence of the Dark Ages in Europe, following the fall of the Roman Empire, even though modern historians don't like to use the term Dark Ages. It was dark. Following these Dark Ages, we get a new rise of a new dominant civilization, which was the Greeks, the Greek city-states, the classical Greek that we're all, I think, familiar with.
- 3.27** The ruins in Athens, that if you go to Athens, you can see some of the ruins on the different Greek islands, or if you've ever been to Ephesus in Turkey, which was a thriving Greek city on the Mediterranean in what is now today Turkey. You also get a thriving Persian Empire. Those two, of course, clash. If you've ever seen the movie 300, you know that there's war between those two civilizations.

- 3.28** The Greek economy, which is Greece, we talk about Greece, but there was no Greece. There were city-states. And each city-state, they were all—trade was crucial. I mean, think about the very idea of a city. A city requires food. You don't grow food in a city. So you have to trade for food. You have to bring food in. A lot of times, parts of the countryside are not politically the same as the political entity that is the city. So trade has to happen between different political entities in order to get the kind of food that you want. Greece is generally, if you have ever been to Greece, it's very rocky, hilly. It's not the greatest agricultural place in the world. The Greeks have to be tough in order to grow anything there.
- 3.29** But it's nice just across the Mediterranean. You have Egypt with the Nile and flat and very fertile soil, and they can grow food, and you can trade with them and provide food. And you can send them stuff that you produce, maybe weapons, swords, shields, other things. So, throughout the Mediterranean, in particular, there's a massive amount of trade during the Greek period. Let's see if we've got a map of Greece.
- 3.30** But this is where most of the trade is happening, within the Mediterranean, between these different places. You can see Spain over there to the left is where the Phoenicians have set up some colonies. The Greeks set up colonies as well. There are colonies of Greece in Italy, in Sicily, in a number of different places. And there is intense trading throughout the sea. You've got, in the Aegean, the Ionian, the Mediterranean Seas, and Egypt, Phoenicia, and the various city-states all are participating.
- 3.31** Some of the things that Greece, for example, is exporting—olive oil. Still to this day, a big product of Greece is olive oil. Wine. Greece is very strong in wine, pottery, and silver and metalwork. So they're pretty good with that. And of course, they're receiving a lot of agricultural products, food that they need. So imports into Greece, you get grain, primarily from Egypt, but also from Sicily. One of the reasons the Greeks colonized Sicily is for the grain, for the ability to grow stuff there.
- 3.32** But you also, into Greece, you get luxury goods, perfumes, dyes, fine textiles. Some of those might be coming across the Silk Road from India and from Egypt and coming here through Persia. In spite of the fact that Persia and Greece are often at war, there's tons of trade going on around and in between the battles, so trade never stops.
- 3.33** And one of the items traded a lot during this period, and will continue to be traded really until the 19th century, are slaves. So slaves are being traded here. So typically, slaves in this period are people who were defeated at war. Those who survive a battle are rounded up. They're turned into slaves. Slavery is quite common in all the Greek city-states. It's common everywhere where trade is happening. And slaves become a significant source of revenue, if you will, and source of trading.
- 3.34** And this carries on into European history well after this. The Vikings are good at defeating other people, enslaving a bunch of people, and going and trading them for goods that they want. And the Vikings become major trade players in Europe, in the

Middle Ages. Now, as part of all this, it's not just trade that's spreading. But of course, Greece is well known for its philosophy. Greece is well known for its science, math, plays, culture. Greek is the highlight, one would say, of culture, certainly during this period, and maybe one of the highlights in all of human history.

- 3.35** That culture is exported. That culture spreads throughout the Mediterranean. So you find amphitheaters all over the place that are putting on Greek plays. You find the beginnings of libraries, where they are trying to build up libraries of Greek philosophy and Greek literature. So culture is spreading, writing, ability to write, the ability to read. These cultural institutions are spreading. And again, they're helped by colonies that we set up in different places.
- 3.36** But they're also just helped by the fact that Greeks are traveling around the Mediterranean, trading with other people and exposing other people to their way of thinking. And their way of thinking is unique enough that people pay attention and endorse and absorb some of these attitudes. Trade is ultimately essential for the prosperity of the Greek cities, as it is essential for the prosperity of any cities anywhere in the world. It enables economic growth. It enables cultural expansion and the ideas of Greek that really start spreading.
- 3.37** Maybe one of the strongest influence that ideas of Greek have is on Rome. And ultimately, the Roman Empire will take over all of this space. But Greek influences all the cultures of the period with its philosophy, its technology, its science, and so on. And of course, this sets it up for Greece to influence the West, at least to this day. Greek philosophy has a huge impact on us.
- 3.38** I'm just going to skip ahead to Rome. This is an outline. This is a simple map, so I don't think it's got anything too wrong. Rome is there. Alexandria is there. And now there is an Alexandria. So this is the Roman Empire, and now the Roman Empire fluctuates. It grows and it shrinks over time. It's not always the same size. And if you think about Rome, and you should actually have a little bit of England there, it should be part of Rome as well, because all the way up to Scotland, the Romans occupy England.
- 3.39** Rome occupies all this territory. It establishes Roman law on the entire territory. It's a standard one rule of law on the entire territory. It also brings to this part of the world Roman technology. Much of that technology is built around trade. Now, maybe the original use is not trade. So for example, there are roads now being built for the first time, roads—and sophisticated roads, roads that you can really run wagons over and run chariots over.
- 3.40** And the reason to build roads for the Romans was to be able to move their armies. The primary focus of Rome was warfare. It was expansion or defense, as the barbarians start chipping away at this empire. But because they built the roads for defense to move the army, the roads are then used for trade. Trade becomes much faster, much more efficient, and much more widespread, because now you have basically one rule of

law over all of western Europe, all the way into Persia, down into Africa, across the Arabian Peninsula.

- 3.41 And what you get is an explosion of trade. And it's not just roads. They build bridges and they build ports. And they have a navy that controls the Mediterranean. They defeat Carthage, which is the main opposition that they have in the region, in the Mediterranean region. And really, you have one rule over the entire Mediterranean for a while, which makes it so easy. You don't have trade barriers, you don't have tariffs, you don't have war because the war is at the edges. In the middle, you can trade freely.
- 3.42 And you really see an explosion of trade, and at the same time, an explosion in wealth. Also, money becomes more sophisticated. Roman coins are now circulating throughout this area. Rome even has trade relations with China. So they are known. China knows of Rome. I don't know if any Chinese had actually been to Rome, but China knows of Rome. Rome knows of China. Goods are being transported between the two. You can find Roman coins in India. You can find Roman coins along the Silk Road.
- 3.43 So the ability to have one political entity controlling vast quantities of land is hugely beneficial to trade. A great example of that is the United States of America. We have a big piece of land, massive, right? One political entity. And we have free trade between the states. Why do we have free trade between the states? I'm sure some states have trade deficits with other states, but we don't care. We just have free trade across the United States.
- 3.44 And it's in the Constitution. You're not allowed to charge tariffs because what was happening before the Constitutional Convention, where states were barring trade from one another, they were creating all kinds of trade barriers with one another. And when Congress, when the Constitution says, to regulate trade is to make trade regular, is to have no trade barriers across all the states, the intercommerce clause, interstate commerce clause, are all there to make sure that there's free trade in America.
- 3.45 If it works inside America, why doesn't it work outside America? Of course it does. But you can see, the EU, big chunk of land, lots of countries. They don't exactly have one political entity, but they have coordinated, they're coordinated. And what is the feature of the EU? Free movement of goods. No tariffs, no trade barriers. It's good inside the EU. Why isn't it good outside the EU? Nobody really has an answer. Free movement of people. Free movement of capital. That's what characterizes the EU.
- 3.46 So in Rome, same thing happened. Once Rome conquered all this land, free movement of goods, free movement of people, for the most part, and free movement of capital, coins. And this explodes. Rome is importing grain again from Egypt and North Africa. North Africa, particularly the area around Tunisia and other places, very fertile land. And they can grow a lot of agriculture, and it's very close to Italy, so they can import it quite easily.

- 3.47** Olive oil. They can import olive oil from Greece and from Spain. And wine, wine from France. They still import wine from France, from Greece. Of course, today, you get wine from within Italy itself, from other parts of Italy, away from the city of Rome. They export coins, glassware, weapons, pottery. Pottery, god, there's pottery all over the place. This is a period in which pottery booms. All kinds of pottery. Pottery that's decorative, pottery that's useful, a pottery made from some varying materials, pottery that's decorated, pottery that's simple.
- 3.48** When you go into ancient Roman ruins, what you find is massive quantities of pottery. And there's a whole supply chain of pottery—where it's exported to, where it's produced, where they get the raw material. It becomes a massive industry and a trade network just around the whole idea of pottery. The other thing you find when you excavate a Roman site is you find lots of coins. You find pottery, you find coins. Rome is really a pretty sophisticated civilization. They're fairly rich.
- 3.49** They have tiled roofs. They have mosaics for floors. A lot of craftsmanship goes into what they have. If you go to the place that got run over by the volcano south of, south of Naples—Pompeii. If you go to Pompeii, you'll find water pipes, pipes with water and faucets. They had running water in Pompeii, in the Roman Empire. Europe didn't have running water. After Rome fell, there wasn't running water with faucets until the 19th century.
- 3.50** Rome was a city at its peak, over a million people. Over a million people. After the fall of the Roman Empire, Rome shrinks to 30,000. You don't get another city in Europe with over a million people until London in the 19th century. So this is a time of great prosperity and great wealth. Most of that wealth, or a lot of that wealth, is achieved through plunder. Rome is a plundering society, a militaristic society, that is out there conquering and taking, but also from production, production of things like pottery, production of different types of glassware and things like that.
- 3.51** One of the things that they also have an advantage in is a common currency. You don't have to exchange. You don't have to figure out what a particular coin is worth because the whole system is using the same Roman coins. You don't have currency exchange risk, like you would today or like they will once Rome falls, and you get a multitude of different little kingdoms. So Roman Empire, massive trading within the Empire.
- 3.52** And then once you go outside, they're trading with India and Arabia and sub-Saharan Africa and China. They're trading spices and silks and incense and gems and exotic animals. So you get into Rome, you get, for the games—you've probably seen the movie Gladiator and stuff. They get tigers and they get elephants. And the elephants become a big deal because elephants become also a vehicle of war. It becomes like the modern tank.
- 3.53** And they use, the Carthaginians, used elephants. The Romans then adopt the elephants for their military. But they're getting that from Africa, or they're getting that from India.

They built ports to facilitate all this. And of course, the Silk Road continues. The Silk Road is kind of a constant that is bringing uniquely Chinese goods into the Roman Empire. And again, we find Roman coins in India and Chinese silk in Roman territories.

- 3.54** So again, civilization, wealth, economic and military success tied to the ability to trade and related to the ability to trade and make trade possible. Again, self-reinforcing. The more territory you put under the same rule, the more trade you can facilitate, the easier it is to trade. Rome is about as connected as we get, certainly during this period, because of its size and its scope.
- 3.55** This is also a period, again, where ideas are spreading. So initially, Greek ideas are spreading throughout this area. But then at some point, Rome becomes Christian, and the Roman Empire becomes a vehicle for the spreading of Christian ideas. And Christianity is spread through the mechanisms of the Roman Empire. And the footprint of the Roman Empire shrinks quite a bit during that era. But it is the way in which Christianity is spreading. It spreads so far. And again, globalization, if you will.
- 3.56** The German barbarians who ultimately sack Rome, by the time they are sacking Rome, they're all Christian. They've been converted before by Roman missionaries, by Christian missionaries who come from Rome, to Christianity. And they're the ones who sack Rome. So when we think about barbarians, when we think about Romans, we think, "Oh god, completely different cultures," and yet they shared a religion. And that religion is being spread through the forces of globalization, through the forces of trade and interaction with foreigners and the exchange of ideas and the exchange of thoughts which was going on throughout this period and everywhere.
- 3.57** Rome falls, and you really do get a period called, or used to be called—it's not politically correct to call it anymore; I still do it because I'm not politically correct—the Dark Ages, because it was dark. The population of Rome goes from a million, over a million, to 30,000. But that's just symbolic of the fact that this is a period in which the technology is lost. All the technology that they had in Rome, they don't have anymore. They can't build tall buildings. They can't build domes. I said they had tiled roofs. No more tiled roofs. They're gone. Thatched roofs again. You put wood and you put stuff. None of these tiles. Mosaics disappear.
- 3.58** I mean, literally, life in Western Europe reverts backwards a thousand years. People were relatively literate in the Roman Empire. No literacy. Very limited. I said that you find, in Roman ruins, you find a lot of coins. No coins. If you go into ruins in this period, there are no coins. Money disappears. You don't need money when you don't trade. Trade disappears, so very, very limited trade. Infrastructure collapses. The roads fall apart. The bridges aren't fixed. The ports are gone.
- 3.59** Cities are shrinking, so you need less trade. Remember, one of the reasons for trade is to feed the cities. Well, cities don't exist anymore. People are out subsistence farming, farming for what they need. They don't have any excess to trade. So there's a huge shift

towards subsistence farming and bartering on a local basis. That's why you don't need money because you're bartering. Very simple. It really is a dark period.

- 3.60** Now, there's some trade still going on, obviously. Churches, a lot of times involved in this. Monasteries, a lot of times involved in this. There's, goods are moving—salt, wool, tools, wine, glassware. Luxury goods are out because nobody can afford them. The Silk Road is active. And there's still a lot of trade going on in China, India, the Middle East, but that's not Western civilization. Rome is gone. What that is, is Byzantine for a while. Byzantine is like the Eastern Church, still survives in the East, and it's still trading.
- 3.61** And then what you get is the rise of Islam, and Islam dominates the whole region. So where you had a Roman Empire before, now you've got an Islamic Empire—an Islamic Empire that reaches all the way to India, all the way to Spain. Most of Spain is Islamic. 90% of Spain is controlled by the Muslims, at least for a while. All of northern Africa, all of Arabia. And they're slowly taking land away from the Byzantines, who are still somehow surviving, still survive into the 15th century.
- 3.62** But the Islamic Empire is thriving. It's built on trade. Muhammad was a trader. That was his profession. He was a merchant. Medina and Mecca, where Islam is born, are a part of the, kind of the end part of the Silk Road. They are merchant communities. The Bedouin who travel through there with their camels, they are all traders and merchants. And as Islam spreads, establishes, in a sense, the rule of law over all the territories it occupies, you get a thriving trade going on within the Muslim world, and again, connecting Spain to China all across the—
- 3.63** The only part that is not participating much is Europe. It has descended. It has nothing to trade for. It produces very little. You're getting soap from China, spices from India, grain from Egypt, again. Jewelry and textiles are being exported. The major cities of the world in those days are Baghdad. Baghdad is a thriving city. Largest libraries in the world are in Baghdad, up until about the 12th century. You get Damascus, Alexandria, Cairo. These are the big cities that are thriving. Of course, in Spain, you get cities in southern Spain.
- 3.64** China is very dominant and very prosperous at this point. They have a Silk Route overland. We've seen some of the pictures of that, but also by sea. They're trading with India, they're trading with their various islands, and they're trading with the Muslim empires to their west. There's trade in Africa, again primarily between the Muslim Empire and down into sub-Saharan Africa. Gold, for example, is discovered in Africa, and they're shipping it north from there. There's ivory.
- 3.65** And of course, as always, everywhere, slaves. Slaves are definitely being traded throughout. This is a period in which the Vikings become quite active. The Vikings are still pagan. They're still very warlike, but they're reaching out. They're leaving Scandinavia. They establish networks in the British Islands. If you've ever seen the show

Vikings on Netflix, I think, you know that they take over the British Isles, or they take big chunks of the British Isles. They establish agriculture communities there.

- 3.66** Ultimately, Scandinavia is not a very pleasant place to live, and it's not a very fertile place to live. It's just too damn cold. And a British island is much better weather and much more fertile ground. And they start trading among themselves by establishing bases there. But they also invade France. They make it all the way to Paris. They have settlements in that part of the world. They go down the rivers in Europe and they make it all the way to the Byzantine Empire. They trade with the Byzantines or with the Muslims.
- 3.67** The Vikings are very active throughout this period. They're very active with the slave trade. But those are trading furs and amber, and they're importing things like silver and silk and spices. It's funny, the extent to which spices become such a big part of trade. As we said about British food, I think this is true generally of European food. If you take away trade, European food tastes horrible, or it has no taste, I think, is the better way to phrase it. It has no taste.
- 3.68** So when they discover spices from Asia, this becomes more precious than gold. This becomes the thing that everybody wants. And some of those spices are coming overland through the Muslim empires into Europe. The Venetians become the big traders of the Middle Ages. But the Venetians who sit on the Mediterranean become big traders. They have good relations with the Muslim Empire. So the Christians are fighting the Muslims constantly, but the Venetians are playing both sides, and they're trading with everybody. And they facilitate a lot of the trade via sea.
- 3.69** But you've also got all these different trade routes that are bringing goodies into Europe and different parts of Europe. Middle Ages are a time of, the beginnings of some activity, some economic activity in Europe, primarily. They're importing a lot of knowledge and a lot of information from the Muslim empires. They're rediscovering certain roots that go back to Greece and become Rome. This is just the era, just before the Renaissance, when they fully flourish and they fully embrace, I'd say, Greco-Roman roots.
- 3.70** But this is kind of the precursors to that. You see down there, you see trade routes going into the Pacific islands there, Philippines and Indonesia and places like that. And it really is interesting that if you think about the bottlenecks of trade today, if you think about the places that are often in the news around trade today, well, the Houthis are blocking the entrance into the Red Sea. Even back then, that was an important trading route.
- 3.71** You didn't have the Suez Canal, but you could go up that route. There were canals that were built from the Nile to the Red Sea. You could go up the canals to the Nile and then take it from the Nile to the Mediterranean. So even back then, that little strait there

that the Houthis are now blocking, even back then was an important trade route to connect India and the Far East with the Mediterranean.

- 3.72** On the other side, you see another little trade route there, the Straits of Hormuz, which the Iranians were threatening to block just when the war between Israel and Iran was going on. That is a very narrow passage. Today, I forget what the number is, 80% of all the oil and natural gas in the world flows through that narrow passage. That is the Straits of Hormuz that, even back then, were really crucial to feeding Baghdad and feeding that part of the world with trade. And of course, ships went from there down into Asia to go get spices.
- 3.73** And then if you look at where Singapore is, Singapore is at the end of that long tongue that starts in Thailand and stretches down at the end of Singapore. Right there, there's a very narrow passage. And if you think about it, pretty much all the trade that goes from China to Europe has to go through that passage. Singapore sits in one of the most strategic places on the planet. And it's very narrow. Pirates love it there because they can take ships, or at least used to be able to take ships.
- 3.74** And yet that's Malaysia, Indonesia, Singapore, Philippines. They're all around there. And of course, there's a lot of fighting right now between the Philippines and China around some of the passages there around the Filipino Islands, around on the South China Sea. But right between China and these other countries is the South China Sea. And there's huge disputes over there. So nothing's changed.
- 3.75** And the disputes today are about trade. Who controls the sea lanes? Who controls the ability to move goods? And it's basically the same choke points that existed in the 12th and 14th century still exist today. So this is a period where you get some agricultural innovations. There's increased fruit production. Urban cities are growing. Venice, Genoa become major trading hubs. They are ports on the Mediterranean, and they become the main trading hubs.
- 3.76** Again, imports, silk, spices, silk and spices. That's what Europeans love. That's what they want from the Far East. And what they export is things like wool, timber, metal, glass, weapons. There's a growing trade with northern Europe, so cities like Riga, Hamburg, Lubeck become more significant in the northern area in Europe. And they are exporting fish, dried fish, furs, salt, timber, grain.
- 3.77** Of course, Scandinavia with the Vikings is now linked to the rest of Europe, to Great Britain, to France and Spain. And there's growing international trade. I mean, part of what makes the Silk Road very effective during this period is basically the Mongols have just conquered all of it. The Mongols control vast areas of Asia all the way into the Middle East and into eastern Europe. And again, you get one empire, not exactly the same rule of law because different Mongols, rulers, but a common set of traditions, a common language.

- 3.78** And you get, the Mongols now facilitate a massive growth in trade between Asia and the rest of the world. And much more money, money grows. As trade grows, money grows, and people start to use more and more money in order for the trade. So this is the period really, going into the next period, where you really start seeing a dynamic international trade environment, where people are really expanding vast resources and engaging in trade on a massive scale, leading us to this period, which is the Renaissance and the age of discovery.
- 3.79** Now we're talking about real global trade, because now we're talking about the discovery of North and South America. We've got routes going from Europe to North America, back and forth. We've got the Portuguese establishing colonies in Brazil. So you've got an active trade between Brazil and Europe. The Spanish, establishing trade routes all over South America, but also in the Philippines. So you see the lines coming in from the left to right. Those are trade routes that are coming from Asia to South America, to Mexico from Asia, across the Pacific.
- 3.80** And of course, you're still seeing, now Europeans have discovered a way around the Horn of Africa. So now they can go around Africa to Asia. This is a period in which the Christian world figures out how to trade with Asia without the Muslims, so cutting the Muslims out of the trade. Right before, you had to go through the Middle East, you had to go through Egypt, you had to go through the Nile, you had to go through Baghdad. Yet somehow you had to cross Muslim territory.
- 3.81** Once you can circumnavigate Africa, once you can go around Africa, you don't have to deal with the Muslims anymore. You can go straight to India, straight to the Philippines, straight to Indonesia, and start trading with them. So this has geopolitical, historical significance. It's certainly the beginning of the decline of the Ottoman Empire. They now are less connected. Europe can isolate them more, because they could trade without them, which was not true up until now. The Muslims had been an integral part of the global economy.
- 3.82** I had a couple of other cool maps here. Here's a map just to show you all the traveling. Remember, this is by sailboat. This is hard stuff. A lot of these boats never make it back. They never succeeded. But this is where you're now getting the establishment of big trading companies in Europe, whether it's a Dutch East Indies Company or whether it's the British equivalent, East India Company, that have monopoly over the trade routes into Asia.
- 3.83** They're going through the Horn of Africa. Ultimately, this will motivate the British and the French to build the Suez Canal, so they can cut the travel time significantly, but there's no Suez Canal yet. But they're going to India, and they're going to various ports in India. They're going to Arabia. They're going to, again, the Spice Islands, where all the spices come from, and then to China and to get the silk. And everything else that China has.

- 3.84** I mean, China is a much more advanced, technologically at this point, much more advanced civilization than Europe is. Europe only takes over China and the Muslim world, it only takes them over, starting in the 17th, 18th, and certainly in the 19th century, it blows through them. But by this point, the Muslim world is richer still, and China is richer than the West. The West doesn't really become fully civilized until the 17th and 18th century, and rich and wealthy.
- 3.85** So you can see there, some of the various trade routes that are getting around. A lot of this is led originally by the Portuguese and the Spanish. And then as we move into 17th century, they fade, and the Dutch rise, and then they fade as the British rise. And then Britain turns out to be the dominant trader in the world of the 18th and 19th century. It's another cool map. This is a more global map that shows you all the routes that are happening.
- 3.86** And again, this is nothing as compared to the routes today. Today, it's much more extensive than this. But think about the ships. No airplanes, all ships, sailboats. These are months and months and months at sea. Many of them sink. Hostile territories, pirates all over the place, not just in the Caribbean, but all over Asia there are pirates, and there are lots of little kingdoms that are fighting one another. And the Chinese have a big fleet for a while, and they embrace trade for a while, and they shut themselves off.
- 3.87** Japan opens up to trade and then closes off for a long time. And Japan is completely isolated from the rest of the world for a long time. There's a lot going on in terms of globalization here. But this is really the first era where you really see global trade in the sense that the whole world is involved, is engaged. And it's not just, again, not just—think about it, it's not just ships. It's ideas, it's culture, it's habits, it's people are learning from one another. And this is pretty intense.
- 3.88** All right, one more topic, and then we'll call it. During this period, particularly towards the end of this period, now started rising economic theories that are trying to understand trade and trying to position these different empires, British Empire, the Spanish Empire, the Portuguese Empire. You've got France's empirical ambitions. They're trying to figure out, what does all this mean, all this trading? They're more conceptual. They're not just trading because it seems to, on the spot, make sense. They want to have a theory about this.
- 3.89** And a dominant theory around trade from the 18th, sorry, from the 16th to the 18th century, becomes a theory of mercantilism. Mercantilism is a theory that basically says, wealth is measured and is established by how much gold and silver you have, how much money you have. It is a money-centered system. It also, there is the belief that the world is a zero-sum game. That is, there's a global pie, and it's just a question of who gets what. The pie is not growing. There's no additional production happening.

- 3.90** There's no win-win. There's only win-lose. The goal is to win as much as possible, to hell with the other parties. Remember, this is a period of violence, a period of colonization, imperialism. It's a period in which you take. You don't trade, you take. Trade is fine if you can win in that trade. But what you want to have, and this is like the core of mercantilism, is you want to export and you don't want to import. So you want to import as little as possible. You want to export as much as possible, because when you export, money comes to you. Gold, silver comes into your country.
- 3.91** And what's the measure of success? How much gold and silver you die with, in a sense, as a country. How much gold and silver do you have in your vault. And when you import, gold and silver leave the country, and that's no good. Again, zero-sum game, and what matters is gold and silver. It's also a period in which there is strong state control of the economy. All economies are basically run for the sake of the king and by his command.
- 3.92** So they are monopolies. All these countries have monopolies. France has their own East India Company, but they also have a Mississippi Company that controls much of the land west of Mississippi in the 18th century. And they're all monopolies. There's no competition. They don't believe in competition. They believe in controlled, regulated markets. Tariffs on imports are very high because you want to protect domestic industries, but also you don't want imports. You want exports. Imports mean gold leaves. A lot of subsidies, a lot of regulations, and as we said, a lot of monopolies.
- 3.93** Also, these countries, all these empires, are establishing colonies all over the place. And the whole idea of colonies is, you want to export stuff to the colony. You might import from them raw materials because you need those. British Isles don't have a lot of raw materials. So they want the raw materials, but they only want raw materials. They want to be able to produce everything in England and then sell it out, so that they accumulate the gold.
- 3.94** And you see empire-building. Spain, Portugal, France, Britain, they're all doing the same thing. They're all trying to do the same thing. The British happen to be the best at it. Raw materials coming from the colonies, manufacturing goods go to the colonies. The colonies are economically dependent on Britain. Part of this leads to the American Revolution. The American revolutionaries don't like being told who to trade with and how much to trade.
- 3.95** And they don't like tariffs and regulations and controls as part of their British Empire. They don't like the fact that Britain wants to suck up raw materials, but all the manufacturing happens in Great Britain. None of it happens in America. The system is very much zero-sum. What matters is gold and silver, hate imports, love exports. And that is really what rules trade ideology until Adam Smith. Until Adam Smith's *The Wealth of Nations*, that is the common.

- 3.96** Now, about 100 years before Adam Smith, there are already some economists saying, "Wait a minute, this doesn't make sense." And they're suggesting things. But Adam Smith is the first big name, the first person who has an impact, who's not just suggesting new ideas, but these ideas are resonating now in the world around mercantilism.
- 3.97** So, the period leading up to the late 18th century, the period leading into the industrial revolution, is a period of mercantilism, of state power. It's inefficient, it's rigid, and ultimately Adam Smith eviscerates it. To talk about Adam Smith and about the era that comes after this, the era after mercantilism, we'll do that next time.

Lecture 4: Tariffs Through Time

At a glance — Tariffs through time.

- 4.1 So last class, we talked about the fact that trade has been part of human civilization almost from the beginning of time. Indeed, civilization, civilizations encourage trade, and they're made possible by trade. You can't form cities without the ability to trade and be able to feed those cities and bring products into those cities. So trade is an essential feature of civilization.
- 4.2 And when civilization disappears, that sometimes happens, trade declines. So trade and civilization are heavily related. And yet, even with the growth of trade, and there was significant growth of trade in Europe and really, global trade all the way out to China, and of course, the Americas, into the 16th, 17th, 18th century, there really was a lack of understanding of the value of trade, both economically and even philosophically.
- 4.3 There was definitely a perception of trade as zero-sum. No winners and losers, you're just equal value, or even negative sum. And the ideology of mercantilism, an ideology really focused on the idea that what was important was accumulating money, gold, silver in those days, rather than accumulating stuff, rather than standard of living, quality of life, really dominated. So the dominant ideology with regard to trade was mercantilism.
- 4.4 It was very much central-planned, government-oriented, and in terms of trade itself, very much this perception that imports were bad, exports were good, because exports brought money in, imports, money left. Didn't like that. You wanted to become as self-sufficient as you could. Everything should be produced locally to the extent that you can. And the consequence of that, ultimately, were, relatively speaking, stagnation.
- 4.5 This is a period of vast economic growth, particularly once you enter the 19th century, but it could have been faster. It was investment in industries that one could benefit, have benefited much more by just importing the stuff and investing in the comparative advantage. We talked about comparative advantage in a previous class, focusing on what you had a comparative advantage in.
- 4.6 Now, this intellectual confusion really changed over the 18th century. A number of economists, thinkers, were coming to the conclusion that, wait a minute, trade is win-win. There are enormous benefits here. There's no real superiority to exports, to imports. It's not money that matters. What matters is quality of life, standard of living. And there was a beginning of a change, and people were writing about this, but really, nothing really resonated until the writings of Adam Smith in the late 18th century.
- 4.7 Adam Smith really changes the debate. In the Wealth of Nations, the real, in many respects, the first full-throated defense of markets, of at least somewhat capitalist economy in terms of economics, an explanation of how markets actually work, and an

explanation of the value of leaving markets free. A big chunk of the book, really, the biggest contribution of the book, really, is an explanation of why free trade beats out mercantilism, why controlling trade is a net negative, not a positive.

- 4.8 Wealth of Nations is really the first big manifesto in the name of free trade. And the focus here is, Adam Smith focuses on unspecialization and division of labor, which we talked about. And he points out that what free trade allows you to do is now it allows you to dramatically expand specialization and division of labor, because instead of just being limited to the British Isles, to Great Britain, and a number of population then, a number of people who can specialize and benefit from the division of labor, now you can expand globally.
- 4.9 Now, you can benefit from the specialization of people all over the world. And if there's benefit to specialization locally, then there's benefit to specialization globally. The same economic logic applies. Division of labor is good. You want more labor then. You want more division, not less. More people, more benefits; more trading partners, more benefits; the richer and wealthier you get.
- 4.10 Now, he clearly argued that any kind of trade barriers, any kind of trade barriers, whether it's tariffs or other forms of trade barriers—quotas, just outright embargoes, limitations— hurt consumers, hurt competition. So they actually hurt producers because they weren't incentivized to compete, and they were economically inefficient.
- 4.11 Now, these ideas were pretty revolutionary. There'd been some precursors, other economists and other thinkers had been talking about this. But for Adam Smith, who was a prominent figure among the Scottish Enlightenment, and was widely read throughout the British Island, this was, for most people, this was a real revolution. And it took a long time for this revolution to basically get into and be established into law.
- 4.12 At the end of the 18th century, in the beginning of the 19th century, Britain was still largely protectionist, not completely, but to a large extent. It had tariffs. It still had monopolies, where monopolies here meaning government granted monopoly, where the government prevented competition. They had things like the Navigation Acts. Navigation Acts were acts that basically said that if you're going to transport goods to and from Britain, the ship had to be a British ship, British-manned, carrying a British flag.
- 4.13 You couldn't have foreign ships coming into British ports. And this was the dominant view. And indeed, during the Napoleonic Wars, 1793 to 1815, Britain's trade policy grew even more protectionist. Wars generally lead to protectionism. Wars tend to cause people to hunker down, not to open up to the rest of the world. So the idea that we need to be self-sufficient, we can't rely on these other countries because we might go to war with them, was elevated, became more prominent, and Britain turns towards more protectionist.

- 4.14 Also, at the end of the war, what happened was, you get agricultural surpluses. Suddenly, there's the potential for imports of grain. The sea lanes are open. And there was a, you know, food prices drop. And of course, if you own land, if you're an aristocrat, remember, this is not a period of free farmers. This is a period primarily of aristocrats who own vast estates, who have peasants living there and growing food. You're losing money.
- 4.15 So aristocrats were not happy with falling grain prices. And in 1815, they passed a series of famous laws called the Corn Laws. That was in 1815, which placed high tariff on imported grain, keeping food prices high, landlords happy. Remember who the landlords are? They're the aristocrats. They're the old wealthy. They're the old money. The new money in Britain is coming from the industrial revolution. The old money are the people who hold those estates. They have farmers, and they're dependent very much on the price of food.
- 4.16 Almost immediately after that, there already are a lot of voices and a lot of pressure to start liberalizing trade. And starting in the 1820s, you start seeing a slow shift occurring. The industrial revolution is transforming Britain. It's becoming a lot less dependent on sale of grain and a lot more dependent on industrial production. Britain becomes the workshop of the world. It's where industry is born and where industry is thriving.
- 4.17 And the politics change. Industrialists want cheap raw material that's coming from overseas, and they want to be able to export freely. They have huge advantages, globally. They are at the forefront. They want trade restrictions eliminated. So you start seeing the different pressure groups starting to align in terms of who's in favor and who's against.
- 4.18 But also there's a continued intellectual revolution. David Ricardo, in 1817, publishes his comparative advantage theory, which we talked about a couple of classes ago. And he publishes this in support of free trade, in support of Adam Smith, and this is gaining more traction. In the Importation Act of 1828, they have a sliding scale of grain duties, offering partial relief as a consequence.
- 4.19 Other reformers, this is a time of intellectual fervor in the UK. This is also a time, by the way, that they abolish slavery, that they commit themselves to abolishing the trade in slaves. It's Great Britain that sends its navy out into the seas to stop slave trade. They spend a huge amount of money to stop slave trading. I mean, this is a massive innovation and intellectual revolution and political revolution.
- 4.20 Slavery had been a part of human existence forever. For the first time in human history, a state, it commits itself not just financially, culturally, but militarily, to stopping the trade in slaves. And we think this is vastly underestimated, the importance of this. Indeed, you hear a lot about how awful the slave trade was, and it was. It was horrific, and how much Britain was responsible for it, and it was, but you almost hear nothing

about what they did afterwards and how they basically changed the world by ending slavery, by ending the slave trade.

- 4.21 Slavery, sadly, still exists today. But they ended the slave trade, or most of it, followed by, of course, the American Civil War, which ends a vast majority of slavery. And then those kind of ideas spread into other parts of the world, and by the end of the 19th century, for the most part, in most parts of the world, slavery is gone.
- 4.22 But anyway, at the same time, there's also a reform movement called the Anti-Corn Law League. So these are people committed to getting rid of the Corn Laws. And intellectually active, politically active, they go out there to try to get rid of them. Richard Cobden, John Bright, are some of the leaders in this. And they're arguing, look, we've got poor people in England that are struggling to feed themselves. We're holding food prices artificially high. We could import grain from other countries much cheaper than what we are producing here in the Great Britain.
- 4.23 And this is only favoring rich aristocrats at the expense of the poor. I mean, this is a time where the United States is becoming a massive producer of grains. And because of the railroads, those grains are relatively cheaply, relatively easy, taken from the Midwest to the ports and could be shipped to England. But you can't because of the Corn Laws. Other parts of the world are able to produce food a lot cheaper than Great Britain.
- 4.24 And indeed, Great Britain's comparative advantage is not in production of grain. It's in the production of, in those days, manufacturing stuff. So in 1846, after this growing pressure, there is a, the Irish famine happens. So people are really suffering from the inability to access food. Robert Peel, who is the Prime Minister, breaks from his Conservative party, who is very pro-tariffs, and repeals, votes for repealing the Corn Law, and the Corn Law goes away.
- 4.25 I mean, this is a turning point in British economic policy. From this point on, you get a steady repeal of other laws, ultimately leading to the United Kingdom, from 1846 or 1850-something, basically embracing unilateral free trade. No trade deals. They're not demanding that other countries reduce their tariffs to zero. United Kingdom reduces tariffs, reduces trade values to zero, and says, "Send your goods to us. We'll take them."
- 4.26 And this is, results in a massive boom from 1846 to the beginning of World War I. Great Britain economically thrives, as a result of the industrial revolution, but also as a result of the fact that it becomes a free trade island. It becomes this hub for free trade, and it absorbs massive amounts of imports, but it's also exporting, because it is the major manufacturer in the world. Unilateral free trade dominates the period following this.
- 4.27 One of the interesting things about the Corn Laws and the appeal of the Corn Laws is really to try to understand the politics of it, because we'll see it play out again in 19th-century America, and of course, we'll see it play out again today, in today's America.

How do different groups align in terms of, in terms of free trade, in terms of wanting it or fighting against it? The Corn Laws are a great example because it's quite clear.

- 4.28 The landed aristocracy is, wants the Corn Laws. They generate all their revenue from the fact that they're growing stuff. They are not the industrialists. Industrialization is something that new people are embracing, not the old aristocrats. The old aristocrats are fixed in the same way they've been making money for hundreds of years. Maybe there's no more serfdom, but there's something very close to that.
- 4.29 So there is a clear group of people in the country with a lot of political power, not because there are lots of them—there are actually very few of them—but they have political power, the aristocrats, the landed aristocrats. The industrial revolution is primarily new money. It's primarily middle-class people. It's primarily poor people rising up. It's not the rich aristocrats suddenly discovering industry. They actually don't like industry that much. It's uncouth.
- 4.30 The Conservative Party very much supports the Corn Laws because they're conservative. They're trying to conserve the past, and they're very much aligned with the landed aristocracy. And generally, the Conservative Party in those days— again, what conservative means changes over time—but conservative in those days, or Tory, were generally for a policy of economic nationalism, of build it all locally.
- 4.31 And of course, some farmers and agricultural workers supported this. They were afraid that they would lose their jobs and they would lose what income they had to foreign competition. They probably were right. They probably would and did after the repeal of the Corn Laws. And then finally, more broadly, economic nationalists were supportive of this.
- 4.32 Who opposed the Corn Laws? Well, industrialists and manufacturers, certainly the nouveau riche, but people in Manchester and Liverpool and Birmingham, which were the booming cities of 19th-century England, where industry was happening, where people were moving from—indeed, farmers were moving from the farms into the cities to go work in the factories. And these cities were booming.
- 4.33 They, of course, wanted the Corn Laws repealed. They wanted cheap food, and they wanted to be able to bring in cheap raw materials and export their stuff. They wanted free trade. The middle class, the urban bourgeoisie, as you would call them, who are growing in influence at this time because of the industrial revolution, supported market liberalization generally and free trade, both as economic principles, but also as moral principles. They thought this was right.
- 4.34 Protecting certain groups at the expense of others seemed morally offensive. Something to remember. We'll talk about morality of trade in a future class. And then there were intellectuals. There was a big intellectual campaign. Almost no intellectuals are supporting mercantilism. There was no basis for mercantilism. Everybody, after

reading the Wealth of the Nations by Adam Smith, got it. From an intellectual perspective, mercantilism, restrictions on trade, didn't make any sense.

- 4.35 So the intellectuals generally were aligned with a free-trade, anti-Corn Law perspective, and many of them joined the Anti Corn Law League that worked hard to do that. And then, of course, what you'd consider liberal or progressive— but again, liberal and progressive in the 19th century is very different than what we call liberal today—liberal, generally even today in Europe, liberal is a term that refers to people who believe in free markets.
- 4.36 So it's only in America that liberal became leftist. In the rest of the world, liberal has always been associated with free markets. So the liberals of the time, the people who were embracing the industrial revolution, who saw this as a significant, massive, positive change, were all for the repeal of the Corn Laws. So at the end of the day, whether you had Corn Laws, Corn Laws or not, became a question of politics, not a question of economics. The economics was settled.
- 4.37 Now the question is, politically, can we get the right coalition together? Can we get enough votes, given that the Conservative Party was the one who kept winning elections, can we get enough votes to repeal them? Can we convince enough people within the Conservative Party to switch sides? And in 1846, that happened with the Prime Minister himself, Peel, who was a Conservative, switching sides. And as a consequence, the laws are repealed.
- 4.38 Following that, you get in 1847 the Navigation Act, which repealed the core of the old navigation rules. So this idea that you had to, anything imported to Britain or its colonies for that matter, had to be on a British ship, that goes away. And by 1854, the Merchant Shipping Act cleaned up any remnants of mercantilism; any remnants of lack of free trade were cleaned up, and literally, the UK established itself as an island of free trade, unilateral free trade.
- 4.39 And that was, importantly, they didn't seek anybody else to become free trade. They viewed free trade as a benefit to themselves, where the other countries didn't do that or did do that, was irrelevant to the benefits that they received from it. Of course, if other countries became free traders as well, that would benefit the UK. But they were going to benefit from unilateral anyway.
- 4.40 So, this is a victory, right? So we get into, the 1850s is the real victory from 1846 all the way to repeal of everything. This is the real victory of Adam Smith, the manifestation of his ideas, his economic ideas, in Great Britain in the 19th century. And again, it leads to a period of incredible flourishing and incredible wealth and incredible success.
- 4.41 Britain ultimately loses its empire, loses its colonies, and becomes the number two most important country from an economic perspective in the world, not number one. Not because of free trade. Free trade is actually helping it along in terms of wealth creation and so on. But just because of the fact that—two facts, really. One is,

colonialism and imperialism, it turns out, in spite of what you hear, again, the prevalent wisdom out there, is super expensive. The United Kingdom is actually losing money on its colonies.

- 4.42 Maintaining a large military and a large military force in places like India and South Africa, and really all over the world, was super expensive for the Brits. And if you take the colonial period, add it all up, the United Kingdom spent more money than it received. It actually was a losing proposition. And if you understand trade, you would get that. It would have been a lot cheaper for the Brits not to colonize all these places, but to trade with them.
- 4.43 Let them manifest their comparative advantages and then trade, which would—you'd get all the benefits from being in these places, from the production in these places, from the raw materials, from the silk, the spices. Of course, industry is now changing. Silk and spices are less valuable than they were in the past. Now it's more about cotton and textiles and manufactured goods and raw materials. But you can trade for those, rather than take them. Taking them is a lot more expensive than trading for them.
- 4.44 Which, again, is a shift. The mercantilism mentality is, because it's zero-sum, is, we take. And colonialism was a project of the mercantilists. And Adam Smith was very much against colonialism. He got it. He understood the costs and the benefits from shifting to just a trading relationship with all these other countries. But even the 19th century, Britain continued its colonizing strategy and behavior, and the colonies only—that only starts dissipating between World War I and World War II.
- 4.45 So one of the reasons England slows down is certainly colonialism and the cost. Actually, there's three reasons. I'm going to add a reason. Second reason is, England is living in a place that erupts into war in First World War, and it is right there, and it suffers the consequence of the war in ways that, let's say, the United States never does. We're not invaded, we're not bombed.
- 4.46 Yes, we send troops to Europe, but relatively late in World War I. We don't suffer anywhere near the number of casualties the Brits do, certainly not as a percentage of population. I mean, there's essentially, a whole generation is wiped out during World War I in Britain. And one of the questions always is, for what? Nobody quite knows why that war was fought and why Britain participated. But also from World War I, through the end of World War II, there are a lot of challenges in Europe, and Britain is suffering from those.
- 4.47 Of course, World War I, you get a Great Depression, you get a World War II, not a great time for the Brits. And of course, the third reason is the rise of the United States. In the United States, in many respects, not in respect of trade, as we'll see, but in many other respects, the United States is freer, the United States is bigger, the United States has more of the entrepreneurial energy and ultimately more people, and ultimately takes over, and becomes significantly richer than the United Kingdom.

- 4.48 So let's talk about the U.S., and we'll talk about the U.S. in terms of trade. At the American founding, there was a lot of discussion about trade, and there certainly were founders who had a strong sentiment towards free trade, towards basically as much trade as possible. Indeed, some of the complaints against the British, some of the reasons for the revolution, had to do with trade and trade restrictions.
- 4.49 A lot of the new taxes being placed on the Americans in the colonies were taxes on trade, on tea coming in, right? Tariffs on tea coming in. And the inability, for example, of America to trade with the rest of the world. It had, everything had to go through Britain, right? Because remember, mercantilism said everything goes through, everything comes to Britain.
- 4.50 So, a big part of why there was a revolution had to do with the fact that Americans wanted to control their own fate in terms of trade. They wanted trade to be freer in America. And indeed, that was the dominant, I think, ideology at the founding. But there were counter voices, in particular Hamilton. During the Constitutional Convention, they're considering, how do we raise money for this new federal government? We're creating a new federal government, and what are the sources of revenue going to be?
- 4.51 And you have to remember, this is a time where the mechanisms of tax collection, if you will, and monitoring income and things like that, they don't really exist. So they can't, even if they wanted to do it, and they didn't, but even if they wanted to do an income tax, how do you collect it? How do you know what people are earning? How do you control something like that over the space of the United States? Even a sales tax would be very hard to implement.
- 4.52 Everything is cash, gold coins. People don't really keep that many records. It's not that important. We're not filing taxes. There are no corporate taxes. So there's no record-keeping. This would be really, really hard to implement any kind of sales tax. So they're struggling to try to figure out how to raise revenue for the government. And one idea is tariffs.
- 4.53 And to a large extent, the thinking is, tariffs are relatively easy. There are certain points in which goods enter the United States, and those are well established, the ports, primarily. We put a custom house. They probably already existed because they're British. We're collecting customs, and we just charge a certain percentage off of everything that comes in.
- 4.54 Madison, in 1789, introduces the first law to impose tariffs. And to a large extent, Madison is shocked, and I think upset, by the response. Because immediately, he and Congress are inundated by pressure groups who want higher tariffs on certain goods, lower tariffs on other goods. They want their industry protected, other industries not protected.

- 4.55 The factionalism that Madison talks about a lot in the lead-up to the Constitutional Convention, and then later on, the idea we don't want political parties, and we don't want factions, and we don't want pressure groups—it manifests itself in tariffs in a way more so than in any other topic. I mean, sadly, they had resolved the most contentious topic at the American founding, which was slavery. They resolved it in favor of slavery, sadly. And the next big contentious issue turned out to be, of all things, tariffs.
- 4.56 So tariffs become, for the next 100 years, really more than 100 years, until really well into the 20th century, they become the most contentious issue, other than slavery, until the Great Depression. So this is what Americans fight about. They fight about tariffs more than anything else. So in 1789, they set tariffs, and in the years following that, the tariffs are being set, primarily in order to bring in the revenue.
- 4.57 And they're set low, because low tariffs generate a lot of imports, and then you get to tax a lot of goods, and that generates revenue. Now, remember, and this is really important, particularly given the debate going on today in Washington, federal government in between the founding and really World War I, never spent more than, with exception of Civil War, more than 2.5 to 3.5% of GDP, maybe 4, but that's max.
- 4.58 Federal government was tiny. It did very little. Didn't have a, really, a standing military. No welfare, no redistribution of wealth, no health care benefits. None of those things existed. So 2.5 to 4% of GDP, that was it. Today, just in a sense today, we spend anywhere between 21 to 25% of GDP at the federal government level, not even talking about the states and local.
- 4.59 So the government in the 19th century was 80 to 90% smaller than it is today. I mean, not a little bit smaller, not a half. 80, 90%, no social security, no Medicare, Medicaid, no welfare, no interest payment on the debt. Well, there was some when they took on debt, but there wasn't that much debt because there wasn't that much spending. So you could fund the government with tariffs— something that, as we'll see, is impossible today. It's just mathematically impossible to do.
- 4.60 But if your government is only spending 2.5 to 3%, yeah, you can fund it using that. So the only one of the founding fathers who seemed to be pro-tariffs, and generally pro what you'd call industrial policy or any kind of central planning, was Alexander Hamilton, who wanted America to be self-sufficient. He wanted America to produce everything, both because of security concerns, dependence on Britain.
- 4.61 But also, he was pro-protectionist. He believed that tariffs would help infant industries, that if we shielded those industries with tariffs against competition, they would thrive, they would be successful. So he didn't want one tariff rate across the board, which is Madison. Madison just wanted one tariff rate across the board. No favors, no goodies distributed to anybody. Hamilton wanted differential tariffs for different industries, but still low. He still understood.

- 4.62 Hamilton was really smart. He understood that high tariffs would be destructive, so low tariffs just differentiated to try to help certain industries along as they started out. And that's basically the policy in the United States until the early, until the 1820s, late 1820s, was basically that—low tariffs just for revenue. Hamilton did not win the day in that respect, just for revenue, and that's how tariffs were set.
- 4.63 In 1829, a senator by the name of Henry Clay started agitating for high tariffs. High tariffs that would fund a government spending on infrastructure. High tariffs that would fund, potentially, a central bank. And high tariffs that he believed now not just would be revenue, but high tariffs now to protect American industry. Remember, this is now the beginning of the industrial revolution in America.
- 4.64 And all these nascent industries are coming about, and they're facing competition from overseas, particularly from Britain, but other places as well. And the United States has fought a war with England, the 1812 war, and we don't trust those guys. And so now we want tariffs that do more; they actually protect industries. They block inputs. There's a big debate going on.
- 4.65 Madison and Jefferson, they're pretty old by this point, but they're still around, and they're fighting against this. They argue against the high tariffs. They argue against government getting involved. And it becomes a major dividing line in American politics — for tariffs, against tariffs. And you start seeing regional alliances being created. In the United States, agricultural interests are going to be anti-tariff. Agricultural interests are primarily exporters. They want to get their stuff sold all over the world.
- 4.66 They worry that tariffs, if we impose tariffs, other people will impose tariffs on us, and we won't get to send the goods out. And this is primarily southern states and the Midwest. And this is true during slavery and after slavery, those interests hold. And by the way, to this day, while you could argue that farmers right now are supporting tariffs by supporting Trump, they are generally concerned because what happens every time tariffs are placed, the countertariffs are almost always on American agriculture, to make them hurt.
- 4.67 So what does the government do now? It subsidizes them. So what we do is, we raise tariffs on China. China raises tariffs on soy beans or whatever, and then we give money to soybean farmers to compensate them for the tariffs, so we're all subsidizing American farmers. The Big, Beautiful Bill they just passed has a huge increase in farm subsidies, to compensate them for the fact that other countries have made tariffs on agricultural goods as a response to Trump raising tariffs, or we, America, raising tariffs on them. So you can see how these cycle through.
- 4.68 Mid-Atlantic states were primarily protectionist. These were heavy industry states, so Pennsylvania, New Jersey, places like that, heavy industry. And America didn't have to worry about importing raw material, right? Because America is so rich in raw material, all the raw material could be found in America. So they didn't worry about importing.

They just wanted to protect their finished goods from competition from foreigners. So a lot of industrialists opposed or supported tariffs.

- 4.69** New England was split. The textile mills were pro-trade because they were doing very, very well and they were starting to export and they wanted open markets. Others were against it. And oftentimes, the western states were swing states. So you see it already geographically, as it often does in America, red states, blue states, yellow states, green states, whatever, you start seeing different states have different interests based on the lobbying of pressure groups within those states.
- 4.70** And really, the issue most lobbied about early on in American history was tariffs, was trade. So between 1824 and 1846, what you get is competing regimes. When one party is in charge, tariffs go up. In 1824, we get what's called the tariffs of abomination, where tariffs shot through the roof, 60%, over 60%, on goods coming in. And immediately, what happens? Some industries go, "Oh, lower it a little bit for us." And other industries, so immediately this hustling and pressure groups and try to manipulate the tariff rates.
- 4.71** And this is what Congress lives for, right? You get these politicians, suddenly everybody's coming to them asking for favors. And don't think the politicians don't ask for favors in return. This is how it works. So this is a period of a lot of pressure group politics, a lot of favors being given, a lot of exclusions and benefits and so on to different groups in different industries. But average tariffs are 60%, which means some tariffs are much higher and some tariffs are lower. No one set of tariffs.
- 4.72** This actually leads to a constitutional crisis in 1833, where some states are saying, "No, we're not going to abide by these tariffs." This predates the Civil War, but there's already conflict between the south and the north over tariffs in the 1830s. As a consequence, during this constitutional crisis, there was an agreement in the end to lower tariffs to the level of 1816, which is a lot lower. But then politics changed, and in 1842, they're raised again. And 1846, they're lowered again and streamlined.
- 4.73** This is about the same time as the Corn, exactly the same year as the Corn Laws. So there's an influence, an intellectual influence, coming from England about, "No, this is not a good idea." And indeed, from 1846 until the Civil War, we have a relative period of relative free trade in America, and indeed, on both sides of the Atlantic. So this is a period where America is shipping a lot of grain to the UK because of the Corn Laws are gone away. So America can now feed Britain.
- 4.74** In Britain, farms are shutting down. People are going to work in industry. It's becoming the workshop of the world. But at the same time, America is becoming an industrial country. So America is creating competition to the Brits and starting to export industrial goods as well. So if we step back, so between the ratification of the Constitution and the Civil War, tariffs accounted for between 85 to 90% of all government revenue. This was the predominant source of government revenue.

- 4.75 And again, primarily because it was easy to collect. For Madison, certainly, it wasn't ideological. I think Madison understood the benefits of free trade, but they needed something to tax, and you couldn't tax sales, you couldn't really tax income, and I think they were generally opposed to taxing income. They philosophically did not believe in taxing work. But they couldn't, even if they wanted to. They just didn't have the mechanisms in place to be able to do that. Those mechanisms only developed much later.
- 4.76 Now, during the Civil War—by the way, in the Civil War, there is an income tax, right? In the north, Lincoln does impose an income tax. At that point, they have a little bit more of a technology to do it. It's ultimately ruled unconstitutional several times. The U.S. tries to pass an income tax, and try to get the revenue from income throughout the late 19th century, and it's ruled unconstitutional by courts.
- 4.77 Ultimately, we pass a constitutional amendment, an amendment to the Constitution, to make income tax legal, and that kicks into place just in time for World War I. In 1914, we get the beginning of an income tax. It's interesting. When it first passes, just to give you a sense again of politics and war and everything else, when it first passes, the income tax rate is 7%, and only the richest 5 % of Americans pay it. Nobody else pays. That's 1914.
- 4.78 By the time America enters the World War, or when America enters the World War, 1917, the top marginal tax rate is raised into the 80s, and pretty much all Americans are now paying income tax. So within three years, we went from, "We promise only the very, very rich will pay it. Nobody else will pay it. And it's going to be a tiny little amount. Nobody will even feel it," to, "You're all paying it. And you rich guys, we're taking a majority of your income." So when they promise, "It'll be small. We won't—", don't believe them.
- 4.79 So, following the Civil War, and here you get a clear political split now, Republicans become the pro-tariff political party. Democrats are the anti-tariff political party. Sound a little familiar these days? But that flipped in the post-World War II era and now flipped back again. Republicans were the pro-tariffs. And what we got was, early on after the Civil War, we got something called the Morrill Tariff Bill, Morrill, M-O-R-I-L-L. Who was that? A Republican.
- 4.80 An economist by the name of Jevons called this, he said, "The most retrograde piece of legislation to ever emerge from the United States." This is the Tariff Bill. Again, you get the same pattern, corrupt pressure group politics, trying to figure out who should pay which tariffs, which group pays the most, which pays, doesn't. It also antagonized the British because, again, they were free trade. We have no tariffs. And the United States for a while had no tariffs, and it was mutual free trade. And suddenly the United States raises massive barriers to trade.

- 4.81 Probably the most famous act of the period, because President Trump mentions it all the time, is the McKinley Tariffs of 1890. What I don't think people know is, in 1890, McKinley was not the President. The President was Harrison. But he was the chair of the House Ways and Means Committee, and he wrote the legislation. So it came out of his committee. That's why it's remembered as the McKinley Tariffs.
- 4.82 Basically, what McKinley was trying to do was to protect his Ohio constituents by raising tariffs and protectionist tariffs to protect them from competition. It was a lot of pressure. What's interesting is, once McKinley became President, I think in 1897, he clearly started shifting away from a protectionist policy. He wasn't representing Ohio anymore. He was now representing the United States of America. And the idea was, from a United States of America perspective, tariffs made no sense.
- 4.83 And indeed, he was instrumental in starting to find ways to lower tariffs. And he came up with the idea that really was new and that starts dominating from then on, really dominates to this day, of trying to establish, instead of lowering tariffs all at once, which would have been ideal, but create a lot of political pressure, a lot of political force, to establish reciprocal trade deals with different countries. He's the first President to go around and try to negotiate these reciprocal deals.
- 4.84 And as a consequence, he managed to open up imports and export markets for the United States, and increasing trade significantly. He said in 1901, that commercial wars, trade wars, in other words, were unprofitable. Now, a lot of people say, look, the United States had really high tariffs from the Civil War until 1913. And yet, this was the golden era of American economic growth. The economy grew faster than it's ever grown, faster than it's grown since then.
- 4.85 It was industrialized. It became the largest economy in the world. It became the most powerful country, economically and militarily, in the world during that period. And they had high tariffs, so tariffs must be good. Now, there's an obvious fallacy here, which you've probably heard many, many times, the fallacy of correlation is not causation. And in this case, it's that fallacy on steroids. It's definitely not causation.
- 4.86 Indeed, most economists who look at that period and analyze that period have shown that what tariffs actually did is retard economic growth. They slowed it down. It could have been faster, it could have been greater, if we hadn't had the tariffs. The fact that tariffs coincided with rapid growth, according to an economist by the name of Douglas Irwin, "That tariffs coincided with rapid growth in the late 19th century does not imply a causal relationship."
- 4.87 And indeed, he argues, and many other economists argue, again, that the opposite happened—that most of the growth in the U.S. economy at the time happens in industries that are not exposed to trade, that tariffs probably resulted in significant misallocation of capital. Went into industries that we probably didn't have comparative

advantages in and didn't have to invest in, instead of going into industries where we clearly had a comparative advantage.

- 4.88** So what tariffs did was distort the allocation of capital, distort employment. But there was so much energy, there was so much economic growth; and trade, because America was so big, still is big, and still this is true still to this day, because the American economy is so big, the place is so big, there was so much entrepreneurial energy, the tariffs just didn't have that big of an impact. They didn't retard the ability of the economy to grow that much.
- 4.89** This is also a period—I found this fascinating—that a lot of the costs increase in tariffs were offset by a plummeting cost of transportation, primarily shipping. Shipping becomes a lot more efficient. We'll see this again when we get to the 20th century. Shipping becomes a lot more efficient. Rates plummet. You get steamboats, you get much faster shipping.
- 4.90** And the consequence of that—but even the sailboats just become more efficient, more productive—and a consequence of that, whatever effect tariffs have, technology is mitigating it. And indeed, there's a lot of trade going on. There's a lot of trade. This is the 19th century, particularly the second part of the 19th century. This is considered the first era of globalization. Goods are moving all over the world.
- 4.91** Some places are trying to resist it, like China. The British don't let them resist it. They kind of force the issue on them. Indeed, Britain, its navy, its primary goal of the British navy during this period is the protection of the sea lanes for trade. They're all over the world, making sure trade is open. And countries that block trade, the British had a term that they used for dealing with those countries. It was called gunboat diplomacy. We send the gunboat into the ports. We lob a few bombs into the city, and suddenly the port opens up, and trade flows.
- 4.92** So the primary function of foreign policy and military policy for the Brits during the late 19th century was free trade, bring free trade to the world. And of course, the United States benefited from that because it was starting to export in significant numbers during this period. Tariffs are sustained; for most of this period, until 1913, high tariffs are sustained. The consequence of that are multiple.
- 4.93** We exported less than we could have, because there were retaliatory tariffs, primarily by mainland European countries. We had high prices for imported machinery and capital goods that, actually, if you look at its own industrialization, America could have grown faster in terms of industrialization. Americans had higher consumer prices than they needed to and should have. Again, retarded consumer well-being.
- 4.94** And again, productivity, as I mentioned earlier, grew much faster in areas that weren't affected by trade. So in industries that weren't affected by trade, that's where productivity really grew. Overall, theory and evidence suggests that the growth in the 19th century happened in spite of the tariffs, certainly not because of them. And of

course, one of the real prices that were paid for the tariffs leading up to this point is political corruption. Call it cronyism.

- 4.95** Businesses got used to lobbying government. Businesses could benefit by lobbying government. Government didn't have a lot of tentacles into the economy, very few. America was relatively a free market. But the one tentacle they really had were tariffs. And in many respects, this is the first big cronyism. Since then, cronyism runs rampant, because the more government intervenes in the economy, the more businesses push back. In those days, government didn't intervene in the economy, except for this one thing. This is where business pushed back.
- 4.96** There was a lot of cronyism around the railroads. That's the other area in which the government was very involved around the building of the railroads, and there was a lot of cronyism there. But generally, government left business alone, particularly at the federal level. And there were constant attempts to eliminate lobbying, though often ruled unconstitutional. You have a right to speak. You have a right to go and argue your point. But of course, that always has the danger of slipping into actual corruption.
- 4.97** One of the reasons that were argued for an income tax, one of the reasons that were presented to the American people why we should change the Constitution and embrace an income tax, was the idea of reducing cronyism. We would get rid of pressure groups, because we could lower tariffs, because our revenue could now come from income tax. So the idea was, we raised income tax, now we don't need the tariffs. We can eliminate those.
- 4.98** So, the United States grew in the late 19th century because it was economically free, because it had a massive market internally, because it was not that reliant on trade, and it hampered itself. It hurt itself by imposing tariffs. Now, once the income tax is introduced in 1913, tariffs are lowered. You have a Democratic administration, which is generally anti-tariffs, and tariffs are lowered significantly at the same time. Again, this was part of the excuse for why pass an income tax.
- 4.99** So from the rates of 40 to 60%, they are lowered to under 20% by 1920. And indeed, revenue from the income tax grows dramatically, particularly after, what I said, the top rate was actually 77 % in 1918, and after World War I, the top rate was 50%. So there's a lot of revenue coming in from income tax. They didn't need the tariffs for revenue anymore. And indeed, the discussion about tariffs changes. It's completely dropped. The idea of tariffs for revenue is eliminated.
- 4.100** And instead, what you get is, now whatever argument for tariffs is, is a pure protectionist argument. We need to protect industries, particular industries, from foreign competition. In 1922, not that long, we now have Republicans dominating. They reinstitute tariffs with the idea that this will stimulate manufacturing. This had just been a very deep recession, '20, '21. And the idea is, we need a boost, we need to

stimulate manufacturing, we need to stimulate industry, and they raised tariffs in order to do it.

- 4.101** The economy is strong in the 1920s, so the tariffs don't, again, you can't observe any real damage done because the economy does well during this period for a variety of other reasons, at least until, as we know, 1929. 1929 happens, stock market crashes. We're entering into a pretty deep recession that ultimately turns into, of course, a depression. And the tariffs, again, are being considered.
- 4.102** We want to protect the whole market. We're shrinking. People are unemployed. People are losing their jobs. We need to protect them from all these inputs that are coming in and destroying jobs even further. So the question is, how much? And it really is presented as, again, a stimulus package. This is going to stimulate industry. This is going to help grow. And we're heading towards a depression, so this will help us.
- 4.103** And this is the infamous Smoot–Hawley Tariff Bill. Tariffs go up again to 60%. Lots of pressure groups again, in favor, moderating. It leads to a trade war. Throughout the world, people respond, react. They raise tariffs on American goods. Exports plummet. And world trade declines from about \$3 billion in 1929 to under \$1 billion in '33. So you get a reduction in global trade of about a third because of the trade war that Smoot–Hawley instigates.
- 4.104** Now, there's quite a bit of debate—the extent to which Smoot–Hawley was responsible for the Great Depression. I think everybody agrees that it wasn't. But the extent to which it made the Depression worse, I think there's generally an agreement that it made it a lot worse, that Smoot–Hawley had definite negative implications for the United States, but really the global economy.
- 4.105** Remember, the Great Depression was very much a global phenomena. Even though it lasted longer in the U.S., it was dealt with in a poorer fashion in the U.S. than in other places around the world. It was a global phenomenon. A big chunk of that was the fact that global trade collapsed and U.S. trade with it. In 1934, the FDR administration comes to conclusion, particularly the Secretary of State, a guy named Cordell Hull, comes to conclusion that these tariffs are not good.
- 4.106** This is bad policy. They're really hurting the United States. We're trying to get out of the Great Depression, and this is not helping. Now, FDR does a lot of things that sustain the Great Depression, make it worse, prevent us from getting out of it. The Great Depression lasts until really the end of World War II. But one of the few positive things that the FDR administration does during this period is, Cordell Hull comes to this conclusion.
- 4.107** Now, they can't repeal Smoot–Hawley. They don't have the votes for that. And we're in the midst of a depression, and people still believe that the tariffs are helping, in a sense, protecting us from foreign competition. So what happens is, he gets Congress to shift the focus, and this is still true to this day, shift the focus from Congress voting on

bills that raise or lower tariffs, like Smoot–Hawley and like McKinley, to giving much more power to the executive.

- 4.108** The idea is this. Congress is dominated by special interest groups, by pressure groups, by different industries, and every state has its own pressure groups. So congressmen and senators can never be objective, can never take their national interest into account. They are too localized. They're too influenced by the local pressure groups. We saw that with tariffs from the beginning, how much they are.
- 4.109** The President, because he's elected by the whole country, is going to be a lot more favorable to free trade. He's going to be a lot more open to free trade because he's taking into account the interests of the entire country, not any particular group. We all know free trade is good for the country. So we're going to give the President the power, more power, not all of it, more power, to now establish trade agreements with other countries, bilateral trade agreements, where Congress gets to ratify these agreements, but they do it with a simple majority.
- 4.110** No 60 votes in the Senate. This is one where you can pass them all by 51 votes. And the deal, Cordell Hull goes out there and starts making deals with different countries to lower tariffs with the United States and to establish more free trade. He's bypassing Smoot–Hawley. There's a sense in which Smoot–Hawley to this day is on the books. It's never been repealed. But tariffs have been lowered on a case-by-case basis, based on treaties and bilateral or multilateral agreements that the United States have signed. We'll get to that in a little while.
- 4.111** So what you get is a period, starting in 1934 really, that goes through World War II, where you're slowly seeing liberalization of trade, and tariffs come down from an average of around 60% to about 30% by the at the end of World War II. And this is an international phenomenon now. Granted, there's not a lot of people to trade with during this period. Everybody's at war. It's a World War. And we'll see how trade relates to the World War in a little bit. So there's not a lot of people to trade with.
- 4.112** But by the end of the war, of course, the rest of the world is decimated. And all the United States is really interested in is exporting. It doesn't need much. Or put it this way, nobody's producing anything for the United States to be importing much stuff. Europe is in ruins. Asia is in ruins, not just Japan, but all of Asia is really in ruins. They fought a war. Latin America is the only other part of the world, together with the United States. Africa, but Africa is not producing anything. Latin America is the only other part of the world that hasn't been devastated by the war. But most of the productive capacity in the world has now been destroyed.
- 4.113** So, we are shifting towards an era of lower tariffs, of lower trade barriers, because tariffs are not the only trade barriers. We'll talk about other trade barriers in a little bit, and other protectionist things that government can do in order to stop trade. We'll talk about those in a little while. So next class, we'll pick up on the history from the end of

World War II. We'll talk about the end of World War II and how tariff regimes and protectionist regimes evolved from there.

Lecture 5: Globalization Accelerated

At a glance — Globalization accelerated.

- 5.1 So last class, we talked about the history of trade and of tariffs, and we got all the way up to the end of World War II. And we saw the century, the 19th century, where Britain moved towards a free trade country and unilateral free trade, lowered tariffs to zero. And at the same time, the United States, in a sense, moved in the opposite direction, of increasing tariffs, but a constant battle, and constant back and forth, and constant lowering and increasing. Until there came a time where there was a general consensus, I'd say, coming out of the Great Depression, that certainly, tariffs were not the way to go.
- 5.2 I want to step back a second and just make another point, and that is, tariffs are not the only way to stop trade. They're not the only way to restrict trade. There are lots of other barriers to trade that can be experienced. We saw that in Great Britain with the Navigation Act. The Navigation Act were not tariffs. They just said, if you want to move products into Great Britain, you have to use a British ship. Well, there are only so many British ships. That limits trade, and it limits competition around seafaring stuff.
- 5.3 Well, the United States also has quite a few non-tariff trade barriers. And I'm just going to mention, I'll mention two that are worth talking about again because they tend to be in the news. And the first one is one that is very similar to the Navigation Act. That is the Jones Act. Not quite as oppressive as the Navigation Act, but still quite oppressive. So the Jones Act, which you might hear quite a bit—I don't know if you've heard in the last few years, there's a lot of talk about it. Still nobody's offered to repeal it yet.
- 5.4 But there's a law that passed in 1920. It was called the Merchant Marine Act of 1920. Basically, the idea was, look, during World War I, we very much relied on the merchant marines, that is, commercial ships to transport troops, weapon systems, to Europe. Couldn't have won the war without a robust shipping industry, without robust ships that transport troops and everything else to Europe. So the law passed in 1920 that basically governs domestic shipping, shipping in the United States between American ports. So this is not international, but it has big implications for international.
- 5.5 So the whole idea was, we need to beef up the shipping industry. We need to make sure that we have American-built ships, American-owned ships, American-crewed ships. We need to make sure we have people who work on ships because there might be a war and we might have to transport stuff overseas. So here's the rule. Here's the law.
- 5.6 If you're shipping stuff between two American ports— I don't know, Baltimore to Boston, Houston to Puerto Rico, Houston to Jacksonville, Los Angeles to Hawaii—you could only do so on American-built, American-owned, American-flagged, and American-

crewed ships. American-owned, American-built, American-flagged, American-crewed ships. Crew must be at least 75% U.S. citizens.

- 5.7** And all this in the name of national security, to make sure that we have enough ships and we have enough people who know how to use the ships, man the ships, to be able to deal with another crisis like World War I or World War II. Of course, there's an industry here to protect, an industry to protect with the shipbuilding industry. The United States built a lot of ships during World War I and then World War II. Huge number of ships, aircraft carriers, battleships, but also merchant, marine ships.
- 5.8** United States was probably the largest shipbuilder, certainly during World War II, in the world. And we wanted to make sure people had jobs. Job creation. It's always good for a Congressional bill to pass if you can phrase it as job creation. We want to create jobs in the merchant marines. And we want to maintain our infrastructure. And of course, this was backed at the time by the marine, the maritime union, the people who work on ships, by the shipbuilding industry. Shipbuilders and defense contractors and defense industry generally all supported this bill. And again, national security as the primary.
- 5.9** Now, the reality is, over the last, what is it, 100 years, the United States has lost its comparative advantage in shipbuilding. Maybe because of the Jones Act, because it doesn't face competition, or maybe because other countries have a comparative advantage in shipbuilding. We make almost no ships in the United States, some for the navy, but in terms of merchant marines, there's no shipbuilding industry in the United States.
- 5.10** Korea, South Korea, Japan became dominant shipbuilding countries. And of course, China today builds over 50% of all the ships that sail the seas. In spite of the Jones Act, we have no shipbuilding, or maybe to some extent because of the Jones Act, we have no shipbuilding capacity in the United States, or almost none.
- 5.11** At the same time, it's very, very difficult to ship products between two U.S. ports because there are very few ships in the world today that are American-owned, American-flagged, American-crewed, and were built in America. Actually, there were only 72 such ships. So let's say you're in Boston, and it's a very cold winter. And you've basically exhausted the amount of natural gas coming in through the pipelines that bring natural gas to Massachusetts from the oil fields in Pennsylvania or wherever. And you need more natural gas.
- 5.12** And we have today LNG. LNG is a way you liquefy, you take natural gas, you turn it into liquid, you put it in a ship, you take that ship and you bring the natural gas that's needed. There are special terminals that turn the liquefied natural gas back into a gas, and you could supply Boston with more natural gas. So you'd think that Boston would be bringing in natural gas from probably Baltimore. Baltimore has an LNG terminal. It's a quick ride up.

- 5.13** But there are no Jones Act LNG ships. So it would literally be illegal to use a foreign-owned or a foreign-flagged. 75% of the crew has to be American. How many Americans work on ships? Very few. So it turns out that, before the war in Ukraine, most of the natural gas that Boston imported from beyond what they brought in from pipelines was from Russia, not from Houston, not from Baltimore, from Russia.
- 5.14** I live in Puerto Rico. Puerto Rico, you can't get stuff from the United States, can't get natural gas from the United States because there's no oil and gas ship. Puerto Rico is an American port. You can't leave America on a non-American-built, owned, all the rest ship and come to Puerto Rico. So everything is more expensive in Puerto Rico because there's only so many ships that can make that trip.
- 5.15** Natural gas cannot come from the United States. Puerto Rico, in the past, used Russia. Now it has to scrounge around South America and find somebody who's willing to sell them natural gas in order to fund natural gas turbines to produce electricity. After Maria, for a little while, the Jones Act was suspended because they needed to get goods to Puerto Rico. So it's funny how the government suspends these laws for a short period of time because they know they hurt, but they won't suspend them forever. They won't take them away forever.
- 5.16** To ship stuff from, I don't know, Jacksonville is a big port, and to ship it up to New York and other places on the east coast, it makes sense just to load up on boats and ship it up. We don't. We don't even have Jones Act boats. So what do we do? We put on trucks. So if you go on I-95 from Miami all the way to New Hampshire, it is loaded with trucks. And by the way, trucks are what kill people on highways. If you get casualties in car accidents, it's almost always from trucks.
- 5.17** And we have clogged up I-95 with trucks, when they should be going by boat. Would be cheaper, safer, more effective, more efficient, but we don't have enough Jones Act boats, and we don't have a shipping industry. So we can't really build any Jones Act boats. We got 72, and that's pretty static. Now, once in a while, people get all excited, like me, get all excited about this and make a big deal out of it and so on. And nobody pays attention. Nobody cares.
- 5.18** It's basically economists who seem to be affected, except when something like Hurricane Maria comes about and then you suspend the Jones Act for a little while, but then reimpose it later on. And Puerto Rico, of course, and Hawaii, Hawaii is very expensive to live in. One of the reasons are, it's a very—you can't use normal ships, and they're much more likely, by the way, in Hawaii, to import stuff from China, Japan, or Korea, than they are from the United States, because there they can use any ship.
- 5.19** And shipping, by the way, shipping outside of Jones Act stuff is super cheap. We'll talk about this. Super cheap. You got these massive container ships. The marginal cost of shipping is very low, except in the United States between two American ports. And the Act is still in force. It's really interesting when Trump came in office, one of the things

they were going to do is they wanted to reestablish a bunch of marine shipbuilding capabilities in the United States.

- 5.20** Trump was very upset by the fact that, or people in the administration were very upset by the fact that China was building over 50% of the ships in the world. What are we going to do? This is a national security issue. People were appointed, positions created, in order to launch a big shipbuilding capabilities. You know who lobbied against it? People who own Jones Act ships. They don't want more ships to be built. That'll lower prices, and it'll create competition for their 72 boats.
- 5.21** And by the way, that whole initiative is now being sidelined. It's too expensive. Nobody in America knows how to build ships. We're not going to build ships. It's just a fantasy. We've got allies who build ships. If we really need ships, let the Koreans and the Japanese build it for us. Again, trade is better than doing it all yourself. We don't have a comparative advantage there. Every aircraft carrier we build takes gazillions of dollars more than budgeted and way, way longer than what we expect.
- 5.22** So we can't even build the stuff we really need, like military boats, never mind the merchant marine stuff that we can buy cheap off the rack from anywhere in the world other than here. Jones Act is on the books. It's not going anywhere, in spite of the irrationality. Nobody out there can really defend the Jones Act because it hasn't lived up to its purpose. We don't have merchant marine capabilities to—if we had to transport lots of troops and everything else to Europe, we'd have to rent boats from other countries.
- 5.23** Which is, so what? So we rent boats from other countries. We couldn't do it ourselves. So it's failed to achieve its goal. But you see, nobody lobbies for that. Think about how politics work. Who's going to go to Congress and pound the table and demand and ask and bribe in subtle ways? Our politicians, to get rid of the Jones Act? It's like, nobody, a bunch of intellectuals like me. Nobody's actually going to do it. There's no special pressure group. There's no pressure group that is going to get excited about this. Like, Puerto Ricans, maybe, but nobody cares about Puerto Rico. So nobody.
- 5.24** But is there somebody who's willing to pound the table and really get excited and really lobby and really fight for the Jones Act? Yeah. A tiny little merchant marine industry. It's very small. It doesn't count for much, but they're in a few key states. And their senators and their congressmen will fight for them. And this again, it goes to the point that we the people have really no representatives in Congress. The representatives in Congress are for the little districts or the states.
- 5.25** And the people who have most influence on those districts and the states are, to some extent, us, what we vote for or against. But nobody's going to not vote for the congressmen because of the Jones Act. And we're not going to lobby, because lobbying is something you need to spend money on. It's costly, it's expensive, and not a single

one of us loses enough on the Jones Act to make it worth their while to spend the money. This is how politics works on everything. It's just a little example from trade.

- 5.26** I'll give you one other example, non-trade barriers. In 2022, Abbott, the food manufacturer, had to shut down a plant that produced baby food. And the consequence of that was a massive shortage in baby food. It turns out there were only a small handful of baby food manufacturers in the United States. And for a variety of reasons that have to do with the structure of the industry we won't get into, small number. A lot of the sales are sales through the government because a lot of the baby food is distributed through the welfare system.
- 5.27** But anyway, small number of baby food manufacturers, one plant going under, completely destabilized the industry, and there were massive shortages. People relied on specialized baby food, like non-allergenic baby food, and couldn't find baby food. Women who weren't breastfeeding, couldn't find substitutes. It was a real—I mean, people were panicking. This was a real crisis. There were a lot of news stories about this.
- 5.28** And of course, the obvious solution to this is what? Well, Abbott closed their factory, yes, but there's a lot of capacity for baby food all over the world. And indeed, there are countries, Europe, for example, that have very high regulatory restrictions on quality. They're just as safe as American baby food. They're just as healthy as American baby food. Their FDA equivalent is just as rigorous as an American FDA equivalent.
- 5.29** But you cannot import baby food into the United States unless the FDA has tested it and approved it. And none of the European companies bothered to get the FDA testing and approval before this crisis because it was expensive. It takes a long time, and it just wasn't worth it in terms of competing. It's a competitive industry. Prices are low. They're not making a lot of money. So American companies dominate the American market.
- 5.30** So, even though there's completely good, healthy baby food all over the world, couldn't import it into the United States because none of them had that stamp of approval. And of course, once the crisis hit, you couldn't get the FDA to approve it quickly. There's no way to do that. You have to go through all the tests and run tests and everything. You can't just accept the approval of the equivalent of the FDA in Europe.
- 5.31** Now, ultimately, after weeks and parents panicking and everybody being very, very upset, the government actually did. They said, temporarily, we're going to suspend the requirement that these get FDA approval, as long as they got an approval by an agency similar to the FDA in their home country. They also, it turns out, do tariffs on baby food. They eliminated the tariffs, and baby food flooded into the United States and solved the problem. You got it from Mexico, from Ireland, from the EU. Plenty of places that have baby food. They can produce more, and they can get it in, and the crisis went away.

- 5.32 Now, why was it temporary? Why not for good? If this is such a benefit, if this smoothenes out supply, and maybe even lowers prices, could be competition does that. But no, as soon as the crisis and the Abbott factory opened up again, the tariffs were raised, the trade barriers were raised. If you go to Europe and you find a brand of baby food you really like, I don't know, the super non-GMO, whatever, organic, super-duper, the Europeans are very good at these things, and you buy a bunch and you put in your suitcase and bring it to the United States, you will get into big, big trouble.
- 5.33 It's not that you pay a fine; they will confiscate it from you and you will pay a fine and you're committing, you're breaking the law. You cannot bring baby food from Europe into the United States. It's not like Europeans are dropping dead because their baby food is poison. But of course, Abbott, the people who got us into trouble, are probably lobbying heavily to keep those barriers in place because, for them, this is in the short run a profitable deal.
- 5.34 So you've got lots of trade policies, lots of different types of trade policies, that restrict supply chains, restrict importation into the United States, create distortions and perversions. And we all pay much higher prices on a lot of the goods that we get. We pay higher prices. We have less variety, as much variety as we have. We could have had even more, because of tariffs and non-tariff barriers, non-tariff barriers that are negotiated.
- 5.35 In the 1980s, when Intel was losing market share, significant market share to Japanese memory chip producers, Ronald Reagan basically forced the Japanese to establish a quota system where they would only agree to sell so many chips into the United States. It wasn't a tariff. It was just, you can't sell more than X number of millions of chips into the United States, trying to resurrect Intel—again, trying to resurrect a particular small industry.
- 5.36 In the meantime, Intel had pivoted and was starting to make other chips, which were much more profitable and much more valuable. So the whole thing was not that—it didn't really save Intel. What saved Intel was a strategic pivot. Right now, Intel is struggling again. It happens every, I guess, 30, 40 years. And again, we're subsidizing Intel and trying to bail them out. That was in the CHIP Act that President Biden signed. So that happens all the time.
- 5.37 Same thing happened with automobiles. There was a period in the 1980s where, obviously, the Japanese were kicking our butt in terms of automobiles. They were selling much better, much cheaper, less maintenance, just better cars, and also smaller cars, which Americans were demanding at that point. Detroit couldn't keep up. So what did they do? They went to Ronald Reagan and asked really, really nicely, and Reagan imposed quotas on Japanese cars. And the consequence is, fewer Japanese cars entered the United States.

- 5.38 Ultimately, what the Japanese did is, they came to the U.S. and built cars here in order to get around the tariffs and the restrictions. But they didn't build them in Detroit. So they didn't build them in the Rust Belt because who wants to deal with American auto union workers? They built them in Alabama and South Carolina and places like that that are not unionized, are more efficient and more productive. And a lot of the Japanese cars you buy today, and a lot of the cars generally, BMWs, Mercedes, they're all built in the southeast United States.
- 5.39 So let's go back to our story, back to our history. So following World War II, the United States, the world really faced this issue of the destructive capacity of most of the world had gone. The United States was the only one standing. The dollar now was going to be the global currency, the reserve currency of the world. The British pound had been before World War II. Now it was going to be the dollar.
- 5.40 And the intellectual mindset had completely changed, really globally. Finally, Adam Smith was listened to. And the whole orientation was now towards establishing as much free trade as possible, and bilateral or multilateral free trade, so free trade across the board. And what you see is, starting from the end of World War II, really until the first Trump administration, you saw a steady, systematic, uninterrupted really, decline in tariffs in the United States and across the world.
- 5.41 Initially, in 1947, they created an organization called GATT, the General Agreement on Tariffs and Trade, which was basically, all these countries got together once every few years and negotiated trade deals and negotiated to lower tariffs and to lower trade barriers. And there was a series of these multilateral agreements that basically, globally, lowered tariffs and lowered non-tariffs. And global trade expanded dramatically.
- 5.42 The Carter administration, during the 1970s, negotiated something called the Tokyo Round, of these GATT agreements in 1979. That resulted in average tariffs reduction of 34% across the United States. And again, if you look at the graph, tariffs are just going down throughout the late 20th century into the 21st century.
- 5.43 GATT is then replaced by the World Trade Organization. The World Trade Organization again is a place where you're supposed to come together, and multiple countries, to negotiate lowering of tariffs and lowering of not just tariffs, but non-trade barriers as well. The idea with the World Trade Organization is it also had mechanisms for dispute resolution. "Oh, country X raised tariffs and country Y in contradiction to the agreement they had. What do we do about that?" And the WTO would get the parties together, they would negotiate.
- 5.44 And there was a potential even to impose restrictions on a country that had violated the WTO agreement. And countries brought suit against other countries, in a sense, at the World Trade Organization. And again, this basically led to a steady global reduction in tariffs. Tariffs were reduced by another third as a consequence of this. And indeed,

again, if we enter into the pre-mid-20-teens, tariffs into the United States were in the low single digits. I mean, they were very average tariffs, very, very low. Some goods were higher, pressure group again.

- 5.45 But for the most part, tariffs were incredibly low, getting close to zero, getting close to the situation of global free trade. And in spite of what people believe, this is true globally. That is, China had reduced its tariffs dramatically during this period, as had the Europeans and the Japanese and the Koreans and everybody. The entire globe was moving towards a regime that came very close to zero tariffs. Tariffs across the board for most of the major trading partners were in the single digits, average tariffs.
- 5.46 The United States also, during this period, negotiated some bilateral agreements or trilateral agreements. Of course, the most obvious case of this is NAFTA. NAFTA was a deal. It was negotiated in 1994 between the United States, Mexico, and Canada. Now, NAFTA is supposed to be a free trade, a deal, and a lot of tariffs were lowered to zero. So there's a lot of goods under NAFTA that you can import and export across the United States and across Canada and Mexico at zero tariffs.
- 5.47 For example, auto parts. This is why, for example, in Michigan, in Canada, you have a lot of auto parts and assembly and pieces of the automobile are being built in Canada and some in the United States. And sometimes, in a sense, an automobile goes back and forth and back and forth seven times between Canada and the United States. Zero tariffs. They're very close geographically. The Canadians, Canadian companies, have a comparative advantage in certain products, and Americans have comparative advantage in certain products.
- 5.48 These are all American cars, American-labeled, American-designed, American everything, but parts of them are produced in Canada. And then, of course, the huge advantage is to building certain cars and certain things in Mexico as well. So American car companies have—and they can come in and out with basically no restrictions. There are no non-trade, no barriers, no tariffs, nothing. They can go back and forth. And that's how things worked until very recently.
- 5.49 NAFTA was then, in the first Trump administration, Trump came in, and there was a lot of angst about NAFTA. A lot of stories, we'll get to them in the next class, probably. A lot of stories about NAFTA's destroying jobs, unemployment, this is bad, this is horrible for America. Trump came in, "I'm going to renegotiate NAFTA. NAFTA is no good. NAFTA is a lousy deal." And indeed, in 2019, Donald Trump signed the USMCA, which was NAFTA 2.0, with Mexico and the Canadians.
- 5.50 There were some, you could argue, improvements to the deal, particularly for the auto industry. I think they really went down to zero at this point, but there were other things that were improvements. And that was the situation until a few months ago. It is true, though, that every time we talk about these deals, they're not really free market deals.

There are a bunch of industries. There are a bunch of industries that are protected. There are a bunch of restrictions that are placed.

- 5.51** Like, if we're going to import stuff from Mexico and it's going to be at zero tariffs, then we demand all kinds of environmental regulations that Mexico follow, and that raises the cost of the good, but they might not follow them. So then we do have a tariff. There are quotas on certain things that we do not want them flooding our market with—I don't know, trucking, for example. A Mexican truck can't drive into Texas and deliver its good because drivers are less reliable. I don't know, their trucks are more polluting and all that.
- 5.52** So you have to unload the truck on the Mexican border, load it up to an American truck, and drive it on, which raises costs. So there are a lot of limitations and restrictions, but overall, NAFTA liberalized trade. That is, it increased the amount of trade between Canada, U.S., and Mexico. It lowered tariff significantly, and it increased trade dramatically. That is, trade pre-NAFTA and post-NAFTA, much, much higher post-NAFTA.
- 5.53** And it allowed Mexico and Canada and the United States to each specialize in what they have a comparative advantage in and increase trade efficiency dramatically. But it's not free trade. These deals are almost never free trade. They're always, everybody's trying to protect the thing that they think needs protecting, or the industry that lobbied the most.
- 5.54** The U.S. came very close to another deal. This was a multilateral deal called the TPP, Trans-Pacific Partnership. This was an interesting deal because this was going to create, again, not perfect free trade, lots of non-trade barriers and lots of restrictions and a lot of phasing-in and stuff like that. But it was focused on the Pacific. It was going to create a trading agreement between countries in the Americas, that were on the Pacific side, and countries in Southeast Asia. And strategically, it kept China out.
- 5.55** So this was a great way to trade with a bunch of allies, because most of these countries are American allies, while, in a sense, creating a bloc of trade to counter the influence of China, which we view as a potential enemy, and therefore we wanted a little bit of distance from. And this was negotiated over many years. The United States was a key partner, and Japan, and Korea, and the Philippines, and Indonesia, and Chile, and other countries in South America.
- 5.56** And it came time to sign the deal. Well, Trump became President, and Trump nixed the deal and refused to sign it. So there is a TPP. All these countries are part of a trade bloc. They all have very low tariffs among each other. They're all participating, except for the United States, which refused to sign a deal and enter into this trade deal. In effect, that was great news for the Chinese. Great news for the Chinese.
- 5.57** There are a lot of these trade deals. We've engaged in quite a few of them. Again, particularly NAFTA and then the GATT and WTO agreements. The net result of all of

that is a dramatic reduction in tariffs across the board and across the world. The world has become a lot more free in terms of trade than it ever was, maybe ever in history. It's maybe even freer than what it was in the late 19th century when Britain protected the sea lanes.

- 5.58** Of course, really, since World War II, the United States has taken that job. Our navy, what's its main job? The main job of our navy is to protect the sea lanes. It's to protect shipping. If you look at where we have bases, if we look at where we put ports for our navy—Southeast Asia, in the Indian Ocean, in the Middle East, to protect what ships from the Middle East is oil. Today, we're maybe less dependent on that Middle East oil, but we are, in a sense that oil is a global commodity, so it's priced on a global scale.
- 5.59** If something happens in the Middle East to oil, even though we produce oil in the United States, prices will go up because prices are set globally, they're not set locally. So we protect that. And of course, Europe, we protect the Atlantic, although there weren't very many threats to the Atlantic. Russia, arguably now. But U.S. has a navy fleet stationed in Mediterranean. So we are there to protect the shipping lanes of the world. We've taken over what the job of the British navy was in the 19th century. And that's what we do, more effectively and less effectively.
- 5.60** It's one of the reasons the idea that the Houthis are blocking the strait that dramatically restricts trade is so offensive. And the idea that this primitive, barbaric tribe can get away with that, and the U.S. will just turn the other cheek and not really do anything about it, is so offensive, because the reason we have a navy is to make sure that goods travel around the world, and that goods can get where they need to be. We benefit enormously from that ability for goods to move freely around the world.
- 5.61** But Suez Canal is now mostly dormant. By the way, Suez Canal was built in the 19th century during the height of free trade, by the British and French, in order to facilitate trade from Asia being faster. So they cut this big canal in Egypt that separates Egypt from the Sinai Desert. And a big chunk of world shipping goes through the Suez Canal. It used to go, before the Houthis, through the Suez Canal, primarily from Asia to Europe.
- 5.62** But this was, again, the height of British free trade, the idea of the benefits of free trade. They're willing to invest in it. They're willing to build this canal and then protect it. They had troops there basically until 1956, when the Egyptians kicked them out and nationalized the Suez Canal. And when the British tried to take it back, the British and the French tried to take it back, Eisenhower told them to go back home. And it became completely Egyptian.
- 5.63** The one downside to these free trade agreements is, as I said, they're not really free trade. They lower tariffs, but they have all kinds of other non-tariff barriers. The tariffs are not uniformly lowered. I'm going to argue in a future class that the ideal trade policy any country should embark in, any country, is that of late 19th-century Britain.

Unilateral zero tariffs. Other people want to tax their own citizens and destroy their own industries and create barriers to trade, go for it.

- 5.64** But we, particularly the United States, that is such a huge domestic market, we can tolerate some competition. We're good at it. We can handle it. And the benefits to Americans are massive, and generally, there's pretty much no tax that I would not cut. And here's a tax that is discriminatory, that hurts consumers, and it has no upside, no benefit. Yeah, take it to zero. Take it to zero. And I think we benefit, the rest of the world benefits. It's generally a positive.
- 5.65** The rest of the world, while we are busy right now raising tariffs and creating more trade barriers, the rest of the world is shifting in the other direction. Adam Smith is still alive and well elsewhere. So TPP just keeps expanding. They keep on adding countries to the TPP. I think the UK now is a member of the TPP, even though they're not in the Pacific. They can trade with all those Pacific countries with very, very low tariffs.
- 5.66** Africa. Africa has created a massive free trade zone, where between African countries, there are no tariffs. And this is going to be a huge boon for Africa over the long run. Hopefully, Africa is moving in the direction of more free markets and, as a consequence, industrialization and rising standard of living, we'll see. But free trade is an important part of that. They recognize that, and they have done it.
- 5.67** The EU, of course, is a free trade zone. That's what it is. There are no tariffs between European countries. Zero tariffs. There's no restrictions on movement of goods, movement of labor, movement of capital between European countries. So Europe has embraced, in that sense, free trade. They have high barriers for goods coming into European Union. But even those are slowly being lowered. And Europe now is talking about joining the TPP and other things.
- 5.68** What do you think the UK and Europe try to protect the most? What industry is the one that they're most protective of? It's interesting. Agriculture. I think for historic reasons, because not a lot of people work in agriculture, because of increased productivity. But agriculture is the one industry that lobbies the hardest and is most protected in places like Europe and the UK, because they realize, the reality is that if Europe lowered tariffs on agricultural products to zero, if the UK did the same thing, lowered tariffs on agricultural goods to zero, they'd be flooded by really cheap, really good food from Africa.
- 5.69** It would boost incomes in Africa. Africa would actually become quite wealthy because they would be selling food into the European Union and the UK. I mean, some of the food would come from the U.S., but Africa is right there. It's close. They would, by the way, solve some of their migrant problems, because some of the migrant problems is the fact that Africans, having lots of babies, lots of kids, and they can't feed them, there are no jobs—suddenly, you'd create jobs. So Africans would stay in Africa instead of going to Europe.

- 5.70** But what's the cost? The cost is that a lot of people in the agricultural industry in Europe would lose their jobs. A lot. And they would all find jobs in something else, something that Europe have a comparative advantage in. Europe has no comparative advantage in agriculture. Maybe wine in France, maybe certain specific speciality, organic, whatever products. But mostly Europe, Europe, agriculture is an industry for up-and-coming countries. It's not an industry for developed countries.
- 5.71** I mean, indeed, if we got rid of all the corn subsidies, we subsidize corn in this country with no end. I mean, so much of your tax money goes to corn farmers. It's hard to believe. And we don't do it directly. We don't just pay them. I'll give you an example. There's a big issue now about Coca-Cola. You know Coca-Cola? They use corn syrup in Coca-Cola, not sugar. And Trump has now got Coca-Cola to agree that some Coca-Cola, later this year, will be produced with cane sugar.
- 5.72** Why, in America, do we use corn syrup instead of sugar for Coca-Cola drinks? I mean, people should really ask this question. If you go to Europe, you go to Mexico, all the Coca-Cola has cane sugar in it. Only in the United States do we use corn syrup. Why do we use corn syrup? Because we have very, very, very high tariffs and indeed quotas on importing sugar from outside the United States.
- 5.73** Sugar, the simple sugar that you buy in the supermarket, costs exactly double in the U.S. what it costs anywhere else in the world. Now, why? Sugar. Why are we doing that? Like, there are five sugar farmers in Florida that, you know, I get that they lobby and they've got two Florida senators that are sugar senators, and they'll do anything to preserve sugar. But that's small. That's only Florida. Who the hell cares about Florida? So why is sugar so powerful?
- 5.74** Because who else lobbies for sugar tariffs? Who wants sugar tariffs? Who loves sugar tariffs? The corn farmers. And corn is big business. We're talking about Iowa, Nebraska, Kansas, and the whole Midwest. It's all corn. I mean, Bob Dole used to be called the ethanol senator. Why? Because that's the other thing we make from corn, ethanol. Why do we make ethanol? It doesn't really clean anything. All our gasoline has to have some ethanol in it. The only reason that's the case is because they lobbied for it.
- 5.75** But we pay double for sugar because of the corn farmers. If we eliminated the sugar subsidy, we would destroy much of the corn farming of the Midwest. If we eliminated the ethanol, we'd destroy—I mean, how much corn do we actually eat? We don't make that many corn tortillas here, and we could import them. I think Mexico has a comparative advantage in corn tortillas. But imagine if you freed up all that land where we grow corn for nothing, for no purpose, and feed it up to grow something else, or to build housing and lower housing prices in the United States. I don't know. Something crazy like that.
- 5.76** Or just grew something different or planted forests for recreation. I don't know, whatever you want to do with that land. But the least productive use of that land is to

grow corn. But we've created the structure of incentives, of regulations, of tariffs, where it's very profitable to grow corn, and then we subsidize the corn farmers. On top of that, we have the largest farm bill ever in American history right now just passed under the guise of the Big, Beautiful Bill. Massive increase in farm subsidies, many of them going to corn farmers.

- 5.77 And corn farmers are represented by big business. We got Cargill, and I forget the second big food manufacturing. They're both private companies, and they're big, and they have figured out the lobbying game. They know exactly how to do it. It's one of the most strongest lobbies that we have in this country. Maybe the strongest is corn farmers, the corn interests. So that's why we don't have sugar. And if we just lowered tariffs to zero, across the board, among the many benefits we'd get is, we'd get sugar cheaper. Not that we need more sugar, generally, but I think it beats corn syrup.
- 5.78 One of the accusations, significant accusations, during this period against trade was this idea that there was something called a China shock. So in 2000, U.S. law granted China permanent normal trading relations. It's called PNTR, where we traded freely with China, and China benefited from the relatively low tariffs the United States had. And then in 2001, the United States helped China get entry into the WTO, into the World Trade Organization. It became part of this multilateral trade deals. And again, tariffs went down.
- 5.79 Now, tariffs went down in China as well, but they went down vis-a-vis Chinese goods into the United States. And the argument was, and there was a series of academic papers that were published, that this constituted a shock to the American industry. You've heard those stories. The middle class has hollowed out, people lost their jobs. The industry shut down, the rust belt was destroyed, and that all of this was caused by the China shock.
- 5.80 Some economists estimated that in the communities that were affected by it, not generally in the United States, but in the communities affected by it, something like 2 million people lost their jobs in national industries between 1997 and 2011, so that's over 14 years, either directly or indirectly, because of this increase in inputs from China. Of those, just under a million were manufacturing jobs.
- 5.81 The most effective subsectors were going to be where you had low-skilled workers like apparel, textile, footwear, but even computer and electronics parts suffered as a consequence. And this constituted a significant harm to American workers in the communities. Of course, the reality is often different than what those initial first impressions happen to be. The reality is that trade with China had been increasing dramatically starting in the 1980s when China opened up, when China abandoned its official communism and started opening up its markets.
- 5.82 Trade had already been increasing well into the 2000s, before the 2000s. Indeed, Chinese imports to the United States increased more than sixfold in the decade before

the United States gave them normal trading status. So there was no shock. There was just a continuation of a pattern that already existed. Now, this accelerated because they joined the WTO and they got this normalization of trade status. So there's no question that accelerated the amount of trade.

- 5.83** But the reality is that 2 million jobs is not a lot. I mean, it sounds like a lot. But how many jobs are lost every month in the United States? If you take in the United States, how many people are fired or leave a job every month in the United States? Over a million. They go find other jobs. It's not the case that we saw a massive spike in unemployment. People left the jobs that were now out-competed. But if they wanted to, and this is a big if, they could find other jobs.
- 5.84** Maybe not in the same region. That's part of the issue. A lot of these studies looked at these small communities where the jobs were lost. And there's a real question, important question, I think, that we have to answer as Americans, is why people are not willing to move states and move to other places. The American people, historically, have been the most mobile people in the world. We are a people who are quite happy to get in our cars and drive to a new city and start over. Not anymore. Last 30 years have seen a dramatic decrease in mobility among Americans moving from place to place.
- 5.85** If you really look at it, it has to do with welfare. It has to do with states' licensing laws. It's like, I'm licensed to be a nurse in this state, but I can't get a job in that state because they have different licensing laws. There's a lot of government bureaucracy that keeps you in place. It doesn't allow you to leave. You might qualify for welfare or some child benefits or something here, but then you won't qualify there because it's on a state basis.
- 5.86** There are lots of reasons, but the reality is that people don't move as much as they used to, a lot less than they used to. I mean, 1 million jobs lost in manufacturing, but the reality is that over the same period of time, those 14 years, 5 million jobs were lost in manufacturing. And tens of millions of people left their jobs or were fired from their jobs. It's a fraction. China didn't have that much impact on what happened in the United States.
- 5.87** And then the question is, are we better off or worse off? Are people better off or worse off? We'll get to that next class in terms of how we've done since then. And look, I'm not saying people weren't hurt. Of course, they were hurt. But if I introduce a new technology that automates an assembly line, people are going to lose their jobs. Every step in automation, in technology, some people lose their jobs. But overall, more jobs are created, always. Employment generally goes up.
- 5.88** People lost their jobs in COVID and from the great recession and from all kinds of reasons. OK, so we got more inputs from China. People lost their jobs. But the question is, how do we recover from that? And what happens to those people? And as we'll see,

overall, we're doing pretty well. The China shock produced winners and losers like any other shock, any other change, anything else that shifts.

- 5.89** If another state, for example, there was a lot of people lost their jobs in the auto industry in Michigan. And if they were willing to move to South Carolina and leave the union, they could have got a job in the auto industry in South Carolina. So the shock was a South Carolina shock. Suddenly, South Carolina was attracting auto manufacturers. Some jobs are destroyed. By some estimates, for every manufacturing job lost to Chinese input—this is an astounding number—for every manufacturing job lost to Chinese input, Americans gained something around \$411,000, net positive.
- 5.90** We'll talk about this in the next class. The whole question of a Chinese shock, the whole question of losing manufacturing jobs, it just, if you look at the numbers, it just doesn't add up. It does not add up. It's a huge benefit to us as Americans, as individuals, as people who are willing to work. Anybody willing to work is better off for China, not worse off. Now, we can put aside the national security issue. We'll talk about that. That's a separate issue.
- 5.91** Economically, China is a massive boon for America. We are so much richer, so much better off. Our stuff is so much cheaper, so we have more money to buy other stuff or to go on vacations or go to restaurants or do whatever we want to do with our money, because the Chinese are willing to produce stuff for cheap and willing to subsidize. Remember we did the subsidy example in the first class, second class? They're willing to subsidize some of the industries. We benefit from their subsidies. We get stuff cheaper.
- 5.92** And people say, yeah, I see. Again, so you get T-shirts, it's cheaper. Big deal. Oh, it is a big deal. The more stuff I get cheaper, the more money I have to get to live, to buy other stuff that I want. And by the way, to invest in my comparative advantage. What is America's comparative advantage? It's this. Even though this is not built in America, I don't know where this is built, probably China. But who makes this possible? Apple, in America. Designers, programmers, engineers in America make this possible. They add most of their value to this and to the iPhone and to everything else, even though it might be assembled, the least value added in China.
- 5.93** Anyway, the consequence of all this is a massive increase in trade. Trade is a percentage of world gross domestic product, world GDP. More than doubled, from 25% in 1970 to 57% in 2021. Now, not all of this is a consequence of just lowering tariffs. There's something else that drives globalization beyond the policies of our governments. It drove globalization in the 19th century, and it's driven globalization in the 20th century. And that is technology.
- 5.94** Technology makes globalization dramatically cheaper. And you know what the most important technological innovation of the late second half of the 20th century, when it comes to trade, was? The most important technological innovation of the second half

of the 20th century was the container, the shipping container, the 20 by 40 box. It reduced costs dramatically. It made the United States a superpower and a titan of trade.

- 5.95** It was invented by an American owner of a trucking company, who could see all these boxes loaded up on a truck, and then they would be unloaded at the ports, and then they were loaded onto a ship, and it was never quite fit, never quite—and he knew something was wrong here, and there had to be some better way of doing it. And then in the mid-1950s, he came up with this, which had been used in the past, but never systematically, of a container. He couldn't get people to embrace this. He bought a shipping company, and he built a container ship.
- 5.96** Then he basically started loading everything up into containers, standardized size. They just stack on top of each other, secure them in place, and off you go. You go, you bring the truck into the port, you unload it, you stack it, store it, then you load it onto the ship. It reduces time, costs, effort, dramatically. Indeed, nothing is shipped today on the oceans if it's not containerized, I mean, other than oil and natural gas. But if you're talking about goods, it's all container ships.
- 5.97** So in 1956, in the port of Houston, he had a boat that carried about 58 20-foot containers. And he basically showed that this could be done. Ultimately, they built special cranes for it. Today, you look at a port, all the cranes are there, special cranes, just to lift containers. In old days, you guys probably have never seen an old port where they have all these different size cranes, all these different size cranes that do it differently. Those are all gone, they're all replaced by a standardized container. It reduces loading, shipping, unloading costs dramatically.
- 5.98** And in 1956, to hand-load loose cargo onto a ship in a U.S. port costs \$5.86 per ton. That was in money back then, right? Today, in that money back then, the same thing cost 16 cents, 16 cents a ton. In today's dollars, it cost \$65, \$65.50 dollars, to load a ton onto a ship. Today, it costs \$1.79 in today's dollars. That's a revolution, and it makes trade so cheap. Shipping is irrelevant today for stuff. I mean, it's almost free when you do it in quantity. And this is why there's so much trade going on in the world.
- 5.99** So now you have an almost zero cost opportunity to engage in comparative advantage. Whereas before, yeah, a distant country might have had a comparative advantage versus us. But if you take into account the shipping cost to get here, and the time, forget it. It's not worth it. You're paying too much on the shipping costs. Now, with shipping costs driven almost to zero, everybody can do what they have a comparative advantage in doing, and everybody can trade, and everybody becomes wealthier as a result.
- 5.100** As a consequence, ships have become much bigger. So ships now can carry, I think the largest ship can carry 24,000 containers. These things are massive. And very little, very few crew—like some of these large container ships have 13 crew members. Just to give

you another perspective, in 1890, it cost \$200 per ton. So a \$2 t-shirt per ton, it would cost, I don't know, \$200 to ship it for a ton of T-shirts. Today, every t-shirt costs less than two and a half cents. The shipping price in the t-shirt that you buy, two and a half cents is the shipping cost. It's nothing. In a Nike shoe, it's 35 cents per pair is the shipping cost.

- 5.101** So you see it's irrelevant. Now all that matters is comparative advantage. And the next step is just in front of us, hopefully. The next step in this is autonomous ships with no crew. Self-driving ships, Waymo for the high seas, a lot safer. So people are already experimenting with autonomous ships. At this point, they're still too expensive, but we're heading there. We're going to get there. It's just a question of time, AI driving the ships over the sea. Cranes with no people in our ports, loading them onto trucks, the self-driving trucks that are taking them where they need to be.
- 5.102** Now, all of that said, what is going to stop that? Politics. I mean, there was this big, recently, there was a big negotiation between port owners, shippers, and the unions. And one of the demands of the union was, we don't want mechanization. We don't want robots. We don't want autonomous cranes because otherwise we'll lose our jobs. We want to keep digging with shovels. We don't want any of those big heavy machinery stuff. Why shovels? Teaspoons. You create a lot of jobs if you do all your digging with a teaspoon.
- 5.103** And there was a strike, and ultimately President Trump came in and basically sided with the union, and forced the shippers and everybody else to sign an agreement with the union that basically says we're going to slow down automation dramatically into the future in our ports. While in China, their ports are becoming automatized, very few people, it's all going to automation, our ports are still going to have union employees lugging things, operating cranes, and so on. We are stuck in the past.
- 5.104** By the way, just a few other technologies that have made trade so much more efficient. Air freight is a lot cheaper today than ever before. You got big planes. They're incredibly efficient. Revolution started by FedEx and then DHL and UPS and everybody else. There's real competition over this. A lot of the goods we get from Amazon, stuff like that, all delivered by airplane. They're not by ship, and internationally as well. iPhones, for example, when iPhones are assembled in China or in Vietnam or in India, they're loaded up to a Boeing 777, and they are shipped to the United States by plane.
- 5.105** Again, that would have been unimaginable even 50 years ago. So the technology, the costs have shrunk dramatically to the point where we can ship a lot of stuff, ship a lot of stuff by airplane. Of course, the improvement of air freight and the cheapness of air freight because of competition has also driven something like tourism. We move around. That's also part of globalization, the fact that we can travel anywhere in the world at a pretty reasonable cost. You can get a flight to Europe for 300 bucks if you're willing to sit in coach. I mean, airplane tickets are unbelievably cheap.

- 5.106** And that's all competition, and that was all a consequence of doing away with regulations that you just set. Airplane routes and airplane prices. And that was true until the 1970s. Only in the late 1970s was the airline industry deregulated. And as part of that, FedEx was allowed for the first time to use big planes. FedEx used to deliver everything it delivered in prop planes because regulations didn't allow them to use jet planes. And it's only after the airlines were deregulated, FedEx was deregulated as well, and air freight was made possible, and has become now international.
- 5.107** You can travel in Europe for pennies. For a few dollars, you can get on a flight from country to country. It's pretty amazing how cheap flying is, flying is today, if you want it to be cheap. And of course, a lot of what makes all of this possible is the revolution of telecommunication. We can coordinate, we can order, we can sell, we can buy instantaneously, we can transfer money instantaneously. If there's a problem, we can call people up. Think about in the old days how to use telegraphs. Maybe then we had to use phones. Now we can use the Internet to communicate instantaneously, 24/7, all over the world at a marginal cost of zero.
- 5.108** I remember when I moved to the United States in the late 1980s, I never called my parents. Not because I don't like them, but because long distance phone calls were, like, it was, nobody could afford it. I couldn't afford it. I was a student here, had nothing. My parents really couldn't afford it. They lived in Israel, which was not a very wealthy country. We'd call once in a while; every three months, we'd do a phone call. And it was by the minute. They charged by the minute, long distance phone calls.
- 5.109** Today, for my iPhone, which is not on me, I can make the call anytime, any place, marginal cost of zero. It doesn't cost me anything to call anywhere in the world. I mean, that is an incredible, not only improvement in our quality of life, because I can video conference with them, as well as just call them. A huge improvement in quality of life, but it's also a form of globalization. I am now better connected to the rest of the world, interacting with people all over the world, engaged, doing business with them, or just vacationing with them, or just chatting with them in ways that I couldn't even imagine. Just 40 years ago, couldn't imagine.
- 5.110** So, globalization technology has made globalization so much more possible today. And think about again, people coming to America in the 19th century, never seeing a photo of America, never seeing a video of America, maybe reading stories about it, and knowing they would never see their relatives again because they crossed the ocean, not going to cross back. And today, before you go anywhere, you can get videos, you can chat with people locally, you can get restaurant recommendations. It's not a mystery.
- 5.111** Even if you go to, I don't know, Ulaanbaatar in Mongolia, you can pretty much know exactly where you're going to go, and you can use the Internet over there, because I have, and communicate back home, and everything's fine, and you can get on a plane

and leave anytime you want. Anywhere, anywhere in the globe, pretty much. So we live in a pretty magical time in terms of the benefits globalization has provided us.

5.112 We're done with the history, at least up until this last administration. We're going to talk a little bit about that as well in a future class. But next class, we're going to take a step back and say, OK, wait a minute, Yaron. All this good stuff you're telling me about globalization, but people have objections. So we're going to take the objections. We're going to look at them. All the objections that people have made about globalization, particularly about trade, all the supposedly harmful effects of trade, and we're going to see whether they stand up to reality. Thank you.

Lecture 6: Made in America

At a glance — 'Made in America'.

- 6.1 We have talked in the last couple of classes about the history of trade and its progression from really ancient times until today, and how trade generally has elevated human well-being, wealth, and the culture and civilization throughout the world. But how it's gone in ebbs and flows, in a sense of our attitude towards trade. Often periods of time where we seem to resent it and hate it and want to stop it, and other times where we're more open to it.
- 6.2 And of course, we're in a period right now where there is real hostility or real suspicion of trade, certainly free trade in the United States, more so, I think, than the rest of the world, but certainly in the United States. And there are a number of arguments that are being made by opponents of trade that I think are worth taking one by one and discussing and seeing if they stand up to scrutiny.
- 6.3 One of the things to realize is the perspective that they take. Basically, free trade means free markets. We have a relatively free market in the United States. A lot of regulations, a lot of controls, but relatively—and we tend to generally respect market answers to issues, market solutions, and respect what the market determines as efficient and good for all of us. All free trade is, is extending that to our borders, extending that beyond our borders, acknowledging that markets function pretty well for Americans as individuals.
- 6.4 Well, one perspective that a lot of the critics of trade take is a different perspective. This is Marco Rubio, for example. He says, "What the market determines is most efficient may not be best for Americans." So this is sort of a skepticism about efficacy of markets, about where the markets work. Now, we've always had this, particularly on the left. The left has always been suspicious of markets and wanting to control them and wanting to regulate them.
- 6.5 Or if you take the current candidate for the mayor of New York seriously, actually, the government should own the means of production. That's what socialism demands. That's what socialism wants. We don't trust the market to deliver the goods. We don't trust the market to provide us with what is right. We'll do it. We know what's best for America. And of course, whenever you hear somebody say, "What the market determines is most efficient may not be best for America," you've got to ask a really good question, which is, who gets to decide what's good for America?
- 6.6 And by the way, who is this America anyway? America is a label for a political entity. But is every single person in this room aligned in terms of what's good for each one of us? Should we be determining what's good for the other person? And how do we aggregate the goodness in this room? How do we do what's really good for the most

people in this particular room? How do we measure that? Who gets to decide it? Who gets to determine it?

6.7 I know what's good for me, I think. Do you know what's good for me? Am I going to just delegate you to decide what's good for me? But that's what Marco Rubio wants us, or to delegate to somebody, what's good for America—and what is America? Me, you, and every one of us, as individuals. One of the beauties of the market is, I get to decide what's good for me, and I go into the market and try to manifest that. I might succeed, I might fail, but I get to do it.

6.8 And nobody can stop me as long as I'm not punching in the face and I'm not committing fraud on you. The beauty of a marketplace is, we each get to express what we think is good for us. Some people like steak, some people like sushi, some people like Mexican food. Hey, eat where you want to eat. And if nobody likes sushi anymore, sushi will go out of business. But isn't that what's good for America? To have that variety and that choice and the ability to exercise that choice the way we want to?

6.9 I mean, one of the manifestations of this idea of what's good for America, they get to decide, was Trump's comment a few months ago about, well, "Maybe this girl won't have 20 dolls. Maybe she'll manage with one." I don't know if you remember this, but in regard to people complaining about toys being, tariffs restricting the importation of toys. And it was like, you guys have too many dolls already. Who gets to decide that? Shouldn't we each decide how many dolls or how many toys we want to play with?

6.10 Why is it politicians are going to get to decide? Because it's good for America? Well, who is channeling America? I don't think that's what we elect politicians to do; at least, it's not what we should be electing politicians to do. It's certainly not what our founding documents suggest that we should be electing politicians to do, to channel what is good for all of us. But they've decided, for example, so there's now this attitude that what trade policy should be doing is not promote what consumers want, which would be low tariffs and cheap goods, supposedly. But instead, trade policy should be geared towards maximizing jobs.

6.11 Oren Cass, who is one of the leading thinkers around this new attitude, contends that an economy should emphasize "a healthy labor market, rather than merely rising consumption." Or Robert Lighthizer, the former trade representative of the United States, says, "Americans are producers first, consumers second." Yeah, right. But how do you maximize consumption? Let's say what I really want to do is make it possible for everybody to consume a lot. What do I do? How do you get to be a consumer? Where does consumption come from?

6.12 You have to produce. You have to go to work to make money. So consumption is a consequence of production. Every producer is a consumer. It's why we produce. Why do we go to work? Well, I mean, yes, we enjoy it and we get some self-esteem from it, but ultimately, so we can get money to buy goods to consume. Try living without

consuming. Now, consumption is destruction. We're destroying stuff as we consume it. There's an ice cream there. We consume it. There's no ice cream. We've destroyed the ice cream.

- 6.13** But it's destruction for the sake of life. It's the only way in which we live. We live by consumption. And in order to consume, we have to produce. So there's a sense in which there's no such thing as consumers. We're just producers. We produce by working. We produce the stuff that we ultimately consume. But consumption and production are this flip side of the same coin. You want to maximize consumption? Maximize production.
- 6.14** So it's true, Americans are producers first, consumers second. We're still consumers. And the only reason to produce, ultimately, what's the ultimate purpose of production? Consumption. The only reason to produce is to consume. We produce food so we can eat it. We produce computers so we can use them, consume them. We produce cars so we can drive them. We produce homes so we can live in them.
- 6.15** The point of production is consumption. Consumption is living. Producing is living, too. They're interwound. They're not in conflict with one another, is the point. It's not like America is obsessed by consumption and doesn't produce. To the extent that we consume, it means that we're producing. We can't buy Chinese goods unless we make money to be able to afford to buy Chinese goods. Just like you can't go to the grocery store unless you produce something and get money to consume in the grocery store.
- 6.16** Now, you might be on welfare and get money that way. But somebody had to produce and then get taxed. Before every single act of consumption, there has to be an act of production—actually, two acts of production. One is the money to make it possible for you to consume. And second, the something that you are consuming, somebody had to produce. Yes, the driver of economics is production. That doesn't mean that we need central planners to dictate who should produce what, where, and when. That's the beauty of the market. It does a phenomenal good job at that.
- 6.17** And if somebody in Vietnam is better than us at producing a particular good, why would we stop them from doing that? If they're more efficient, if they have a comparative advantage, then that frees us up to do something else. I don't know how many Americans actually want to go back to producing textiles in the United States and working in a textile mill. My guess is almost nobody. We have much better jobs now, because you can be a social influencer on TikTok, make a lot of money. Beats sitting around on a weaving machine all day.
- 6.18** And it's also worth noting that people's perspective of what production is is very narrow. Working in an assembly plant, that's production. But being a waiter in a restaurant, that's production. That's a productive job. You're satisfying the needs of customers who come in. You're making their lives better in some way. Service jobs are

just as much production as manufacturing jobs. Indeed, most of the most lucrative jobs in America today are service jobs.

- 6.19 Programming, service job. Chip design, like at NVIDIA. NVIDIA now is worth \$4 trillion, the most valued company in the world. What do they do? They don't produce anything. TSMC produces their chips. They have the comparative advantage. All NVIDIA does is design. That's a service job. Intel, service company, half service, half production. They also produce some chips. Apple, Apple, service company. They don't assemble these things.
- 6.20 And yet, service is bad because we think flipping burgers at McDonald's. Production, that's good, because we think those great union jobs in the 1950s. Really? Nobody wants to go back to those great union jobs in the 1950s. And as we'll see, they weren't that well remunerated. So one of the big issues with trade, we are told, is deindustrialization. America has lost industrial jobs. America doesn't produce, manufacture anything anymore. Maybe services, but nothing. We don't manufacture stuff anymore.
- 6.21 We're told this, over and over and over again. Manufacturing is way down. Indeed, it's true. Manufacturing employment peaked in 1979, I think, and is down quite a bit since then. In 1975, 22% of working Americans worked in manufacturing, 22%. In 2023, sorry, 2025, 8%. That sounds horrible. We went from 22% of the workforce doing manufacturing to 8%. But 300 years ago, pretty much 95% of all people who worked, worked in agriculture, including in America. How many people today work in agriculture? 1%. And how much food do we make, more or less? A lot more. Same is true of manufacturing.
- 6.22 There's a graph of employment over the last 100-plus years. Agriculture has plummeted. So has manufacturing. Why? Why has employment in manufacturing gone down? Well, because we're more productive, we're more efficient. Same reason as agriculture. I just read a story about, in agriculture, they've got now these tractors, I don't know what they call them, machines. They drive themselves. No human being is necessary.
- 6.23 As they go through the field, they're sampling the soil and deciding whether they add fertilizer or not. They're doing all this stuff. They're planting seeds, all this stuff, pulling out weeds where they need to, automatically, all based on sensors and AI. So nobody works in agriculture, and yet we'll get more food. Same thing is happening in manufacturing. It used to be that you used to stand all day and do one thing. This was the assembly line, a Ford assembly line. You specialize in this one movement, and that's all you did. Talk about boring and exhausting. And today, a robot does that.
- 6.24 And yet, there are still people working in manufacturing. We had a lot less because we can automate simple tasks. We can automate pretty advanced tasks. And by the way, that line will continue dropping. The number of people working in manufacturing as a

percentage of the total will continue to decline. And it's declining, not just in America. If you look at the percentage people working in manufacturing in Europe, declining. In the UK, declining. In China, over the last 10 years, declining.

6.25 China has these factories. They're called lights-out factories because they're factories where nobody ever needs to turn the light on because it's 100% automated. Robots do everything. People are leaving manufacturing and they're going into those damn services, service jobs. Same thing happening everywhere in the world. And indeed, a lot of manufacturing, simple manufacturing, automated stuff China can do, the simple stuff goes to Vietnam.

6.26 And in 20 years, it'll go from Vietnam to Africa, because nobody actually wants to do the really dumb, painful, physical, boring manufacturing stuff, unless you're so poor that you'll do it because it's a way up the ladder. It's a way to get a little bit richer. Once a society gets rich, it stops doing the stuff that's not, I'd say, fun. Fun is not exactly the right word, but it's not enjoyable. So agriculture sucks. You have to be in the sun. You have to, all of that. No good. You get out of there, you go to the city.

6.27 Cities are much more fun, much more enjoyable, much more interesting, many more people, much more culture, much more stuff to do. You go to the city, you do manufacturing. That's boring and hard work and tough on the muscles. And then you leave that and you, ultimately, your kids are programming on a desk in an open environment where they can chat at the water cooler. And even that programming is now soon going to be done by AI, and the kids will move on to the next best thing that they do.

6.28 There's an interesting fact. A greater percentage of the population in Western countries do entertainment than ever in human history. There are more people in music and film and comedy and just entertainment than ever before, which is a blast. They must have so much fun. I know my son is in comedy, so I know it's a blast. They love it. It's work, not really. They're enjoying it. And why are so many people doing that?

6.29 Because the rest of us are so rich that we can afford to spend a lot of our time watching stupid little TikTok videos or watching stand-up comedians or watching movies or listening to music and all this stuff. A hundred years ago, almost nobody listened to music. Nobody went to a play. Comedy was rare. Maybe at a pub somewhere, informally. But as a profession, 150 years ago, nobody's a comedian as a profession. Nobody would pay them. The richer we are, the more jobs we're going to have that are about entertaining people, about human connection.

6.30 That's not a bad thing. It's a good thing. And less jobs of twisting that bolt or going behind the oxen. And what do you call what you do in agriculture? Plowing the soil. Not fun. Much more fun to be a stand-up comedian. So, yeah, manufacturing jobs as a percentage, total jobs are declining. They will continue to decline. Happens everywhere in the world and will continue to happen everywhere in the world.

- 6.31** And the cause is greater productivity, mostly because of technology, mostly because of mechanization and automation and computerization or whatever you want to call it. At the same time, as productivity rises, what happens to wages? They go up. We'll see evidence of that in a little bit. But the correlation between productivity and wages is amazing. And we know the causal relationships. Here, correlation doesn't prove causation, but is, in the sense, caused.
- 6.32** The reason I can afford to pay you more is because you're producing more. If I don't pay you more when you produce more, what are you going to do? Leave. Go find somebody who will pay you, based on how productive you are. And if you're in a job and they're not paying you as much as you think you deserve, resign. Go find somebody who will. If nobody will pay you as much as you think you're worth, you're probably not worth that much. You probably have a bloated estimate of your own value.
- 6.33** Americans don't want to go back to those manufacturing jobs. Not yet. They don't want to go back to those manufacturing jobs. The fact is that there are 500,000 open manufacturing jobs right now, and nobody's filling them. There are plenty of people who could. There's a huge shortage of people, not jobs; jobs are plentiful. There's a shortage of people to fill jobs. Unemployment rate is very low, and people are not going. Manufacturing jobs today are not necessarily more stable than service jobs.
- 6.34** Again, service jobs are not hamburger-flipping jobs. They're some of the most advanced jobs that exist today. Manufacturing jobs actually tend to be more sensitive to recessions than service jobs. So service jobs tend to be more secure these days. They pay better. Service jobs, by far, pay better than manufacturing jobs right now. Again, think service jobs, think Silicon Valley. Don't think hamburger-flipping.
- 6.35** There's a whole generation of immigrants who came to this country and work in what is left of the garment industry in places like Seattle and Los Angeles. There's still these little factories where they make clothes. It's typically immigrants who work there. And what, of course, is amazing about these immigrants is, they work really, really hard, so they can make a little bit of money, so they can send their kids to college, so those kids don't have to work in a garment industry ever again in their lives.
- 6.36** And when these immigrants, old immigrants, die off, nobody is going to work in these garment industries. Not because there's no industry, but because nobody wants those jobs. Because their kids don't want them. They got an education because the mother's worked really, really hard. And now they're gone off to Silicon Valley or to Hollywood or somewhere else to do something a lot more fun than sit at a sewing machine all day. So our whole perspective on these jobs is distorted and perverted.
- 6.37** At the same time, as we saw, all these things go down, as a percentage of number of people in manufacturing go down. Our industrial capacity, the number of factories we have in the United States with capacity to produce, manufacture stuff, highest it's ever been in all of our history. We have more manufacturing capacity in the United States by

12%, higher than when China joined the WTO in 2001. We have 65% more industrial capacity than when we signed NAFTA in 1994. And we have 148% higher capacity than in 1975, which is the last year the United States ran a trade surplus.

- 6.38** So all these years of trade deficit, our manufacturing capacity has actually gone straight up, with fewer people. Industrial production, actually produce stuff, is about close to all-time high, maybe 1% less. COVID damaged industrial production. It's taken a while to get back to those levels of pre-COVID. It's still 18% higher than when China joined the WTO. So we produce, in terms of stuff, more stuff today than when China joined the WTO. So it hasn't caused production in the United States to disappear.
- 6.39** 55% more than NAFTA, when NAFTA was signed, 1994, and 153% more than in 1975 when we had a trade surplus. Where do we see a big fall in manufacturing? Apparel, leather goods, a lot of other industries that we don't have a comparative advantages, hard to automate. It's cheaper to just find somebody to work it really cheap, like in Vietnam or somewhere like that. So industries that, countries or— seemed to adopt at the beginning of a process of industrialization, are those the ones we lost, because we're at a very advanced level of industrialization.
- 6.40** America started off industrializing in textiles, competing with the British, who started off industrializing. What was the first use of a steam engine? Textiles. Textiles are always the beginning, for industrialization. It was for Britain, was for America, now it's for Vietnam. It was for China, now it's Vietnam, Bangladesh, other places. That's where they start.
- 6.41** It appears that we've lost a lot of automobile manufacturing. No, it turns out we manufacture more automobiles today in the United States than ever before. Indeed, we produce more automobiles and auto parts, 182% more, almost three times as many as we did in 1975. We just don't do it in Michigan. We do it in other states around the country. And it's not all American companies.
- 6.42** Toyota has plants in the United States. So does BMW and Mercedes Benz. All the Chinese companies do. We won't let—all the Japanese companies, I mean. We don't let the Chinese companies come in. We've stopped them from coming, but they would love to come and build automobile plants in the United States. They would kick Tesla's butt like that. BYD makes better cars, cheaper cars, than Tesla. We would all—electric cars shouldn't cost more than \$20-25,000 a car. And yet, Tesla's double that.
- 6.43** BYD could sell under those prices. We won't let them in—trade barriers. They're not allowed to sell electric cars in the United States. Why? Because we're protecting Tesla. And GM, who wants electric cars, and others, we're protecting them. Not because of anything to do with production. They'd come here, they'd open manufacturing plants in the United States, who would employ Americans. So it's not about jobs.
- 6.44** We have an amazing supply chain, or at least did until last few months. We have an amazing supply chain in the auto industry with Canada and Mexico. Just stunning.

Again, parts and stuff moving back and forth with Canada. A lot of our cars being produced in Mexico and coming across the border. But parts might be produced in the United States and moved to Mexico for assembly. Cars are better, cheaper, more efficient today than ever, because of this amazing supply chain that we have created, because of NAFTA.

- 6.45** Made possible, this, in a sense, one economic union, with zero tariffs on automobiles at least, between Canada, United States, and Mexico. Beautiful thing. Being destroyed like that. In 2021, the United States was the second largest manufacturer in the world, second largest after China. But just to give you a sense of how big that second place is, we produce more stuff. We produce more stuff than Japan, Germany, and South Korea combined. Japan is number three, Germany is number four, South Korea is number five. We, at number two, produce more than all three of them combined.
- 6.46** So this idea that manufacturing in the United States has died, it's just nonsensical. It's just not true. Indeed, if you think about value added, like how much a worker in the United States adds to the final product, we're number one in the world by far. American workers add more value to final products than any workers anywhere in the world. We're efficient. We're unbelievably efficient in what we do, and we produce high-end products.
- 6.47** We don't produce garments because it's not a high-end product. We don't add much value. Being in America doesn't add much value to your ability to sew a garment. But it does if you're producing fine machinery, you're producing automobiles, you're producing other goods. In terms of value added, we're at record levels in terms of the value added that U.S. manufacturers produce.
- 6.48** And foreign direct investment in the United States—that is the willingness of foreigners to bring their money here and invest in the United States—is huge. \$2.1 trillion in 2021, in manufacturing, foreign money coming in to American manufacturing to invest here. That's the stock. That's the total that they've invested. In terms of the one year, it's \$121.3 billion in 2021. But 2.1 trillion total has been invested in the manufacturing base in the United States. So they come here. Again, BMW, Mercedes, all manufacture cars in the United States.
- 6.49** So we have not declined in terms of manufacturing. We're certainly not growing. The amount of manufacturing is, seems to be fairly— it's up, but at very low rates over the last 10, 15 years. But to a large extent, that's because we can buy stuff much cheaper overseas. That's absolutely right. But it's not like we've disappeared. It's not like we don't manufacture anymore. You hear all the time people saying, "We don't make anything in America anymore." That's ridiculous. We make more stuff than we ever made.
- 6.50** And the stuff we really make, which people are not even measuring, is the design. And design, I mean, not just aesthetic design, which is also important, but the design, how

to build this, how to make a computer, we dominate that space. How to put it all together. How to maintain it. How to write software. The operating system is a big deal here.

- 6.51** I don't know where the programs might be in India. I don't know where the programs are, but it's planned, orchestrated, managed from the United States. The value added is added here, and those are all service jobs. We make stuff, we produce stuff, and a lot of that comes from the service industry, not from what technically is defined as manufacturing.
- 6.52** All right, so the next story is, OK, so manufacturing is still a big part of our American business. We haven't really declined as much as people think, or we haven't declined at all, really. But we've lost jobs. I mean, I told you the story about the China shock, 2 million jobs lost, and so on. We've lost jobs. The Chinese have taken all our jobs. Well, let's look at unemployment rate. Unemployment rate is in the orange. It's like under 4%. This was before this year, but you know, it's about the same.
- 6.53** Yeah, it spiked up in COVID, but that doesn't count. But generally, it's been around 4%, sometimes 6%. There was a recession in the early 2000s, it got up to 7%. But other than that, it's at 4%, which is pretty low. It's lower than what economists used to think the natural rate of unemployment was. If you had told people, economists, 20 years ago, that we would stabilize over 4% unemployment rate, they would be shocked. They would be surprised. It was not expected.
- 6.54** If you look at the blue line in the graph, that is people not in the workforce because unemployment is a tricky number. They measure people looking for work. If you're not looking for work, you don't count as unemployed, even though you're unemployed. So this captures that—share not in the workplace. It's insignificant. It's a tiny little fraction. And it hasn't gone up a lot. So where are all the jobs lost? Where did they go? People are working.
- 6.55** It just hasn't happened. Use employment to population ratio. How many people employed, relative to the total population? Is it close to all-time highs? If you consider that the largest generation right now is the baby boomers, and we're all retiring, I think I'm at the last year where it counts as baby boomer, or very close, and I'm just that short of official retirement in terms of age. So we're all retiring. So we're out of the workplace, and yet we're still at high employment ratio to population. That's amazing.
- 6.56** A lot of people are working. They have plenty of jobs. And as I said, there's about 10 million unfilled jobs in the United States right now. For all kinds of reasons, partially because there's a gap between the skills people develop and what are needed—one of the reasons immigration might be good. Part of it is, they're jobs people don't want to do. How many Americans want to go pick strawberries in the summer in southern California? Not a pleasant job to do. A lot of manufacturing jobs people don't want to do.

- 6.57** And part of it is, we're so rich, we really are. Americans don't appreciate, until you travel around the world, and Americans don't appreciate how rich they are, that we can afford to keep our children in the basement playing video games. Like, in olden times, they were out of there because we couldn't afford to feed them, and they had to go feed themselves. Now we can afford to feed them. They can stay there for decades. It's reality.
- 6.58** Private sector prime age employment. Prime age is what? 20s into your 50s is at all-time record highs. People are working. If you see people, if you see labor participation rates that are low, that's because so many people have retired. But at the prime age, it used to be, just a few years ago, there were a lot of kids in basements playing video games. A lot of those kids have come out of the basement and gone to work over the last two, three years. Things are actually pretty good. U.S. economy is doing quite well.
- 6.59** All right, you say, fine. People are working. But they're competing now with Chinese and Vietnamese and people all over the world. That must mean their wages have gone down. Wages are down. Everybody knows Americans make less money today than they did in the past. Is that true? So this is average hourly earnings of production and non-supervisory employees. And it's also median, so you don't want to just take average. Median means 50% make more, 50% make less. And it's all adjusted for inflation using the PCE as the deflator.
- 6.60** And what you get really is a graph that shows three periods. You see in the post-war area, wages went up dramatically, and they went up dramatically for obvious reasons in the United States, primarily because we were the only country that had manufacturing capacity, that had the ability to produce stuff, and we were producing it and then exporting it to the world. So a lot of it was trade-driven, for trade, export.
- 6.61** Now, there is a little bit of a mystery why wages stagnated. From 1974 until 1994, wages stagnated. It's not exactly clear why. I think there's some good reasons for that. I think that boom from 1945 to 1970, something was a little artificial, and you pay for it with less growth later on. It's artificial because we're the only country in the world out there producing stuff. Suddenly, we get competition. That puts downward pressure a little bit.
- 6.62** But also artificial because unions were very strong then. And unions push wages up, push wages up ultimately in an unhealthy way, to a point where ultimately you have to pay the price. And 1974 is not just a period where wages start stagnating. It's also a period where you get high inflation in the United States, you get high unemployment in the United States. And then the 1980s, a period in which American business restructured.
- 6.63** 1980s is where we took up our conglomerates; if you're interested in that, I've got a course on the Peterson Academy on corporations and business and the history of business in America. I think you'd really enjoy that. During this period, there's a lot of

restructuring. A lot of conglomerations are broken up. We become much more efficient and much more productive. But in the process, a lot of change has to happen, and a lot of adjustment has to happen. I think some of the stagnation here—

- 6.64 Remember, this is hourly wages of production and non-supervisory employees. These are not hourly wages of programmers, not hourly wages of people working in Hollywood, not hourly wages of people in the service industry, not hourly wages of, I don't know, celebrity chefs. I like celebrity chefs because, in the 1990s, how many celebrity chefs were there in the world? Like, in America. Like, zero? Nobody knew any celebrity chef. What the hell is that?
- 6.65 Like, maybe Wolfgang Puck or Emeril in New Orleans or maybe Keller in Napa. But nobody knew who they were. Nobody went to fancy restaurants. Michelin wasn't in the United States. Nobody cared about this stuff. Maybe in France, there were celebrity chefs. There were no celebrity chefs in America. Now, old-town Scottsdale has celebrity chefs. James Beard award winners. Michelin stars are going to come to Arizona at some point. You're going to have Michelin-starred restaurants.
- 6.66 And all these people are celebrity chefs. They're making a lot of money. That is a whole industry that didn't exist before. And we have, I don't know, 100 times more restaurants now than we did 50 years ago? All of old Scottsdale is basically restaurants. I've been walking around, trying out the different restaurants. If you want recommendations, you can ask me. It's a whole industry that didn't exist and now exists. Not just restaurants, which is new relative to 100 years ago, but now super fancy, cool, interesting, weird, ethnic, all kinds of restaurants. Those didn't exist.
- 6.67 Now, most of you are too young to remember them not existing. When I was growing up, we never went out to eat. McDonald's was special. We just didn't go out to eat. Chinese food, something like that, was super exotic and weird, and you didn't even think about it. And today, I know how often I go out to eat. It's not a measure, really. But people are out to eat all the time. Restaurants are packed. And fancy restaurants, middle-of-the-road cheap restaurants, all of them are packed. And that's an industry that didn't exist.
- 6.68 None of that is captured here. This is just non-supervisory production employees. And the reality is, since 1994, their wages have gone nothing but up. If you include chefs, it would go up even faster. If you include software, it would go up much faster. And what you can see is that the restructuring of American business during the 1980s starts paying off in 1994. Productivity goes up, wages go up, dramatically improved.
- 6.69 And the fact is, NAFTA is 1994. NAFTA does not reduce wages. I don't know that you can say that NAFTA causes the wages to go up. That would be inferring something that's not necessarily there, and we certainly don't have enough information here to say that. But it doesn't make them go down. China joined the WTO in 2001. It doesn't make wages go down. And look, this is a tough period. 2001, we're in a recession, and

we have 9/11. And yeah, you see a little bit of flattening of the curve there because it's a tough time.

- 6.70** And then we have the—you remember the great recession? This was the end of the world, the financial crisis. And yeah, wages go down during that period. And then they start ticking up again. And we're right now at all-time records. Does anybody feel like we're the richest we've ever been? Nobody. Because everybody in the press, in our politics, everywhere, keeps telling us how poor we are and how miserable we are, and our wages are down, and we've lost manufacturing jobs, and we're all unemployed.
- 6.71** And generally, Americans, when you ask Americans about their satisfaction, they say, "Oh, we love our lives. I love my life. But the situation in the country is not good. Other people's lives are not good. My life is great." And you ask this, and this is a general phenomenon, right? Everybody thinks their life is good, but other people's lives are not, because that's what they hear from the media and the politicians. And everywhere, all we hear is doom and gloom and bad stuff.
- 6.72** And yet, now, in my world, if I was running things— nobody will let me—but if I were running things, you see that graph from the '45 to 1970? It would be unstoppable. It would be steeper, and we would be three times richer today than we were back then. Because I know how to make that graph really steep. But OK, I don't get my perfect world. I still can recognize that it's pretty damn good. So, no, there's no basis.
- 6.73** And by the way, if you think this is biased, then I've got a bunch of graphs. Median wages for men and women after adjusting for inflation. So you see that a lot of the benefit went to women. Women's wages are way up, and they don't have stagnation. It's like, you could argue that men did really badly in the 1970s and early 1990s. But women, it's been a steady rise, as women have entered the workforce, as women really dominate today in education. They have done extraordinarily well.
- 6.74** They're still going to professions that pay less than men. That's why you've got a wage gap, because women do, generally, jobs that pay less than the jobs men do, for a variety of reasons. But this is all good news. Even men's wages are way up. Again, could be better. The one way I can tell you it won't be better is by restricting trade. That won't make wages better, make wages worse.
- 6.75** You want wages better? There's one big way to do that, and that's to deregulate and shrink the world of government, shrink the size of government—the opposite of the Big, Beautiful Bill. This is real median household income in the United States, another measure of income. Here again, you see from 1984, you don't even see as much of the stagnation here in the '80s and '90s. Here, you see up and then a recession, so down and then up from there. Again, the decline in the late 2000s is the great financial crisis, but then we're at record highs. Again, COVID creates a bit of a problem there in 2020.
- 6.76** This is another one, real disposable personal income. This is the one that really matters, actually. This is more important than all the other ones. Because this is how much is left

over. How much can I spend? How much do I have to spend on stuff that I want? That's just straight up, it's just amazing. Now, why does it spike up in COVID? Because the government sent us checks. Remember those checks you got in the mail? That's the spike.

- 6.77** But other than that, and by this measure, we're the richest people in the world, much richer than Europeans. I mean, a lot richer than Europeans, a lot richer than pretty much anybody, except, I don't know, some people in Dubai or something, where they have a lot of oil. But again, life is pretty good for Americans in spite of everything else. If we look at real consumption, that is consumption, how much we spend on stuff adjusted for inflation, we're richer today than any time in history.
- 6.78** Yeah, the middle class is shrinking. It's shrinking because they're getting rich. I mean, this is, I think, one of the most astounding graphs I've ever seen, and nobody talks about it. I mean, a few people do; that's how I got the graph. OK, so the dark, the, what is it? Brown or black on the top there—that's the number of people making \$100,000 or more. Now, this is 2018 dollars, so this is all adjusted for inflation. There's no cheating going on here. And this doesn't bring it up all the way to 2025. It's just as good when you look at 2025, this is as of 2018.
- 6.79** In the 1960s, 9% of Americans made over \$100,000. Today, 30% make over \$100,000, in real terms, no cheating. The middle class has gone from 53.8% of Americans to 41.7. We've lost 12% of the middle class has gone. Yeah, that's because it all became rich. Because the reality is, poor people have shrunk as well. Low income, that is people making less than \$35,000 a year, has shrunk from 37% to 28%. Again, this is amazing.
- 6.80** I mean, I'd love to see low income disappear completely, be zero. Nobody should make 35,000 or less. I know how to do that, but still better news than what you're hearing on the TV and from economists and from our politicians. And look how rich people are. 30% of Americans make \$100,000 or more, households. Now, people will tell you some of that is because two people are now working, instead of one person working. That's all true.
- 6.81** It's also true that we live in much bigger homes today than we did in the 1950s. We drive bigger cars. We have much nicer televisions. We have iPhones, we have Internet. We have stuff that people in the 1950s couldn't even imagine. They saw it in the Flintstones, but they couldn't imagine they would live it, and yet they're living it. I don't know where all—and again, you could put in NAFTA in there, and you could put in the China shock in there. It doesn't seem to make any difference at all.
- 6.82** And there are various ways to look at this. You can also look at this. I don't know if you can see this, but this is a graph that shows how many Americans have things like refrigerators and washers and dryers and air conditioning. Air conditioning is a big one. In 1960, very few people had air conditioning. Today, almost over 90% of Americans

have air conditioning. In terms of quality of life, standard of living, that's huge. It makes Phoenix possible. This area wouldn't be possible.

- 6.83** Somebody was living here in the 1960s, god help them. I don't get it. But today, you can have many million people. Microwaves, almost 100%. And look at Internet, smartphone. No technology in the history of man has been adopted as fast as mobile phones, Internet, computers, smartphones. I mean, look at that, straight line up. It didn't exist, not just in my lifetime, recently in my lifetime. I remember resisting mobile phones, not wanting one. It didn't last very long.
- 6.84** Color TVs, look at that. Refrigeration. Everybody has refrigerators. And of course, washing and drying. Just think about the time-saver and dishwashers. The time-saving involved in that is huge. It's one of the reasons we can have two-income families. Nobody has to stay home to do their laundry and to do all this that takes hours and hours and hours. I remember washing by hand. Just one year. I was very poor and no washing machine.
- 6.85** All right. In terms of net worth, the net worth, net worth is—so some people will say, "Yeah, OK, we got all the stuff, but we're paying for it with debt. We're all mortgaged to the hilt. We're living on borrowed time." Hear that all the time. OK, net worth is a good measure of that, right, because net worth is assets minus liabilities. It's what you're actually worth. It takes into account the amount of debt we had.
- 6.86** Well, our net worth today is 78% higher than it was in 2001. It's 140% higher than it was in '94, 232% higher than it was in '75, all adjusted for inflation. The average bottom 50%, poorest 50% of Americans, have a net worth today that is 84% higher than it was in 1989. I remember 1989. It wasn't that long ago. It's grown almost, it's almost doubled, than it was in real terms. In other words, the standard of living, quality of life, is doubled.
- 6.87** I have to admit that it pains me a little bit to be so positive, partially because I want to make real changes. I'd love to see real changes in our economy, in our world. And it's hard to make real changes when people's lives are pretty good. It's why our politicians have to scare us to death to make changes. It's why everything is negative, because in order to control us more, they have to convince us that our life is not that good. It's how politics works.
- 6.88** People complain that, yeah, we're importing a lot of stuff, but the quality is inferior. Really? Have you bought one of these 85-inch thin televisions recently? And I have one, and I've had it for eight years, nine years, I don't remember, and never needed an adjustment. I don't remember a tube television. Last thing I owned, if it did, it was only 20 inches. I mean, this is 85. It's huge. It's amazing. You can buy them now for under a thousand bucks. Eighty-five inches. It's like almost being in a home, in a movie theater.
- 6.89** I mean, the clothes we wear, are they're less quality because they're made in Vietnam, than when they were made in the United States? Is that real? That's not real. We have

higher quality and more fashion in our clothing today than ever before, in all of our history. So no, products are not more inferior. And to the extent that they are, it's because we want them to be, because we replace them a lot, and we don't mind, and we're rich enough to be able to do that.

- 6.90** People say inequality has exploded because of globalization. It's the exact opposite. The gap between rich countries and poor countries has shrunk. That is a fact. Look at the numbers. Even Thomas Piketty, the godfather of inequality, admits that internationally inequality has shrunk, even though in the United States, inequality has increased. The fact that inequality has increased in the United States is dubious probably, but it's certainly true that inequality has shrunk globally. And the reason is, the poor have gotten richer.
- 6.91** So global poverty, extreme poverty. Extreme poverty is defined by the United Nations as \$2 a day or less. Fifty years ago, about 30% of the world population lived in extreme poverty. It's about 40 years ago. What percentage of the world population lives in extreme poverty today? Eight, 8%, and heading lower. And if trends continue, we will have nobody living in extreme poverty within a few decades.
- 6.92** Now, they're still poor. The extreme poverty is extreme. There's still poverty. But just take China. China has gone from a country where everybody was extremely poor, where the whole population was poor, a billion people were poor; and today, China has a middle class of something like 800 million people. It still has some poor people, but it's shrunk dramatically. That's good news. That's globalization.
- 6.93** China couldn't have done that without capital flowing in from the United States and Europe, expertise flowing in, and goods flowing out. So one of the things that we've done through the China shock is help 800 million Chinese get richer. It's a pretty cool thing. And we benefited, too, it turns out. Trade is win-win. You don't engage in trade unless both parties win.
- 6.94** I buy an iPhone for \$1,000 because the iPhone is worth how much to me? More than \$1,000. That's why I'm willing to give \$1,000 up. Apple won; they got a profit. I won; I got something that's worth more than \$1,000 to me. And that's every single trade you engage in. You buy bread at the grocery store. It's because the bread is worth more than the money you're giving up. Grocery store made money, and you made—standard of living went up. Quality of life went up.
- 6.95** Trade is this magic where everybody wins. It's win-win. Globalization has reduced global income and wealth inequality by helping raise people who are poor up, not by suppressing wages in the West. But the reality is, inequality is smaller. And if you look at consumption inequality, which I think is the most meaningful thing, consumption inequality is constantly shrinking. That is the difference between what the rich consume and what the poor consume. It's constantly shrinking with globalization.

- 6.96** Products are cheaper and cheaper. They're better and better. And because they're cheaper, poor people can consume them in ways that they couldn't before. Bill Gates goes to a restaurant, and so do poor people go to restaurants. Yeah, he goes to three-star Michelin restaurants, and they might go to a local ethnic, I don't know, Chinese kitchen or something, but they're all going out to eat. The differences are minor, as compared to the fact that they're going out to eat.
- 6.97** And that's true everywhere. You go to Shanghai and place like that, full of restaurants everywhere. People are going out to eat all the time. They didn't used to. That's a sign of wealth. You go to Europe, go anywhere. Restaurants everywhere. It's a sign of wealth. And everybody is participating, poor and rich. Not at the same level, but consumption equality—
- 6.98** I mean, take flying around the world. Fifty years ago, you either flew for business or you were fairly rich. Like, upper-middle-class people went to Europe. It was very hard to go to Europe otherwise. It was expensive. Cost a fortune. Today, you can fly to Europe for a few hundred bucks. I mean, it's, in coach, you can do it for less than that. It's really cheap.
- 6.99** Well, the first time I came to America, it was 1974 with my parents, and my parents couldn't afford a flight. So my dad got a job as a doctor on a merchant ship, like a container ship. And we came to America by ship, on a merchant ship. We were a family of five at the time. And that's how we came to America, which was great because I got to go by ship into New York harbor and see the Statue of Liberty and all that. It was very cool.
- 6.100** But yeah, I mean, flying was like, rich people did that. You didn't do that. My parents on their honeymoon went on a boat. You went on a boat. You didn't go on airplanes. That was rare and unusual. Today, does anybody think twice about getting on an airplane? No, it's just a question of whether you want to be in business and coach, but everybody flies. And yeah, the super rich get private planes. They don't have to share them with anybody. It's nice, but it's not a life-changing difference. We all fly, almost all of us.
- 6.101** People complain about globalization causing global poverty. But again, there are fewer poor people today than ever before, ever in all of human history. What percentage of the global population was extremely poor, \$2 a day or less, 300 years ago? Everybody, 95%-plus, few aristocrats up here, everybody dirt poor. So yes, too many people are dirt poor still. Agree. Africa, primarily. But we know how to solve that. And the fact is, 92% are not. And of those, many, many, many of them are middle class and above.
- 6.102** We're told that globalization destroys culture. It mixes and merges all these cultures. So if you go to, I don't know, you go to the Far East, there's McDonald's and Starbucks at every corner. And there's American-style music playing, even though it's in the local language, but it's got the same, like rappers, taking over the world. You really have to

be a snob to view that as a negative—people are adopting things because they like them. They like parts of Western culture, and indeed, we like parts of Eastern culture.

- 6.103** It's why we have great Asian restaurants all over here, and we do fusion food and all of that, and all this cultural appropriation BS. I mean, everybody's been appropriating everybody's culture since the beginning of mankind. Trade involves that, involves mergers of cultures, and merging of different values, and learning from other people, and improving your own culture by learning from others. There is no such thing as a pure culture, except for some isolated tribes in the middle of nowhere.
- 6.104** And it's arguable whether you can define them as having any culture, really. Culture involves achievement. So, yeah, some of the best musicians in the world in classical music today are Japanese and Chinese. I think that's a beautiful thing. I saw Rachmaninoff's second piano concerto in London with an orchestra, and the pianist was from Uzbekistan, and he was amazing. I mean, isn't that a beautiful thing? No, no, Uzbeks are not supposed to play Rachmaninoff, although Rachmaninoff is Russian, not Uzbek. I mean, it's meaningless.
- 6.105** So yeah, if it works right, and I think over time it does work right, then the best of each culture rises. And the garbage, we all have garbage, sinks. And yeah, one day we have a much more homogeneous global culture with all the good stuff. So, why we need uniformity is beyond me. Uniformity is boring. And why not take the best? I mean, the melting pot, which is what America used to be, and used to believe it was, is about embracing other cultures, taking the best within them, and acknowledging, asking them to acknowledge, that they leave the worst at home.
- 6.106** And America constantly improves by bringing in the best from all over the world, the best elements. That's part of the beauty of immigration, part of the beauty of trade. So culture is better for trade and globalization and exposure. My wife and I now love Korean dramas, Korean dramas on Netflix. There's some really good stuff on Netflix. And I love, my wife doesn't, but I like the dark Scandinavian murder mysteries. They're really dark. Those Scandinavians are not happy people, based on their TV and based on their novels. They're really dark. But I like it, for whatever reason.
- 6.107** And it's amazing. And they're in foreign languages I don't understand, but everything's subtitled, and I can enjoy them. I know K-pop is very popular in the United States. Is that bad? Maybe, because it doesn't sound like it's great music, but people enjoy it. Who are we to say, "No, it's bad. It's from Korea, so we shouldn't have it"? I'm sure some of the younger people here are big on anime, Japanese animation. Great. Why would anybody view that as a negative, because it comes from Japan?
- 6.108** So, yeah, I don't get the cultural argument. Now, it's true that if we only took the negative elements and dropped all the positives, that would be bad. But that's not what human beings are about. I have a more positive view of humanity than that. Look, at the end of the day, all the objections to trade boil down to the idea that the world—a

mercantilist idea— that the world is a zero-sum. That my gain is your loss. If I'm happy, you have to be miserable. If I'm going to be rich, you need to be poor.

- 6.109** It's constantly expressed by political leaders. This idea that, we trade with Canada, they must be exploiting us because we're sending them dollars and they send us stuff. Never makes any sense. No, the world is win-win. As long as we're trading voluntarily, as long as there's no coercion, as long as there's no slavery, as long as there's no—violence and wars are always lose-lose. Slavery is lose-lose. Starts out with win-lose, but ultimately everybody loses. These lose-lose relationships are bad. Stay away from them.
- 6.110** But trade and globalization and being exposed to one another's ideas and one another's culture, that's all win-win. There's no losing here. If we don't like something, we walk away. Nobody's forcing anything upon us. So there is no argument against trade. There literally is no argument against trade, except for one. I'll get to it in a second. Trade is win-win. It's win-win-win. We do it among each other. And the fact that we grow that to a nation level doesn't change the nature of the win-win nature of trade.
- 6.111** The ideal trade policy is unilateral free trade, as Britain did in the late part of the 19th century. And hope that everybody else imitates you. Now, the one argument against free trade that has potential legitimacy is the national security argument. It's the argument relating to trade with somebody who is or potentially is an enemy of yours. Trade is win-win. Do you want your enemy to win? Next class, we'll start off by talking about that. Thank you.

Lecture 7: The Tariff Trap

At a glance — The tariff trap.

- 7.1 So last class, we talked about objections to globalization and to free trade, and we went through them, the deindustrializing America, and we lose jobs, and income goes down, and all these things. And we saw that there's really, those arguments are empty. There really is no evidence to suggest that any of those things are actually happening. And ultimately, trade, globalization, are all good. They are positives.
- 7.2 And the so-called costs that they involve are really not real. It's more about fearmongering than anything else. And it's all about politics. It's all about special interest groups advocating for a particular thing. If you're a steel company, you'd rather not have to compete with some of the really good steel companies out there in the world. So you lobby for tariffs on steel, and right now we have very, very high tariffs on steel. Steel companies, American steel companies have raised prices dramatically because they don't have competition.
- 7.3 So why not? Raises their profits, the unions support it, and you get real, you get real political buy-in, and we'll see some of the consequence of that when we talk about the consequence of some of these tariffs. And of course, aluminum and copper and all these things, they all have special interest groups fighting for this, just like the Jones Act, that's special interest groups fighting for those.
- 7.4 It doesn't make any economic sense. It doesn't make any real sense in terms of the well-being of Americans. But yeah, you can understand, particularly industries, fighting for those special favors. Right now, Intel is in big trouble in tech, in chip manufacturing. They're on the verge of potentially going bankrupt. They are lobbying heavily for potentially tariffs on TSMC, on chips coming in from Taiwan, because then that will create an incentive. That would create a possibility for them maybe to be able to resurrect the company.
- 7.5 So there are pressure groups, there are special interest groups, the little power points that press politicians, and they have a lot of influence, a lot more influence than we do, and a lot more influence than economists have. So economics get put to the side, and politics triumphs. If you remember our discussion of 19th-century America, same thing there. Tariffs were strongly advocated by certain industries in the Northeast. There were other pressure groups that were against it.
- 7.6 Farmers, for example. So it was constant back and forth, back and forth. There's almost no group today that's organized, that's really organized, around objecting to tariffs. And as a consequence, we're just rushing in to it, following the special interest groups and following a particular agenda that we'll get to in a minute.

- 7.7 I did say at the end of last class that there's one argument that you could make that is, I think, a legitimate argument, that needs to be really thought about in terms of trade policy, and that is the national security argument. The national security argument. There are a number of different dimensions to this argument. First, trade is win-win. Both parties win. Do we want our enemies to win? If we have clear-cut countries that pose a real threat to the United States and are enemies, we can define them clearly as enemies, do we want to trade with our enemies?
- 7.8 Do we want to benefit them? Because they're better off. Now, we are better off, too. So if we stop trading with them, we need to be honest with ourselves that we will be slightly worse off. But that might be worth it if we prevent them from achieving the riches, the prosperity, that makes them a bigger threat to us. Should we have continued to trade with Nazi Germany during the 1930s, before the war? We did. We did trade with them. We continued to have diplomatic relations with them.
- 7.9 Did that make them stronger and, as a consequence, make it more possible for them, ultimately, to launch a war? Probably. So there's thought that needs to be put into, well, how do you deal with an enemy? And of course, the other thinking that has to go into this is, what is an enemy? Who is the enemy? So it's easy ex post to look back at Nazi Germany and say, "Well, they're the enemy. They launched a war."
- 7.10 But a lot of very smart people at the time thought, "Yeah, Nazis are bad. They're horrible. Their ideology is horrible. They're going to do bad things to their own people. They're not stupid enough to go to war." Well, they turned out to be wrong. They turned out to be wrong. And they did go to war. For many, many years, we had very limited trade with the Soviet Union, most of the Cold War. Now, we still traded with them. Indeed, we often gave them foreign aid. We sent them food when their harvest dwindled, and they couldn't feed their own people. So we fed them, our enemy.
- 7.11 Communism was opposed to us. We were fighting proxy wars with them in various parts of the world, whether that made sense or not; for us, it's questionable. But clearly, the Soviet Union was an enemy, and they had ambitions. They said, "We believe in communism to be an international ideology. We want to spread it around the world. We want everybody to be a Communist." It wasn't an isolationist ideology.
- 7.12 You could argue today, Iran is an enemy. They've killed Americans, they export terrorism, and so on. North Korea threatens the United States with building nuclear weapons, as ballistic missiles that could reach United States; you could argue they are an enemy. But then it gets tricky. It gets tricky. Is Russia an enemy of the United States? It's an enemy of Europe, pretty clear. And that's why Europeans are not trading much with Russia these days. There's a lot of restrictions and limitations and embargoes.
- 7.13 Is China an enemy of the United States? Are we likely to go to war with China? I mean, that's a foreign policy question which we're not going to answer in this class, but it's the kind of thing we need to think about and really think about it. Not, "Yeah, we don't like

them. They're the enemy. Or they have nukes. They're the enemy. Or they spy on us, but we spy on them, and they hack us, but we probably hack them as well." Really thinking about what constitutes an enemy.

- 7.14** And once some country constitutes an enemy, how do you deal with it is not easy and needs to be really thought through. And if some countries are clear-cut, if somebody is identified as an enemy, like we've identified Iran, let's say, and we've identified North Korea, the solution to that is not to tariff them. It's not, "You can still sell your goods here. We're just going to tax our own people so they buy less of your stuff." The solution is to embargo them.
- 7.15** Just, you can't buy, I don't know, which I think is silly, but you can't buy Cuban cigars in America. It's illegal, right? Because we don't import stuff from Cuba, because we've embargoed them. Now, why we've embargoed Cuba? I mean, it's a horrible regime. They're horrible to their own people. But is Cuba a threat to the United States today? I mean, again, not to get into foreign policy, but I think that's a joke, right?
- 7.16** I mean, Cuba is no threat to the United States. It's not going to launch an attack on the United States. It's not going to invade the U.S. It doesn't have missiles to launch at Miami. It's not going to happen; and yet, we embargo them. So that has to be rethought again, foreign policy-wise. But notice that once we identify somebody, we embargo them. We don't have tariffs on Cuban cigars.
- 7.17** So I get national security threat, but then you've got to say, OK, no trading, none. Not on some things, and we'll tariff them, like Russia. The solution to Russia is, they've invaded Ukraine. They're a threat to NATO. They're an enemy of NATO. We're part of NATO. Again, put aside the question whether we should be or shouldn't be; we are. OK, no trading with Russia. None of these games that we play, which we do, around trade. No trading with Russia.
- 7.18** Tariffs are not an answer to national security. Embargoes might be. Bans might be. We might also say, look, because China, let's say, is a rising threat—it's not a threat yet, it's not an enemy yet, but they could become one—there's no reason, we shouldn't trade with them in a way that provides them with some of our most sophisticated technology that can be used in weapons systems. So the Biden administration blocked the sale of the most advanced chip technology to China.
- 7.19** Maybe that makes sense, probably makes sense. I get it. So you need to really, again, you need to really think through which technologies are most important, which technologies they could use for weapons systems against us. Is blocking them really effective? It's a tricky question, because whenever you block trade, what happens? You get black markets. So for example, a lot of companies in Singapore are buying a lot of very sophisticated chips. And you wonder, what are they doing with these very sophisticated chips in Singapore? They're not exactly known as this big industrial hub of sophisticated tech.

- 7.20 Well, what they're probably doing is, they're smuggling into China, or some of them are smuggling into China. But you might want to still have those restrictions because you might want to slow down the ability of the Chinese to gain this technology. There might be other technologies. Certainly, we're not going to sell China F-35s. So a certain trade we know we don't do. We don't sell our enemies our most advanced airplanes or any war airplanes or any military equipment.
- 7.21 Well, if a chip is essential for military equipment, maybe we shouldn't sell it to them. That's all reasonable and should be thought about. But again, tariffs are not the solution. A tax is not the solution. Just bans. No sell. If a Chinese equipment manufacturer, and this has happened, is discovered—Huawei, the Chinese telecommunication company, was building a lot of the telecommunication networks in Europe and was actually involved in a lot of telecommunication networks and equipment sales here in the United States.
- 7.22 And then people discovered that they were implanting into the telecommunication networks basically a technology that allowed the Chinese to spy on everything that was being said on these networks, the ability to hack into those networks, the ability to disrupt them. So, yeah, if a country that is somewhat hostile, maybe it's not quite an enemy yet, but it's hostile, is selling you equipment that clearly you've identified as potentially damaging, if you ever did get into a combat, then ban them. And the Trump administration, first term, banned equipment purchases from Huawei. I think they're called Huawei.
- 7.23 And a lot of Europeans now, countries have done the same because they've discovered their entire networks were compromised. All of their telecommunication was compromised by the Chinese. It also could be that you can find software solutions to get around all that. It could be that you could tinker with it and solve the problem. I don't know. But again, these are the kind of things that you need to think about, if you're taking national security seriously.
- 7.24 And then there's one other dimension to this. Let's say, if you ever go to war one day, you're going to have to suddenly start building a lot of, I don't know, missiles, maybe tanks. I don't know where you'd send the tanks to, but airplanes. You're going to have to—like, in World War II, American industry ramped up massive production of all these things. Now, war today is very different than it was in World War II. So you'd have to, again, strategically think, what do you really need in a war?
- 7.25 Drones, for example. I mean, we probably need drones. America does not produce anywhere near enough drones to be able to handle a modern war, where countries are using drone swarms. On the other hand, China can build millions and millions and millions of drones a year. I mean, they have industrial capacity in drone. They dominate the space. DJI alone, the company DJI, it builds millions of drones right now, and most of those drones can be easily made into weapons systems. The Ukrainians are doing it, taking DJI drones, putting a bomb on them, and sending them out. It's relatively easy.

- 7.26** So what do you do? So you need steel in order to build a lot of these things. So the idea here is, you need a robust American steel industry. So you tariff steel all over the world, so that the American steel industry becomes super profitable, makes a lot of money, invests in more steel production. So if there's a war one day, we'll have the steel available. This is a very expensive way to deal with a war that maybe will happen or may not happen someday in the future.
- 7.27** Basically means, everything we make from steel in the United States is going to cost more money. And indeed, the steel tariffs Trump passed in his first term on steel generated significant price increases on stuff made with steel. It makes American auto companies, for example, a lot less competitive, because they have to buy steel from American companies that is a lot more expensive than if they could buy them on the world market.
- 7.28** So the question is, is that the best way to deal with it versus—and I don't know what the answer is, but this is the kind of thinking that has to go into it. The Defense Department could come to the steel companies, American steel company, and say, "Look, we don't think you have enough capacity to deal with a war. So what we'd like to do is, we'll give you money to build additional capacity. Mothball those plans. We'll pay you to mothball them, so that if we have a war, we can get them rolling and get them going and build the steel."
- 7.29** That would be a lot cheaper than tariffing global steel and raising steel prices. And there might never be a war. Or the war might be different. Or the war might not need steel. We might need something else. Who knows? I would do the same thing today with drones. I would say, "Here's X billions of dollars. We, the Defense Department, are willing to buy X million numbers of drones, only if they're made in the U.S. or a friendly country. We don't care. You can make them in Mexico. That's fine. As long as it's easy to get to the U.S., friendly country, easy way to get here, we will guarantee that we will buy X number of drones at 20% above market."
- 7.30** And then let entrepreneurs go figure out where to build them, how to build them, and how to scale. Instead of the U.S. government going and building drones, or deciding who the winners and losers are, or setting up, having tariffs on drones to incentivize people to build drones here. And even with steel, the way to think about steel is not to say, how much capacity do we have in the U.S.?
- 7.31** The way to think about steel is, how much capacity do we have in friendly countries? Like, if there's a war, is Mexico going to be on our side or on China's side? Given the proximity and the difference in militaries, I think they're on our side. I don't think they're risky. So if the steel production happens in Mexico, we don't care. Or Canada. Is Canada on our side or on China's side? Well, up until a few months ago, clearly on our side; not sure today. Or Europe, or Brazil. Brazil is a huge steel manufacturer.

- 7.32 As long as the capacity is with our friends, and maybe we even have deals with them, in a war, will you provide us with all the steel we can consume? Yeah, why wouldn't they? It's another way to cement a good relationship with the world's superpower. So drones, chips. The Biden administration passed this law, CHIP Act; the Republicans support it. Basically, massive subsidies for people to build chip manufacturers in companies in the United States. The government gets to decide who gets a subsidy. They divvied up tens of billions of dollars to different companies to build chip manufacturers.
- 7.33 The Trump administration has just doubled down on that. The Big, Beautiful Bill has a bunch more, tens of billions of dollars, to hand out to chip companies to build in the United States. Why not do it the other way around? Like with the drones, say to companies, "Pentagon, we're willing to spend X number of billions of dollars on the most advanced chips in the world. We'll pay above market. We'll pay your premium above market if you build them in the United States."
- 7.34 Here, the friends showing is a little—making the chips in friendly countries is a little complicated because our friends who build chips are all in Asia. So you have to worry about—I mean, chips is the one thing you really need to worry about because TSMC is in Taiwan. If we ever get into a conflict with China, that's where the conflict is going to be. Samsung is in South Korea, much closer to China than to us. Very easy for the Chinese to block ships leaving South Korea to come to the United States. The South Korean President right now seems to be a little friendlier to China than he is to Trump.
- 7.35 So you never know. But if you say, in the Western hemisphere, if anybody's willing to build chips in the Western hemisphere to spec, best chips in the world, we'll pay above market for them. It's an easy way to, again, change the allocation of resources, but without dictating who the winners and losers are. Go compete. Whoever can produce the chips, we'll buy them from you, if they're really that good.
- 7.36 What you don't want to do is have the government pick winners and losers. They don't do a good job of it. They've never done a good job of it. So national security is a real issue, needs to be thought through. We need to really think through who our enemies are. I don't think Cuba is, for example. I would completely open-trade with Cuba, even though they're a bad regime and horrible and all of that. And I think I would never go to Cuba. I don't want to spend my money in Cuba that goes to the Castros. But trade, if you guys want to trade with Cuba, none of my business.
- 7.37 Same with Venezuela, horrible place, horrible regime. But they're not invading the United States. So why would I stop trading with them? Iran, North Korea, makes sense to embargo them completely. China, you have to really think it through. I'd love to have some inside intelligence from the intelligence agencies in terms of what they're really thinking and how probable a war with them ever is. But you'd have to think about it. Russia, again, same thing. You'd have to really think about it.

- 7.38** And then you'd have to come up with strategies, how to deal with it. But here's the thing about national security. Politicians use it as an excuse. 90% of the times, politicians say, "We're doing this for national security reasons." It is a ruse. They are pulling one over us. Like right now, the auto tariffs, 25% of auto tariffs globally, is a national security issue. What's national security? If we lose the ability to build automobiles in the United States, is that really hurting our national security? BS.
- 7.39** And there's a bunch of other ones where it's a national security issue that we have to—I mean, steel, aluminum, copper. These are all the tariffs on our national—not all the tariffs, specific tariffs, all national security tariffs. And then we restrict sales of all kinds of things. And 90% of the time, it's to benefit some industry. 90% of the time, it's because of lobbying and has nothing to do with national security.
- 7.40** But Congress, which has constitutionally—we'll get to this—the authority to impose tariffs, has delegated to the President the ability to impose tariffs for national security reasons. So he's going to use it to do whatever he wants. And no court is going to say, "No, we don't think that's national security." The courts, that's not their job. Their job is not to evaluate foreign policy. So a President can get away with anything and call it national security.
- 7.41** So I would be very cautious about believing them when they use this as a reason for tariffs in particular, but even for blocking trade, generally. And again, tariffs should never be the solution for national security. There are other ways, I've articulated a few here. Embargo, if it's a real enemy—you don't trade with a real enemy.
- 7.42** And here's the thing. Trade, generally, is a peaceful activity. Countries that trade a lot, typically—there are exceptions. Ukraine, Russia is a good example. They're exceptions to this. But countries that trade a lot don't usually go to war with one another. Embargos and trade restrictions often push countries into war. I'll give you an example. It's a controversial example. And it's not clear that we wouldn't have had a war anyway.
- 7.43** But if you take Japan in the early part of the 20th century, before the 1930s, United States and Japan had fairly normal trade relations, pretty open, especially in the 1910s and 1920s. U.S. exported—Japan's an island, has no natural resources really, has no oil—and the United States exported oil, steel, cotton to Japan and imported silk, toys, and consumer goods from them, early part of the century. And it was pretty seamless trade. And indeed, going into the 1930s, when Japan becomes aggressive and becomes militant, Japan is dependent on the United States for almost all of its raw materials.
- 7.44** By the way, one of the reasons Japan goes to war, in their minds at least, is to access raw materials elsewhere in Asia. They want access, they want direct. They don't want to be dependent on the United States. But even by the late 1930s, the war breaks out in Pearl Harbor. This happens in 1941. By the late 1930s, Japan is getting 80% of their oil from the U.S. Now, in 1931, Japan invades Manchuria, basically China. The U.S. is worried, but trade continues.

- 7.45 1937, the big China-Japan war happens, and Japan is invading the heart of China and doing horrible things. I mean, the massacres, the slaughter, and particularly the Nanjing massacre, really offensive to Americans, and America is really starting to worry, and as a consequence, start real economic pressure. From 1938, you get an embargo, where it's a moral embargo. So Roosevelt basically encourages Americans not to buy Japanese goods and not to sell to Japan.
- 7.46 So discouraging U.S. companies from selling airplane parts and aviation parts to Japan. 1939, the U.S. terminates the 1911 U.S.-Japanese treaty on commerce, but trade still continues. But that starts to change in 1940. So in 1940, have an export control act, where they start to restrict all military materials to Japan. U.S. begins restricting sale of aviation fuel, machine tools, scrap metal, steel. In September 1940, Japan signs its treaty with Germany, to join the German Axis in terms of Second World War.
- 7.47 Of course, America is not in a war yet, but later that year, the U.S. imposes a full embargo on scrap iron and steel export to Japan. After Japan occupies southern Indochina, U.S. freezes Japanese assets in the U.S., imposes a total oil embargo, crippling Japan, which is importing 90% of its oil, not all from the U.S. at this point, but still a big chunk of it from the U.S. So Japan is crippled now. It's getting no natural resources, none of the steel, none of the oil from the U.S.
- 7.48 1941, the U.S. and Japan are negotiating. U.S. is demanding they withdraw from China and Indochina. Japan wants the embargo lifted. The negotiations fail, and we know what happened. Japan attacks Pearl Harbor, and the Second World War, from the American perspective, breaks out. It's only that that drives America to then participate in Europe. So I don't feel—you could say, the trade cornered the Japanese into a no-win situation for them. They felt like they had to do something because they couldn't survive without the steel and the oil and so on.
- 7.49 Turned out to be a pretty stupid strategic move for them to attack the United States. They thought that if they attacked Pearl Harbor, it would end there; it wouldn't engulf the whole thing into a full-on war. The United States at the time was very reluctant to join the war. The American public was not favorable towards joining World War II, even in Europe. And they thought, OK, maybe we do this, maybe then we can improve our negotiating position. And yet, big strategic mistake, as it turned out.
- 7.50 So it could be that by isolating and by embargoing, you ultimately push a country into war. But it's also the case that sometimes you have to do it. You cannot feed your enemy. You cannot feed your enemy. So foreign policy is tricky, but beware of patriotic fervor, politicians riling us all up in the name of national security, and the remedy is tariffs. It's not a good remedy, and it's almost never actually the case.
- 7.51 All right. So let's now turn to talking about what's happening right now in terms of attitude towards trade, because there's no question, since 2016, but even before that, it was beginning. Attitudes of Americans have shifted towards trade as part of, and

many within the intellectual world have shifted about trade, since Trump got elected the first time. It was already, you could already feel it in the air before that.

- 7.52** And ultimately, there's a tradition of this within the Republican Party, going back to Pat Buchanan, who was very anti-trade back in the 1980s, '70s and '80s, '80s when he ran for President. Even Ross Perot talked negatively about trade deficits. And so there's a tradition in American populist politics around these ideas. I call the current trend neo-mercantilism. So what you're getting is a rebirth of a mercantilist mentality in America today.
- 7.53** And it is interesting. It's mostly in America. You don't really see this much outside of America. So, hostility towards trade deficits, so this manifests itself in hostility towards trade deficits. An effort to reduce imports through high tariff barriers. Protect domestic industries at all costs. Domestic industries need to be protected, nurtured, maybe even subsidized, like we're subsidizing chips, and we're subsidizing some other industries right now.
- 7.54** It even goes to pursuing foreign lands. I mean, think about Greenland. Suddenly, Greenland becomes important. We're going to make Greenland part of America. I mean, why? Where does that come from? Greenland has natural resources, but so do we. If we actually were willing to mine natural resources, Montana, Wyoming, probably parts of Arizona, Utah, tons of natural resources, rare earth materials, all the stuff that China, supposedly, is the only producer in the world of. We are plentiful in the United States.
- 7.55** We just can't mine it because of the environmental restrictions. And we don't have the refining capacity, which is pretty dirty industry. We would have to build that, and again, there's a lot of forces against that. But suddenly we want Greenland? It sounds like the 19th century, wanting to go colonize the world. A real concern about, I don't know, the Panama Canal and who controls it. And fundamentally, this view that the world trade is zero-sum, which is, of course, what mercantilism really identified mercantilism as.
- 7.56** There's winners and losers. And if you import a lot and you export a little in comparison, you're losing, because the gold is not coming to you, or in this case, the dollars are not coming to you. They're going out over there. Even though we've shown all the dollars come back, they're not willing to accept that. That's not something that they even think about in the trade itself. You follow the dollar. That's the real value, not the goods, the money. And again, this is pure mercantilism. This is pure going back to the 17th century.
- 7.57** So, the United States, under this assumption, is in this real rivalry with the rest of the world and with their governments. It's a brutal competition, a zero-sum competition. There are winners and losers. Canada is winning. We are losing somehow because it turns out, if you take out oil, we export a lot more to Canada than import. So we have a positive trade balance with them. But once you count the fact that we import a lot of

oil from Canada, it flips. Now we've got a trade deficit with them, and that makes them winners and us losers instead of perceiving our relationship with Canada, of all countries, not an enemy, as win-win. It's essentially win-win.

- 7.58** And for the most part, there are zero tariffs between the two countries. Now, people talk a lot about these tariffs on milk that Canada has, vis-a-vis American dairy products. So they have a 400% tariff or something like that. It turns out that that kicks in after a certain quantity. Tariffs are very low in the quantity. And we've never really exceeded that.
- 7.59** And yeah, it's stupid that Canada has a 400% tariff, or whatever the percentage is, on dairy products coming in from the United States. It doesn't make any sense for Canadians. And by the way, we have a lot of government programs and a lot of tariffs ourselves to protect American dairy farmers. I've talked to dairy farmers from the northern states about free trade, and they are adamantly against it, because they're afraid they'll lose their business. They got a lot of protection from the U.S. government.
- 7.60** So it's a zero-sum mentality and no consideration of the idea of mutual gains, of win-win relationships. What matters is the monetary sum of imports over the monetary sum of exports. And if inputs are higher than exports, that's a bad relationship. We are being exploited somehow. We are being, literally, the verbiage is, we're being taken advantage of. Again, this is like you going to grocery store and buying goods and saying, "The grocery store is taking advantage of me, because they get my money and I only get the goods." It's just bizarre. It makes no sense. It's not economics.
- 7.61** It's a theory that was refuted hundreds of years ago, and there's not a single serious economist who believes this. Not a single serious economist—left, right, center, it doesn't matter; it's not a political issue; it's just plain economics—who believes that tariffs are a good thing, particularly extensive tariffs like we're seeing right now. Not a single one.
- 7.62** Now, there are economists, particularly on the left, who believe you need to protect industries, and we need tariffs to protect particularly interest groups. So tariff here, tariff there. And Biden, when he came in, did not do away with Trump's tariffs. He kept them all because the left, politically, is aligned with selective tariffs. But again, economics, economists, there are just none.
- 7.63** Paul Krugman, who I don't agree with on anything, who I think is a lousy economist on most things—he used to be a decent economist when he practiced economics; now he's become a political hack—but even he gets, he indeed won his Nobel Prize in economics for a paper defending free trade. So it's not left, right, center issue. It's basic economic issue.
- 7.64** All right. So we've seen a resurrection of neo-mercantilism. And the way this has really expressed itself is through a system, an advocacy for a system of tariffs, and an elevation of tariffs as some kind of solution to America's economic problems, a solution

to the fact that the world is taking advantage of us. So let's think about what a tariff is. Again, in spite of what some people in politics tell us, a tariff is just a tax. The founding fathers understood this. Pretty much everybody understands. It's a tax.

- 7.65** It's a tax that importers pay. This is why, in the 19th century, you had tariffs because it was easy to collect it, because the goods showed up in port. You could send a customs officer there, collects the tariffs from the importer, the American company importing the goods. And then the stuff is released. Now, the importer has to make money. So what do they do? They basically mark up the goods with the tariff to the consumer. So it's tax that is paid directly by American businesses and indirectly by American consumers.
- 7.66** It's like a sales tax. I mean, it's the—the grocery store is paying the sales tax to the government. You don't write up everything you bought and then send a check to the government in terms of sales tax, right? But the sales tax is explicitly added to your bill. And while the grocery store is the one who pays it, you're the one really paying it, because it's added to the price of all the goods. And a tariff is added because nobody's going to take a loss on taxes.
- 7.67** By the way, all taxes are like that. It's not like, you know, people love to tax businesses. They love corporate taxes. Yes, let's nail those businesses. But every corporate tax is paid by consumers and maybe employees. But none of it is paid by the corporation because the corporation is nothing. Corporation is there to make money for its shareholders. The shareholders aren't going to lose money because corporate taxes exist. They're just going to price it into the goods.
- 7.68** Since all corporations have to pay the corporate taxes, all corporations do it. So there's no competition to drive that price down. So when we raise corporate taxes, we're just raising the cost of goods and we're lowering wages, because the corporation has to take it from somewhere. There's a lot of mythology around taxes. At the end of the day, all taxes are paid by individuals; and corporate tax is just an indirect sales tax. Tariffs are just an indirect sales tax. We all pay for them.
- 7.69** As a consequence, just like every other tax, taxes place a burden on all of us, on businesses and individuals. And as a consequence, they reduce our standard of living. Every tax reduces your standard of living. Now, you could argue it's, in a sense, worth it because the government then provides me with services that maybe raise my standard of living somewhere else. Fine. But let's at least recognize what it is. It's a tax, comes out of our pocket, goes to government, and then hopefully they're doing something useful with it. I'd argue, one could argue about how useful most of our tax money is being used for, given what the government should be doing versus what it does.
- 7.70** So what does this tariff do? What does this tax do? Well, it protects certain industries at the expense of others. Right now, we're protecting the steel industry at the expense of industries that use steel. It protects companies that cannot compete. And it tries to

do this, and it's doing this by basically raising costs on importation, which basically raises costs on our competitors in other countries, which reduces competition in America, makes American companies lazier, less competitive, less innovative, less productive. And again, the economic proof of this is robust.

- 7.71** It boosts profits, again for some, and losses for others. But overall, it's a deadweight loss for the economy. If you want to raise taxes, if you need revenue, then there are better ways to raise revenue rather than selectively only on products that come from overseas. Now again, if that's the only tax you have, like in the beginning of the American republic, big deal, it doesn't cause that much damage.
- 7.72** But given the multiple levels the government does, there's so many taxes that we have, there are vastly more efficient ways to tax Americans than this, if you're going to tax them. And we're already overburdened with taxes. The problem of our deficit is not taxes, it's spending. And of course, we pay as individuals. Now, there's a mythology out there that foreigners pay. So think about this.
- 7.73** You're a Chinese company. You barely make any money because the margins on— let's say you're selling toys to America—the margins on these things are very small. You're competing with other Chinese companies. You're competing with companies in Vietnam. You're competing with companies in Mexico. Potentially, you're competing with companies that make toys in America. So your margins are small. And now there's a tax on imports of 20%.
- 7.74** Are you going to eat it? Eating it means now you're losing money. Your margins are not 20%. I mean, Apple has margins that are 40, 50%, but almost nobody else in the world has those kind of margins. So you're going to take a loss and now run your business for the next years at a loss, so the Americans don't have to pay a tax? Probably not. And maybe you give them a small discount. Maybe you eat 3 or 4%, maybe, if you've got a margin. But ultimately, it's the importer who does it.
- 7.75** Now, a lot of times you'll buy that toy at Walmart, so Walmart is raising the price. And then a politician will come to Walmart and say, "No, no, we want you to eat it. We don't want you to raise the price." But what are Walmart's profit margins? Walmart does not have 20, 30% profit margins. 6, 7, maybe? You remember there was a period where tariffs on China were 135%. So every good was more than a double.
- 7.76** How can you eat 135% tariff? You'd have to sell it not just below the cost of production, you'd have to literally be subsidizing the cost of production. I mean, that's impossible. Nobody can do that. So the whole thing is just an impossibility. The only people who can pay the tariff are Americans. Yeah, maybe American businesses, maybe Walmart will say, "OK, we'll take 1% off." They can't take the 20. Walmart is not going to lose money.
- 7.77** So if you look at tariffs, for example, if you look at the 2018-2019 Trump tariffs, they were completely passed through to consumers, all of them. Prices went up. I don't

know if you remember this, but the Trump administration imposed at some point tariffs on washing machines. I think it was under the national security guise. Washing machines. Well, washing machines went up in value by \$86 a unit.

- 7.78** Interestingly, because they're sold as a package, dryers went up by \$92 per unit. The manufacturers figured, what the hell? They're being protected. There's no foreign competition. Overall, tariffs on washing machines resulted in an aggregate increase in consumer cost across the United States' economy of \$1.5 billion. Now, that's not a huge amount. \$86 a unit is not a huge amount. We can all survive that. But some people at the margin who are poor, \$86 is a lot of money for them.
- 7.79** \$1.5 billion in overall U.S. economy, not a lot, but what a waste. It's just a tax. And it's a regressive tax. Poor people pay it, more than anybody else, because much bigger part of their basket is consumption than wealthy people. Now, OK if we want a regressive tax; we can have that debate about whether regressive taxes are good or bad. But let's at least have the debate. Let's at least acknowledge what it is. We pretend it's not there. It doesn't exist. It's not happening.
- 7.80** Of course, what happens, too, is the other country puts tariffs on us, on goods, so some of our companies can't export to them. They lose money, they lose jobs. So it's estimated that American jobs saved by, for example, the steel tariffs in the first Trump administration, cost per job, right, about close to a million dollars per job. So we saved the job, but it cost a lot of money.
- 7.81** In the 1980s, when they had tariffs to save jobs, those jobs were costing \$256,000. The job itself was probably a \$50-\$60,000 job or maybe even a \$40,000 job. We were paying \$250,000 to save it. By the 1990s, it was \$800,000, and it's even more today. It's closer to a million dollars for every job saved. Take the steel industry, which is, since the 1970s, has been begging government to help it, help it out. "Give us tariffs, please."
- 7.82** Like, there was one steel company, I mean, Nucor, which today is the largest steel company in the U.S., but Nucor used to have an amazing CEO, like a real free-market guy who really believed in, Ken Iverson. Ken Iverson, a real character, real fighter. And he would constantly say, "I don't want tariffs. I don't need tariffs. I'm going to win this in competition. I am the best steel manufacturer in the world. I don't need your damn protection." At the same time as U.S. Steel and Bethlehem Steel and all the traditional steel companies ran to the government constantly to beg for tariffs.
- 7.83** Of course, when Ken Iverson retired, Nucor became the biggest lobbyist for steel tariffs in the country. And today, it's a huge lobbyist for steel. But over the years, since the 1970s, we've had 304 different anti-dumping and tariff executive orders to protect the steel industry. It's still shrinking. They still can't compete, in spite of all of that. And the consequence of that is, steel produced domestically is significantly more expensive than steel you can buy in the world markets.

- 7.84 So people who use steel in America are paying a lot more for steel than they otherwise would be paying. Steel, of course, is the basis for automobiles, for washing machines, for a lot of industries out there. And companies that use steel employ 80 times more people than company that makes steel. So for every steel job, there are 80 jobs where they are buying steel in order to build stuff with steel.
- 7.85 So for example, the steel tariffs that were passed in the first Trump administration to protect steelworkers saved about 1,800 jobs in the steel industry. 1,800 jobs? Cool. That's really good. And destroyed 78,000 jobs in steel-using companies. 1,800 were saved, in a sense, created. 78,000 were lost. I mean, this is not a hard cost-benefit analysis. This is pretty easy.
- 7.86 I mean, if you cared about the U.S. economy, you'd make steel as cheap as possible, by eliminating all the tariffs and all the anti-dumping and all the restrictions and everything. Let American steel companies compete in, and if they can't, they'll shut down. And all you need to do for national security sense is make sure there are enough friendly countries producing steel, so that you can get the steel if you need it in an emergency and you have to get it.
- 7.87 But of course, steel companies constantly are lobbying. I mean, they are one of the biggest lobby groups in Washington, DC, for more protection, more protection. They're never satisfied, never satisfied. And of course, it's taxpayers ultimately that pay the bill for all of this. It's estimated that the tariffs in 2018 cost U.S. GDP 0.2%. Now, 0.2 % seems like a little, but when all you've got is about 2%, it's about 10 %. 10% of less growth. That's a lot, particularly when you start compounding it.
- 7.88 And that was the tariffs of 2018. The tariffs right now are like 100 times more than that, or a thousand times more than that, if you look at what the tariffs are. All right, we're definitely in a new regime right now. Trump 2.0, very different than Trump 1.0. He's going all-out on tariffs. In his first term, he was very restrained, I think, because he had, the economic advisors he had around him were all anti-tariffs, and they basically restrained him and stopped him. And the tariffs he did, he did in spite of the recommendations of his economic advisors.
- 7.89 He got that through because the one thing you can say about Donald Trump is he's always hated trade deficits. This is not new. If you look at the interviews of Donald Trump in the 1980s, all he is is railing against Japan. "Japan is going to take us over. Japan is going to be the biggest economy in the world. We're losing to Japan," because we have a trade deficit with Japan back then. None of that happened, but it didn't change his mind.
- 7.90 Trade deficits is something, of all the things possible that Donald Trump might believe in, this is one he really does believe in. He hates trade deficits. And in Trump 2.0, in this administration, he has surrounded himself by economic advisors that either agree with him on that or have accepted the fact that they're just going to agree with him for the

sake of agreeing. That is, he's surrounded himself with yes-men who are going to say yes, whether they agree or not. That's just the reality. Every single one of them.

- 7.91** Bessent is probably the one who disagrees with him the most about tariffs, but he's not going to voice that. He's going to try to moderate a little bit, but he's not going to actually come out and say he's against it. Everybody else—Hassett probably knows better, the chief economic advisor, but he wants to be chairman of the Fed when Powell is out. So he's not going to contradict Trump now. He's going to position himself to get the position he wants. Lutnick, I can't explain. And of course, we know people like JD Vance and everybody else bought into this neo-mercantilist view.
- 7.92** So, according to the Trump administration, they believe in tariffs, reciprocal tariffs, high tariffs, on pretty much everybody. They have three goals supposedly for these tariffs, that they have clearly articulated. Three purposes for this. One is, shield domestic industries, but even more than that, reindustrialize America. Force companies to come back to the United States. Make it so expensive to build factories overseas and import into the U.S. that they have to start building factories in the U.S.
- 7.93** For this to be successful, you need very high tariffs. You need tariffs so high that they overcome the fact that it's much more expensive to build in the U.S., and they shut down their factories in Mexico and China and all these other places, and they build in the U.S. So here you need 40, 50, 60% tariffs, and in some cases maybe more, to overcome the cost differences between the U.S. and these other countries. So you need very high tariffs to reshore these goods.
- 7.94** Remember, reshoring means all these goods are going to be more expensive, even when there are no tariffs, because they're going to be made in the U.S. The U.S. is more expensive. If you're going to bring textiles back to the U.S., which would be nuts, they would cost a lot more money. Land is more expensive here, building is more expensive here, and people are more expensive here. So this is why you have to have very high tariffs to achieve this goal. So we, the Americans, are going to pay for it one way or the other.
- 7.95** Also, you have to really think about where all these employees are going to come from. Who's going to work in these places? As I said yesterday, I think, there are 500,000 unfilled manufacturing jobs right now in America. If you reshore all this stuff and you need millions of new people to go work in, I don't know, textile mills, where are they going to come from? They don't exist, these people. They've got much better jobs, much better jobs, that pay more and they're easier, easier on the body. But that's goal number one.
- 7.96** Second is, tariffs are going to replace income taxes. We heard this a lot. There was a period in which you heard this a lot. You're going to get so much revenue from tariffs, we can cut taxes. We can cut taxes elsewhere, significantly. Now, note here that if you want a lot of revenue from tax, from tariffs, tariffs need to be low. If you have really

high tariffs, what happens? We stop buying. That's why people reshore. So we stop buying. If we stop buying, there's no revenue.

- 7.97** So what you want is to find a sweet spot where we'll still buy, and therefore, there'll be revenue. And that's possible. You can find that sweet spot. It just, let's realize, it's a huge tax increase for Americans. There's no way for the government to get revenue if Americans are not paying their taxes. So you can find that sweet spot. We pay the taxes, and yes, maybe it cuts our income taxes, but we've just got a huge spike in our other tax. There's no free lunch here.
- 7.98** Other countries are not paying this tax. We are paying the tax. So it's just a question of, where would you rather pay the tax? When you buy imports and it's hidden, or when it's explicit in your paycheck or in a sales tax or in some other way where you know where it is? Remember Amazon actually, at some point, said, "We're going to add a line that says tariff, and you'll know exactly how much tax you're paying on the tariff." And they got a phone call from the White House, saying, "Jeff, don't do this."
- 7.99** Tariffs were a primary source of revenue in the 19th century. As we said, the government was 80 to 90% smaller. There is literally, I think they've calculated, in order to get the amount of revenue you need to replace the income tax, tariffs would have to be well over 100% on all goods coming into the U.S. But if tariffs are well over 100%, what happens to the amount of stuff we buy from overseas? If prices go up, you buy more or less? You buy less. So as tariffs go up, sales go down.
- 7.100** And you keep raising them to try to catch the total. You can never get there. Just mathematically, it's literally mathematically impossible to replace the income tax with tariffs. You can increase revenue from tariffs. They're doing it now. They're setting it right now. Most of the tariffs are being set at a rate that is not going to stop us completely from buying foreign stuff, and revenue is going to be very high.
- 7.101** And I find it really funny that there are Republicans out there, hailing and cheering and celebrating the fact that Americans are paying more taxes. And at the same time, they claimed they cut certain taxes during the Big, Beautiful Bill, and they hailed that as well. So make up your mind. Do you want higher taxes? Do you want lower taxes? So that's two goals, reshoring and raising tax revenue, replacing the income tax, ideally.
- 7.102** And the third goal, they tell us, is, it's just a negotiating tool. It's a way to force other countries to reduce trade barriers on us, so that we can sell into their markets more. So we want, ideally, a zero-tariff world, but they're screwing us by increasing tariffs on us, so we're using this as a negotiating tactic. Now, it's important here to note that these three are incompatible. You can have one of them; you can't have all three of them.
- 7.103** If you want to maximize revenue, then tariffs need to be low, low enough so that we continue to buy, and you get a lot of revenue. It's not replacing income tax, but it will increase revenue. No question, you're raising taxes. That won't reshore anything. It'll

just raise prices, but not high enough so that we stop buying foreign goods and force companies to come and manufacture in the United States.

- 7.104** If you get companies to come here, if you raise it so high, as the companies have to build in the United States, then there's no revenue. Those two are completely incompatible. And then this idea that we ultimately want free trade, zero tariffs across the board, we just want other countries to lower their trade barriers as well, is incompatible with both of them. Because if you get that, you don't get revenue and you don't get reshoring. So make up your mind.
- 7.105** Now, it seems to me that what they've done or what they're doing, when they keep announcing trade deals as we speak, right now at the end of July 2025, and I think they just announced a deal with India, which India is unhappy with, very unhappy with, is they're setting tariffs much higher than they used to be. The average tariff in the United States before this administration came into being was under 5%. And now tariffs are 15, 20, 25, in that range.
- 7.106** UK got 10%, but they're the only one so far who got 10%. Vietnam, 19. Japan, 15, I think, or 20. So a variety, and in India today, 25 %. And of course, Brazil, 50. What they've decided to do is try to go for revenue. These are rates that are high, means we're going to pay more taxes, but they're not so high as anybody's going to move their factory to the U.S. And yeah, we're getting some opening-ups of trade. None of this is going to change the trade deficit significantly. We're not going to see a plummeting of the trade deficit. We're not going to see exports going through the roof at these levels.
- 7.107** So I think they focused in on the second goal because it's politically easy for them to show. It's easy to measure. It's easy to show. And it's short-term. Reshoring, the idea of companies building factories in the U.S., that'll take years. This has immediate upside, I guess; if Americans value their taxes going up, it has immediate upside.
- 7.108** Now, it's important to note again, or put it this way, right now the Trump administration is using two different laws and criteria to set tariffs. One is what all administrations have used in the past. This is the national security, protecting particular strategic industries. This is the tariffs on steel, on aluminum, on copper, on automobiles, and a bunch of other tariffs.
- 7.109** I guess the fentanyl tariff is part of this. This is the tariff he placed on Canada and Mexico and on China because they're responsible for fentanyl coming into the United States. Although the numbers I've seen, more drugs go into Canada from the U.S. than come from Canada to the U.S. Mexico is a real issue, but Canada, there's no issue of fentanyl. But we're using it. National security is great. So that's one.
- 7.110** And that's one that no court is going to overturn. That one is pretty secure. Congress explicitly gave the President authority to do this. And the reality is that no courts, Supreme Court is not going to challenge national security arguments; unless they're

completely nuts, maybe they would challenge them. But he's pretty secure. I don't think that steel tariffs and those tariffs are going away.

- 7.111** And these are tariffs that have clear identifiable special interest groups, pressure groups that are going after this and demanding these tariffs, and they're being granted them. And then others are getting exclusions. I mean, one of the great advantages of tariffs, we talked about this originally, is that they're great for corruption. I get to give you a favor, not to give you a favor. And if you contribute to my favorite campaigns, then I give you a pass on the tariffs. And if you don't contribute to my campaign or to my friend's campaign, I tariff you. I mean, it's great for different types of political corruption.
- 7.112** But there's another law that he's using. This law is called the International Emergency Economic Powers Act, the IEEPA. You'll hear a lot about this in the months to come. The IEEPA, the law, says nothing about tariffs. It gives the President some authority in a time of international emergency economic crisis to do certain things, but the law does not mention tariffs. And this is all the reciprocal tariffs. So the tariffs on Japan and on Indonesia and on Vietnam and now on India and on UK and EU. We just did a 15% tariff on the EU. All of those are under the IEEPA.
- 7.113** And this one is in the courts. And it went to the first-level court, which is the Court for International Trade, which is an American court that deals with trade issues. And that court has ruled that Trump's use of this law to impose tariffs on all these countries is unconstitutional. The power to tax is with Congress. The power to impose tariffs is with Congress. And that this law, the IEEPA, does not give the President the authority to do this.
- 7.114** Now, as we speak, I think tomorrow, again, this is the end of July in 2025, this is going to be argued in front of the appellate court, and we will see what the appellate court decides. There is no question this is going to the Supreme Court. That is, in the fall or maybe in the winter, the Supreme Court will have to decide whether the use of the IEEPA to impose tariffs by the President unilaterally—I mean, this is amazing. We have a system of government in America that's built on separation of powers. And yet, we're seeing one of the largest tax increases in American history done unilaterally by the executive without Congress saying a word. No intervention.
- 7.115** So the Supreme Court will have to decide whether that's constitutional or not. There is, I think, a 50-50 chance that they rule this unconstitutional. My view is, if that happens, the administration will find a different law, and they won't give up on tariffs. They'll find a different law to impose the tariffs, and you'll have to go through the courts again and fight it all over again. But right now, this is in front of the appellate court, going to the Supreme Court, heading towards the Supreme Court.
- 7.116** So what we've seen over the last 8, 10 years, to some extent because of all the stories that we heard, we talked about last time—stagnation of employment, no

manufacturing anymore, we're losing jobs, the economy sucks, we're getting poorer, all of the stories we talked about last class—there's been a real shift in politics and in the American public towards the need to do something, and really an embrace of mercantilist ideology when it comes to trade and when it comes to economics.

- 7.117** And I think the Trump administration is manifesting that. And you can see that in the fact that the Biden administration didn't retract any of the tariffs that Trump passed because they sensed the mood of the people. The people want this. They really buy in, or at least a significant amount of them, they buy into this mercantilist idea of a zero-sum world. A dangerous idea, a harmful idea. But nevertheless, that is what is prevalent today.
- 7.118** The analysis of tariffs that I just did might sound political, but really this is an analysis that every economist out there is doing. It comes to a particular conclusion, but that's because the science of economics comes to this conclusion. It's pretty much bipartisan in terms of economists. This is the nature of tariffs. This is how they work. This is how taxes work. This is the price we pay, and these are going to be the consequences of this.
- 7.119** So next class, we will talk about some of the benefits to trade. We'll talk maybe about the morality of free trade, and we'll wrap everything up. Thank you.
- 7.120** Is it possible to simply outlaw special interest groups or cronyism, like some suggest, or do you really need a constitutional amendment separating state and economics to really do that?
- 7.121** Yeah, so I don't think it's possible to outlaw cronyism, to outlaw lobby. And I don't think it should be. The reality is that we have given our politicians a lot of power over our economy, a lot of power over business. And I think it would be wrong to tell business they can't defend themselves. And a lot of lobbying is in self-defense. Indeed, most lobbying starts out as an act of self-defense, then maybe turns into trying to manipulate things. But it's a free speech issue. But it's also a self-defense issue. You can't concentrate all the power in politicians. You need something to offset it.
- 7.122** The only way to reduce special interest groups, to reduce lobbies, to reduce pressure groups, is to shrink the role of government. The more powerful the government is, the more involved in our lives it is, the more we will want to have influence over it. I would like to imagine a world in which politics doesn't really matter that much. They're not in my life. They're not telling me what I can and cannot do. They're there to protect me and otherwise leave me alone. And once in a while, I go and vote, but it's not that big of a deal. It's not life or death.
- 7.123** But the more power we give our politicians over our lives, the more it becomes life or death, or life or death for business, certainly. And we have to allow businesses to express their views on that. If you really want to get rid of cronyism, you have to get rid of the power of government over the economy. I'm a big advocate of separating

government from economics. I don't think the government should be involved in economics. I don't think it should be involved in business.

7.124 I think we should have a marketplace where we, as consumers and producers, primarily producers, make choices for ourselves. As long as there's no fraud involved, as long as people are not being abused, physically abused, none of the government's business, how we deal with one another. You'll notice that, as government grows, starting out in the 19th century, all the way to today, as government has grown, as government intervenes in our lives, particularly in our business economic lives, more and more and more, there's more and more and more lobbying. There has to be.

7.125 So if you deregulate, if you shrink regulations, if you shrink the role of government, there'll be less lobbying because there'll be less on what to lobby. If you make politicians impotent over your economic life, then why would I care? I'm not going to lobby them. Why waste my money?

7.126 You said tariffs were never good for the national security purposes. But if we were pursuing free market policy with our Asian allies, which we're not, but if we were, wouldn't tariffs be a good way to disincentivize investing in China, and you get a slower pullout instead of going on a full embargo, which might cause more of a shock?

7.127 Yes. So maybe you could find a way to phase your relationship with China out over time. And maybe you could conceive of tariffs being a means to doing that, assuming you had a real strategy over time of how you would phase it out. I think much more logical in terms of China would be, embargo the things that have national security implications. Don't sell them advanced chips. Don't sell them advanced things that they can use for the military.

7.128 And then, other than that, as they become more threatening, if they become more threatening, then embargo more of the economy. If they become an enemy, embargo them completely. If you have a real goal over a 10-year period and you're going to raise tariffs systematically and you're embracing free trade with all the rest of Asia, then yeah, I guess you could come up with a strategy to actually do that effectively. But you need a strategy.

7.129 And national security is a tricky one to have a strategy. And I don't believe the United States has had a national security strategy since World War II. I don't know what it is if they've had one, because it's incoherent. So you need a strategy. You need to act on that strategy. And I'd say the number one priority in Asia for the United States right now, strategically, is to beef up our relationship and to beef up our cooperation with our allies.

7.130 So we should be zeroing out tariffs with South Korea and with Japan and maybe even Vietnam, because partially because the Vietnamese hate the Chinese. They're always at war with one another. The Philippines, the Indonesians, the Malaysians, the Australians, and India. We don't want to antagonize India. It's the largest country in the

world. They have more people than China today, India does. It's a growing economy, and yet we want to zero out and have a great relationship with all those countries, and maybe even form a defense alliance.

7.131 And if you actually had a NATO of the South Pacific with South Korea and Japan and the Philippines, China would really have to think twice before it thought about invading Taiwan. And then if you embrace Taiwan as part of the defense alliance, and how about diplomatic relations with Taiwan? There would be a strategic move. It'll piss off the Chinese. That in and of itself would probably stop trade with China. So just have diplomatic relations with Taiwan.

7.132 And that would be the right thing to do. It would be the moral thing to do. They're a free country. They're basically a good country. They're an ally of the United States. So there's so many things you could do if you actually had a, if you actually were focused on national security and actually were willing to think outside of the box, and think really in terms of—I mean, I'm a huge advocate for America first, but it has to be an America-first foreign policy. You have to think in those terms.

7.133 And I don't see anybody out there thinking in terms of, what would a real America-first foreign policy really look like and how you would implement it. It takes strategy, and I don't see strategy anywhere. All right. Thank you.

Lecture 8: The Power of Trade

At a glance — The power of trade.

- 8.1 So we've talked a lot about trade in this class. We've talked about how, really, trade is something that goes back to the very beginning of our species. People started making stuff that they were particularly good at. People started then trading for stuff maybe they weren't as good at. This idea of specialization and division of labor and comparative advantage are things that have been with us since the birth of mankind.
- 8.2 And people have been utilizing them without really knowing. All Adam Smith and Ricardo do is identify them and say, "Hey, if we do even more of that, we could be even better off." If we recognize the values of the things that we're doing already and have always done, and then focus on those things and structure ourselves socially, politically, in a way that promotes specialization, division of labor, and comparative advantage—in other words, we structure ourselves in free markets— we will be even more prosperous than we are.
- 8.3 And we've seen that throughout human history. People have traded. People have gone through unbelievably hard times in order to trade. I mean, think about, I don't know, the first traders to go to China and to cross the deserts and the mountains and all the hardships that are involved in making that trip. Not today in a jet plane, but on foot or on horseback or on a camelback, or all of the above. And actually getting to China, not knowing what you will find.
- 8.4 Remember, in those days there was no Instagram photos or videos of what China's like. And Chinese might have never seen a European before. You're risking your life, you're risking your freedom, you're risking everything. And yet, curiosity, interest, and the desire to trade and to make money and to discover new goods and to trade with your fellow man is intense and always has been there. And it's what makes us, partially what makes us human.
- 8.5 The desire to improve ourselves constantly, as individuals and as groups, drives us to trade for something better, constantly, to elevate ourselves, to specialize, to utilize the principle of comparative advantage to our benefit. So, trade has always been there. We've gone through the history from really tribal societies all the way to our sophisticated modern world. We trade for everything. Most of us are very specialized.
- 8.6 We have a profession, we do it well, we get paid in money, and money is the lubricant of trade. It's what makes trade possible. It facilitates trade. That's the primary purpose, reason for money, is as a medium of exchange. It's also, of course, a store of value that allows us to produce and allows us to think about trading in the future, not just trading in the present.

- 8.7** Trade is vital for human existence and human flourishing. I think the standard to evaluate any economic system or any political idea is ultimately, what does it do for individual human flourishing? What does it do for the individual's ability to pursue his own happiness, to be successful, to flourish, to live the best life that he can live?
- 8.8** I think we established a particular political system here in the United States, built on liberty, built on freedom, built on the idea that each one of us has an inalienable right, a right that cannot be taken away from us, to live our lives as we see fit, in pursuit of our own personal values, in pursuit of our own happiness, and free of coercion, free of force, free of people intervening and telling us what we can and cannot do, and forcing us to do things we don't want to do.
- 8.9** That's our Declaration of Independence. That's the basis on which we structured the Constitution. The basic of that is all on freedom. So when we look at trade, we need to evaluate it based on a number of different parameters. What does it do to individual flourishing? What does it do to the ability of individuals to flourish?
- 8.10** And to what extent are different regimes of trade consistent or inconsistent with the principle of liberty and the principle of freedom, which, at least at some point in our history, we believed, were the basis for human flourishing, were the basis for individual success. And if we got freedom and liberty right, everything else would fall into place. That's at least how we used to think about things.
- 8.11** Let's think a little bit about how we've done economically, culturally, in all of these dimensions under different regimes of trade. I think one of the things that we see when we look at the history is the enormous economic benefits, wealth, creating benefits, that we get from trading freely with others, from becoming specialists and trading. And because we can trade, we can become even more specialized. And again, we talked about this as a virtuous cycle.
- 8.12** You trade, you specialize, you specialize, therefore you can trade even more. And the better you become at your specialization, the more you earn, the more you can trade, and thus giving other people an incentive for them to specialize. It's been quite a ride for humanity, in particular over the last 200, 250 years of relative freedom, starting with the founding of America, coming out of the Enlightenment. The recognition of individual freedom is important, individual rights is important, individual flourishing is important.
- 8.13** Global economy has exploded in terms of wealth creation. We are so much richer today than we were 250 years ago. I mean, by measures of income or wealth, it's something like 300 times richer. But that is way underestimates how rich we really are, because that doesn't capture the fact that we have running water, and they didn't; or that we have electricity, and they didn't. All it captures is wealth that we measure in dollars, but we don't really measure our water supply in dollars or our electricity in dollars. We just take them for granted.

- 8.14** We're thousands of times richer in terms of standard of living and quality of life than we were just 250 years ago. What would you rather be? Poor in America today, or the king of a country 200 years ago? In terms of quality of life, standard of living, it's no contest. Rather be poor today. Probably have air conditioning; kings didn't have air conditioning. Running water, kings didn't have running water. Maybe I can afford a flight to go see Europe; kings couldn't fly anywhere, couldn't drive a car.
- 8.15** Probably died very young. Life expectancy 250 years ago was in the 30s. Looking around, most of you are dead already. It's no contest. We live amazing, wealthy lives. And much of that is because we can produce and because of specialization, and we became really, really good at producing stuff. But that is not possible without the other side of that, and that is trade.
- 8.16** And yes, trade within a village is good, and we can benefit from trade within a village, but it's limited. There may be 50 people in this village, and there are only so many trades you can make, and there's only so much you can specialize, because there are only 50 goods potentially there. And we probably consume a lot more than that, so we have to do more than one thing.
- 8.17** But now, if the village is eight billion people, eight billion people, now you can really specialize and now you can leverage. The fact that 8 billion people are specializing, and you can trade with all of them freely, your standard of living, quality of life, your wealth, everything else just explodes. And that's what we've seen.
- 8.18** The population of the world has gone from half a billion to eight billion over the last 200 and something years. Our GDP per capita has gone through the roof. Our standard of living, quality of life, is thousands of times better. And a big chunk of that is because of exactly this dynamics. The fact is that people in Vietnam are benefiting enormously from the fact that they produce textiles that we use.
- 8.19** I'll give you some numbers with regard to Bangladesh in a little bit and how much better off they are. We might as well talk about that, Bangladesh, just some examples. In 1990, it's not that long ago, 50% of Bangladeshi population lived in what's called extreme poverty. I checked. The UN has changed the number. It's now \$3 a day. I thought it was \$2. \$3 a day or less. 50% of the Bangladeshi population lived on \$3 a day or less.
- 8.20** I remember, you can tell how old somebody is by which country their mother used them to guilt them into finishing their food on the plate. When I was growing up, Bangladesh, people were starving. There was massive famines. And my mom used to say, "You got to finish your plate of food. There are kids in Bangladesh who are starving," as if she was going to send that food to Bangladesh. As if there was any relevance to it, too. But that's how mothers often speak.
- 8.21** Anyway, Bangladesh, when it was founded in a bloody, horrible war, was dirt poor. Again, 1990, not that long ago, 50% of its population. Today, or in 2022, three years

ago, the number of people who were \$3 a day or less was 8%, which is consistent with the global number. GDP per capita went from \$2,000 a year, not a lot of money, to \$8,000 a year. Now, still not a lot of money. They're still poor, certainly by our standards, but they're four times richer than they were. These are all adjusted for inflation, not cheating. Four times richer.

- 8.22** Now, none of this happened, none of this happened because of foreign aid. None of this happened because of charity. None of this happened because of anybody's, I don't know, generosity. Bangladeshis got four times richer than they used to be because of markets and trade, because they integrated themselves into the global economy and they started producing things that people outside of Bangladesh wanted to buy.
- 8.23** The country made a huge investment in textiles, in the garment industry. As we've talked about, the garment industry is the industry that almost every country, when it starts out in the process of industrialization, starts out with. It's the simplest, it's the easiest. They start out with that. That's the first one they do. And Bangladesh is exactly the same. Today, it employs about 4 million workers in Bangladesh, 10 million if you count suppliers and people at the periphery of the industry. It's about 10 million people are employed.
- 8.24** And it's a big part, some would say, the main driver of getting out of poverty. And look, getting out of poverty is hard. The natural state of mankind is to be poor. You have to industrialize, you have to work hard, and it takes time. You don't go from poor to rich like that. It takes decades. Every single country. When the UK embraced the industrial revolution, it was poor. They were living on \$3 a day or less. Over 80% of their population was on \$3 a day or less. And it took them time to get everybody to where they're relatively rich today.
- 8.25** And the same thing is true in the United States. It takes time. So we might look at Bangladesh today and say, "Oh, god. They've got sweatshops. These are horrible jobs. They're so horrible." It seems like some want to bring it back to the United States. But these are terrible jobs. They're boring. They're hard. The conditions are not good. They work long hours. And all that is true. All that is true. Children often work in sweatshops. Child labor, all that is, we look at and we say that's terrible. And yet, there's no way for them to get rich if they don't go through that process. You can't skip over it.
- 8.26** No country ever has, in all of history. Every single country in the world, when it was poor, to become relatively rich, went through the sweatshop phase. Every single one, China, Vietnam, right now Bangladesh, United States had sweatshops, UK, every country, every country in Europe. There's just no shortcut. You need these sweatshops. They start paying. And of course, the only way they can pay is if they can sell their goods outside of Bangladesh. Nobody in Bangladesh can afford the goods that they're producing, certainly not in the quantities.

- 8.27** They have to be able to sell it overseas. So you have to have global trade in order for Bangladesh to come out of poverty. They cannot come out of global poverty if they're self-sufficient. Self-sufficiency equals poverty, always, everywhere. Just try it personally and extrapolate. If you would try to be self-sufficient, you would be poor. So Bangladesh specializes. They then sell it out. People start making money. They become slowly more productive. They make more money as they become more productive.
- 8.28** The industry as a whole becomes more productive. As they become more productive, they make more money. People are gaining wages there that are higher than what they get anywhere else in the Bangladeshi economy. So now they're consuming, allowing other people to specialize in other things to service them, because they are better off—and the economy slowly rises.
- 8.29** And now maybe there's some investment capital that is accumulating. They're saving a little bit, and now maybe another industry starts out, and who knows what that industry will be. And over time, one of the things you can guarantee is that textiles will start shrinking in Bangladesh, and other industries will expand, maybe toys, maybe other stuff that move away from China will start expanding there. And the textile jobs will probably go to Africa, which is the next big poor country turning into middle-class countries phase that happens in the global economy.
- 8.30** You can measure these things. You can measure the improved standard of living. So for example, as the sector, as the textile sector expanded in Bangladesh, the probability of neonatal survival for children increased. So fewer deaths in childhood or in childbirth as a consequence. Everything improves once people are working and producing and earning and getting just a little bit wealthier. It makes a big difference. It makes a big difference to health, not just to what they can buy with their stuff.
- 8.31** Different quality of food they're getting. They're getting a better diet. They're healthier, they're stronger because of work. And at some point, a poor country, poor industries will get to the point where they are productive and they're making enough money that now they implement safety, they implement higher wages, they implement all these things, and children disappear from the workforce, all because they've reached a certain level of wealth.
- 8.32** There's a fascinating statistic. Every country who reaches about \$12,000 GDP per capita, so \$12,000 GDP per capita, child labor disappears. That's about the level of income where parents can afford to feed their kids and send them to school instead of working. Below that, in order to feed their kids, the kids have to work. It's true on the farm. It's true in pre-industrialized civilization. We always had kids working because they needed to feed themselves because the parents weren't making enough.
- 8.33** But when you get to 12,000, whether it's the UK in the 19th century, or America in the 19th century, or Indonesia, or Malaysia, or Bangladesh, soon get to 12,000, child labor disappears. It doesn't matter what the laws are. It doesn't matter what politicians do.

The market deals with it. Children are not very productive, so employers are happy to get rid of them once they reach a certain level of productivity, once they can replace them with machines. And parents, as soon as they can afford to do so, pull them out of the workforce and send them to school.

- 8.34** Sweatshops actually are necessary. We in the West can sit there comfortably and pretend otherwise. We can harass them. We can try to stop sweatshops, and we can even bring better working conditions. But what happens when you, from externally, demand better working conditions? Well, the costs of producing the product go up. The cost of the good goes up. The industry shrinks. People lose their job, and the standard of living in the country actually goes down.
- 8.35** The working conditions will improve as they get richer, and enforced from the outside actually will make them poor. So a lot of times, well-intentioned activists make things worse, not better. Retailers often go to these countries and demand better working conditions, and so on. And ultimately, markets will adjust and will get there. And obviously, you've heard of horrible fires and horrible accidents where a lot of people die. It really is horrible, but there's no way around it. That's just life. To get from here to there, that is the price you have to pay. Every culture has done it, every country has done it.
- 8.36** And what makes it possible to get from poverty to success is trade. Couldn't happen without. If we boycott Bangladesh and close, they will stay poor forever. Again, self-sufficiency is a guarantee of poverty at any scale. Now, the bigger the country is, the less poor you'll be, because there's enough division of labor within the country, which is still giving up a huge amount of wealth, that comes from trading with people outside your country.
- 8.37** Trade is win-win. If there's one thing you get out of this is it's win-win. So if you're going to stop trading, you'd better have a really good reason. Like, they're going to kill me. It's national security. That's the only reason I can think of to stop trading. We, as we've seen in the course, have gotten richer. Our income has gone up. In the United States, more people work today than ever before. Globally, more people work than ever before. In spite of all the technology, like technology destroys jobs, it does; but it creates more jobs than it destroys.
- 8.38** Today on the planet, there are 8 billion people. Unemployment is not that high globally. It used to be, we had half a billion people. So we created seven and a half billion jobs. That's pretty amazing. Because every human being that's alive needs a bunch of stuff. So every new person who's alive needs a bunch of stuff. There's a bunch of people now making stuff to satisfy him. And the only way he can afford all that stuff is by working himself and therefore contributing to the stuff being made to sustain his life.
- 8.39** We live today, the United States, in particular today, because, I think, of a unique economy, our entrepreneurial spirit, our ability to produce, the fact that we've

embraced specialization, and we embrace entrepreneurial spirit. We are, on a global scale, pretty amazingly wealthy, significantly wealthier than any other part of the world. Now, again, you'll find little countries that have a lot of oil, maybe they're wealthier than the United States, but no advanced, no serious economy out there of any size comes any way close to the U.S.

- 8.40** Most European countries on a GDP per capita basis, adjusted for cost of living, would rank lower than Mississippi if they were a state in terms of wealth and income. Most of us are amazingly rich and prosperous as compared to anybody else in the world. We live in bigger houses, drive bigger cars. And that is a consequence of the fact that we are really good at producing. And we have been really good at trading, and benefiting from the productive ability of all those people around the world.
- 8.41** And the fact is, one of the great innovations of the founding fathers, one of the great things about America is that we invented the idea of taking a massive geographic area and allowing free trade within it. The founders were clear—no tariffs or non-tariff barriers on trade between the states. So we have this massive geographic area with complete free trade in, with 350 million people. That's pretty good. That's a good starting point.
- 8.42** And then for at least the last, since World War II, we have participated in a global economy in an active way that has enhanced that geographic area. EU learned from us and embraced the free trade zone. Now, the whole world needs to learn and just make the whole world a free trade zone. If you put aside wars, decade after decade after decade, globalization ultimately has increased. It's created an unprecedented standard of living. Global manufacturing has just been amazing in terms of producing stuff that, again, we take for granted.
- 8.43** It used to be, a cost of a dishwasher has dropped 61% since 1979. In terms of how much time it takes us to earn enough money to buy stuff, which is a really interesting way to think about prices and how we consume a standard of living. So the time it takes an average worker to make enough money to buy a refrigerator has dropped 92.4 % since the 1950s. 92. So you have to work a lot less to get the refrigerator, which means you have a lot more time, a lot more work time, to buy other stuff with. So, today, you can get 13.2 refrigerators for the same time you would have worked in 1956.
- 8.44** That's how much richer we are than 1956. I think it's a good way to think about it. And that is not counting the fact that we have these amazing smartphones that, yeah, the value added is produced in the United States, but without the ability of Chinese to assemble it cheaper and the Taiwanese to produce the chips and the Koreans to produce a bunch of other things that's in here, and all the other places that a hundred different countries contribute to this phone, contribute stuff—not contribute, trade—for the stuff in this phone, couldn't happen without that.

- 8.45 Couldn't make it all yourself. There's no self-sufficiency iPhone. Can't exist. And what we have today is more interesting jobs, more jobs oriented towards the values that we have as individuals. There are more entertainers today in the world. There are more jobs that earn a lot of money that don't require a lot of physical activity. We're not eroding our capacity to be alive. That's why life expectancy is going up. Health ultimately is going up. If you take out use of drugs in the United States, and if you take out alcohol, then if you live a healthy life, you're going to live long and you'll live pretty well.
- 8.46 And as we said, poverty is declining all over the world. We've just talked about Bangladesh, where we talked about global numbers. It's astounding how quickly poverty is disappearing. And if we increase trade, if we actually liberalize trade completely, poverty would be eradicated. There'd be nobody in extreme poverty within 10, 15 years. And then the next step is nobody in poverty, period. And all that requires is economic growth. If the world economy grows, then poor people will grow themselves out of poverty.
- 8.47 And that can only happen with free trade. Specialization and free trade. And of course, the next big continent to bring out of poverty, to bring itself out of poverty, is Africa. And Africa will not become rich with foreign aid; will not become rich with charity and, I don't know, the Gates Foundation, or this foundation, or that foundation. Africa will become rich when it embraces markets, rule of law, property rights, and trade.
- 8.48 And if we want to help them get rich, we drop all trade barriers to African goods. We allow them to import, to sell to us as much as they want, whatever they want, that we want to buy, only if we want to buy it. One thing is unequivocal. Economically, trade is good. It's win-win. We know that from, in a sense, first principles. There's no trade if it's not win-win. And if it's win-win, we're better off. And every time we trade, we're a little bit better off. And if you think about the millions of trades you do in a lifetime, every one of them is making your life just a little bit better off.
- 8.49 And by the way, getting a job, having an employer, is a trade. You're trading your time for money, your ability for money. They're paying you money, you're doing a job. That's a trade. Most human relationships are trades, monetary trades or spiritual trades. We interact with one another as traders. Value for value, win-win. I often say, a good life is a life that has maximized your win-win opportunities. Go out there and seek win-win relationships in everything that you do.
- 8.50 And many of those are monetary, and that's great. But it's also a massive benefit culturally. Life is so much more interesting. We can travel around the world and experience different cultures and see beautiful things or interesting things or crazy things. We can, as we said, watch TVs and movies from different cultures. We can read books that are translated and printed and, if anybody reads books anymore, we can experience that. And we can meet people who are different than us, which is interesting and fun.

- 8.51** It's boring to hang around with the same people that are exactly like you. It's interesting to meet people with different experiences who've been in different parts of the world, who've lived different lives. And globalization makes that possible. We can travel, and to some extent, we can immigrate. There's a lot of people, young people now, if they work off of a laptop, they can go anywhere in the world and work. Set up a shop anywhere, and there they are, as long as the local laws allow them to do so.
- 8.52** Then there are a bunch of programmers, American programmers, working in Lisbon or in Athens or in Singapore or in other places around, in Thailand. They get to experience a different culture. They get to live in a different place for a while, interact with new people, and still make a living and get dollars. The cost of living, by the way, in these places is much lower than it is in the U.S. So it's quite a good deal, if you can get a gig like that.
- 8.53** So we benefit enormously from the cultural interactions with all these other cultures, and we learn a lot. Look, America is an amazing place, particularly if you think about industry. We are the world leaders when it comes to business. We're best at entrepreneurship and technology. We're not great at culture. I'm a classical music guy. There are a couple of classical music composers in American history, but they're not that great. They're not up to the standards of the Europeans. We've imported that culture here.
- 8.54** The guy on my Lyft earlier was playing opera in the Lyft, and it's like, yeah, that's something, but that's not American. So we benefit from the fact that there are artists all over the world who produce beautiful things, and we can get it like that. We can trade for it and get it here. That's true of painting and sculpture, and it's true of movies and television. They love American movies, and often we like their movies.
- 8.55** We're very good at making superhero movies. If I never see another one in my life, my life will be actually a little improved. Luckily, other countries make other kinds of movies, like for adults, for actually thinking adults once in a while. And you can actually go watch them. You don't have to be stuck on what happens to be the latest, biggest thing in Hollywood right now. And of course, again, there's a lot of cultural exchange. Moviemakers in Hollywood are very influenced by moviemakers in Europe, who are very influenced by moviemakers in Asia who are influenced by—
- 8.56** They're all influencing one another, and that makes them better. It makes them better. And of course, the same is true of the rest of the world. The rest of the world benefits from Western civilization. And there's a lot of complaining out there, "Well, we're turning all the cultures the same. There's McDonald's and Starbucks on every corner, and it's boring, and it's all the same." I mean, that's so superficial.
- 8.57** Yes, there is McDonald's and Starbucks. I like the fact that—I never eat McDonald's, but Starbucks is great because I can go anywhere in the world and get the same cup of coffee. And yeah, you want to be adventurous and try other stuff as well, and that's all

cool. But you always know that if you really want a safe Starbucks cup of coffee, it's right there and it's going to be the same and it's going to taste the same, and that's really cool.

- 8.58** It's what makes standardization a value out there, is you know you're going to get the same. It's just like, when you're going to McDonald's anywhere, you're going to get the same hamburger. I don't think, I wouldn't eat them, but go for it. But what we ignore is all the other things that we've exported, like political freedom, the idea of individual rights, the idea of individuals pursuing their happiness. Those ideas are part of globalization; they're part of the export of the West to the rest of the world.
- 8.59** The idea of science and technology—now, some cultures had science, but as a systematic approach to studying reality and understanding the world, that's a Western phenomenon, and now it's a global phenomenon. Everybody is doing it all over the world. There are scientists and engineers working on this stuff. It's a beautiful thing. So the technology that we've exported. So, when we think about globalization, all the positives way outweigh any minor negatives.
- 8.60** And look, yeah, are some ancient cultures going to die and disappear because there's no market for them? Yeah, that's called evolution. It's what happens. The better overtakes the old. But some people who are really interested can raise money and preserve those ancient cultures and create museums or create other venues. They can preserve even ancient languages. But that's only possible because we're wealthy and we have the time. And some people are interested in that. Go for it.
- 8.61** Again, globalization is positive. There's very little you can say that's negative. Economically, it's all positive. Culturally, I think it's all positive. Even when it comes to something like war, the reality is that the more countries are interconnected, the more they're trading, the more they have banking relationships, the more they have flow of people and goods and money between them, the less likely they are to go to war.
- 8.62** I think one of the reasons China did not invade Taiwan when Russia invaded Ukraine and everybody was distracted is because, what would China do if the West put an embargo on their goods? They are so dependent on export. They can't afford to piss us off too much. If we stop buying Chinese goods, they're in real trouble. Now, we would be, too. We'd have to find somebody else to trade with. But we're in a better shape than they are.
- 8.63** You know what about all the investments? We brought venture capital to China. All our top venture capital firms in Silicon Valley opened up offices in China. Now, they've all left, but now there are local venture capitalists. The Chinese have learned from us. Again, globalization. China has a thriving venture capital community, and some of the money they get is from American investors. All of that would disappear if they went to war with Taiwan. They would have to really be convinced that it would be a quick, easy war, and that we wouldn't respond, for them to actually do it.

- 8.64** The more connected, the more interwound we are, the less likely it is that it happens. And indeed, I think that some of the steps we're taking towards China are so ill-thought and so in their face that we're almost egging them to do something, pushing them towards more war, rather than less. China is not a good country. It treats its own people horribly. And you should think twice about, as a person, as an individual, buying stuff from China that you don't have to buy from China.
- 8.65** But whether as a political entity, the United States needs to antagonize them, is unclear. Again, you need a strategy. So war is probably reduced when there's more trade. One of the reasons we saw very little war between the Napoleonic Wars and World War I is because the world became so well integrated, particularly Europe became so well integrated. One of the reasons I think we didn't see much war between World War II and today is because of this.
- 8.66** And I'd say, there's a reason Russia felt emboldened to invade Ukraine. And that has more to do with a lack of confidence in the West and a rise in nationalism in the West and a deterioration in this attitude of openness in the West. Now, look, it's no guarantee. Wars can still happen. Trade does not stop wars. It just makes them less likely, which is a good thing given how destructive and horrible wars actually are.
- 8.67** So again, globalization is an unmitigated good from an economic perspective, cultural perspective, national security perspective, if done strategically and right. And from the perspective of poverty, to the extent that you care about the human race and poverty around the world, globalization is a solution to it. It's not the cause of it, it's the solution to it.
- 8.68** And of course, globalization is good for you as an individual. You as an individual have so many more options if you live in a place that is engaged with, interacting with, trading with the world. We have so many more opportunities to do the things that we love. Everybody in Hollywood makes more money today because movies today have a global market. They don't just sell in the U.S. Indeed, if you make a movie that just sells in the U.S., that's like a failure. Half the revenue comes from overseas, which means you now have double the revenue, because of globalization.
- 8.69** I think the Trump administration was talking about putting tariffs on foreign movies or American movies filmed overseas. Like, a lot of American movies are filmed in Canada. Why? Canada doesn't have quite the union restrictions, and they subsidize filming in Canada, so the movies are made cheaper. We get more movies as a consequence. We all benefit from that. So ultimately, globalization is good for the individual. We have more choices.
- 8.70** I like food. I like restaurants. I like the fact that the cuisine pretty much in every city in the United States today is global. You can find food from most significant countries in the world that have food worth tasting. There's no British food in Arizona. But anywhere you go, anywhere you go, you can find a whole gamut. That's great. It just

enhances your life. And then the fact that we can get cheap, really nice, fashionable clothes, shoes.

- 8.71** There used to be a shoe industry in the United States. There's a period in the 2000s where 50% of all the shoes in the world were made basically in one city in China, Dongguan in China, 50% of all the shoes in the world. Now Dongguan makes no shoes. Labor costs are too high. It's all moved to Vietnam, places like that. But we get a huge variety of shoes. I mean, you go to the mall, see how many shoes stores they are and how many variations and how many— that's great.
- 8.72** Just a little bit better, a little bit of enhancement to our lives, the fact that we have all the variety and choice that we have. Do we really want to give that up, for what exactly? What is the thing that we're going to get in its place? More jobs, we're told, but that's not true. We've seen that that's just not true. Globalization does not take jobs away. And as a consequence, if it makes individuals' lives better, then it's enhancing human flourishing. That's good. Anything that enhances human flourishing, that is a good thing.
- 8.73** Of course, there's a freedom dimension of trade. Why is it anybody's business who I trade with? Remember what we said in the first class? Individuals trade. Countries do not. You buy a piece of clothing, you're trading with the guy who made it, ultimately in Vietnam or Bangladesh. Yeah, there's some intermediaries, companies that facilitate the movement of the good, from him making it to you consuming it. But it's the two of you trading.
- 8.74** Why is it anybody's business who I buy my stuff from? Who I buy my steel from, who I buy my car from, who I buy my clothes from? Is there some national interest that we can touch, that we can feel, that is above my own interest, above each one of our interests? I mean, I get national security, but other than that, what is it? My life is important to me. It's my standard. I want to be able to trade with whoever I want.
- 8.75** I don't like what the Chinese are doing to the Chinese people, the Chinese government is doing to the Chinese people. So I'm inclined to avoid Chinese goods when I can. Sometimes you can't because they dominate a particular market. That's my choice. You don't care that much, don't. I mean, we might debate and argue; I might encourage you, but it's again a personal choice. This country is about personal choices. Freedom is about personal choices.
- 8.76** Trade is fundamentally an issue of liberty. It's fundamentally an issue of rights. I have a right to buy what I want to buy. As long as I'm not hurting anybody else physically, as long as I'm not violating somebody else's rights, why is it any of the state's business who I buy from and when I buy it? The whole issue of trade is ultimately an issue of liberty.
- 8.77** If I want to get on a plane and fly to London and buy something there and bring it back with me, why is it your business? Why can a customs official charge me a tax on it? Or

prevent me from being an incomplete—now I get it. If it's, I don't know, a disease-bearing fruit or flower or something like that, OK, I get that. But other than that, there's just no reason. It's an issue of freedom. It's an issue of liberty. It's an issue that our Declaration of Independence and our Constitution is supposed to protect us as individuals from the government exceeding its power.

- 8.78** More people trading with more people means more people producing. More people producing means more people thinking. Thinking is at the heart of all innovation. Thinking is at the heart of all creativity. Thinking is at the heart of all production. 8 billion people thinking about solving problems is much better than 300 million people thinking about solving problems. The more people are engaged in a global economy, the more problems get solved. The more people engage in the global economy, the more production happens—and the richer, the better off we all become.
- 8.79** Production is the source of wealth. Thinking, individuals thinking, is the source of production. Nothing happens without individuals thinking, thought. To think, all you need to do is leave people free; and the more people thinking, the better. Now, the good news is, the world is still globalizing. In spite of what America is doing, the world is still globalizing. We might not be globalizing. We might be shrinking our scope and the amount of trade we do and the freedom in which we trade. But the rest of the world still sees the value of it.
- 8.80** So we've talked about the fact that Africa now has a free trade zone. Europe is trying to establish a free trade zone with the Pacific nations. There are a bunch of free trade agreements being signed by different countries around the world. We have just scraped the surface of what is possible. I mean, one can only imagine a world in which there's zero tariffs everywhere and no restrictions, no baby food limitations on what you can bring in and out of the country. Just free movement of goods.
- 8.81** Based on what? Based on individual decisions, individual preferences. Such a world would be a lot richer than it is today and a lot more varied and a lot more interesting than it is today. And that's what we should be striving for, not more tariffs, but more trade. And to do that, the best thing a country can do economically, culturally, probably not politically, but from every other respect, is just to eliminate tariffs, eliminate restrictions on trade. It's what England did after the Corn Laws, unilateral free trade.
- 8.82** We want our citizens to benefit from the bounty of all the goods that exist in the world. If you guys over there don't want that for your citizens, that's your problem. Our citizens get to enjoy the bounty of the world. They get to be able to pick and choose where and from what they buy. That's the only rational trade strategy to have, if you care about the well-being of your people. That will increase jobs, increase production, increase wealth, increase choice, increase variety. It'll make life dramatically better.
- 8.83** And again, I often say things that are controversial, and I hold a lot of very controversial views. My views on trade are not, surprisingly, that controversial. Again, there's almost

no economist out there who doesn't think that unilateral free trade is the right policy. There was a Canadian economist who, when Trump put high tariffs on Canada, he immediately tweeted out there on Twitter. He said, "The rational response for Canada to do right now is lower tariffs on the United States."

- 8.84** United States has shot itself in the knee. It doesn't make any sense for Canada to shoot itself in the knee as well, just to show America, "You can commit suicide. We can do it, too." No, the solution should be the other way around. We're going to be lowering tariffs on every product coming into Canada from anywhere in the world. You guys want to screw your citizens? Go for it. We're not going to screw ours.
- 8.85** Of course, no politician in the world right now has the balls to do that, although it is interesting that a number of these so-called trade deals that have been signed, America is placing 15, 20% tariffs on the other country, and the other countries are saying, "We're OK with zero tariffs." So a lot of political leaders out there in the world get it. A lot of people out there in the world get it.
- 8.86** Now, they're getting pushback from local industries that are worried about competition, but they should embrace that. You'd expect that in politics. But many of these countries are going zero tariffs with the United States. I don't know of any country that has zero tariffs with the rest of the world. There might be. Well, Singapore, a place like Singapore. Hong Kong used to be like that. Hong Kong never had tariffs. And it went from being relatively poor and small to one of the richest places on the planet, with seven and a half million people in it. No tariffs.
- 8.87** Singapore has no tariffs and has a GDP per capita higher than the United States, and it's a tiny little place. But if you're small, in particular, you want a lot of trade because you need the specialization that other people have around you. What comes out of this, really? If you think about America, particularly America, what's unique about America, one of the things that makes America unique is the idea of the individual pursuing his happiness, the idea of the individual pursuing his values.
- 8.88** If you think about that, what that actually means with regard to trade is that, when you hear the slogans like buy America or buy local, they're super un-American. Buying American is un-American, or at least buying American, discriminating over other places, is un-American. What is American? Buy the best, cheapest, best deal possible for me. And if that's American, or if that's from France, or if that's from, I don't know, Nigeria, or if that's from some other place, it doesn't matter. I'm pursuing my happiness.
- 8.89** We function—what made America as successful as it is, is that we function on the principle of individuals pursuing their own interests, pursuing their own happiness, pursuing their own economic well-being, and the sum total of that is the wealthiest country in the world. Unless there's a moral issue, like I don't want to support the Chinese, or I don't want to support an oppressive regime, buy what makes you happy.

Buy what enhances your life the best. That's the American thing to do. And that's the economically right thing to do. Thank you.